Taxation and Reallocation in the Secondary Market for Capital Goods

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Investment subsidies like bonus depreciation make new investment cheaper, but they make selling used capital—*reallocation*—more expensive. In a vintage model, I show that technology determines whether investment subsidies raise or lower total reallocation. When vintages are sufficiently complementary, subsidies generate sufficiently strong lock-in that reallocation declines. This lock-in can be released by granting full expensing to new and used asset purchases, as happened following the 2017 Tax Cuts and Jobs Act (TCJA). I test the model using evidence from used asset transactions in Texas. When vintages are granted investment subsidies, they subsequently transact approximately 17.5% less. This is accompanied by no evidence of catch-up as capital ages. However, bonus vintages transact significantly more following TCJA, with a larger recovery for machines with more remaining basis, consistent with TCJA releasing lock-in. Altogether, the empirical patterns are consistent with complementarity across vintages, and indicate that reallocation was significantly depressed throughout the 2000s by bonus depreciation.

JEL-Classification: E22, G31, H25, L26

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1 Introduction

Like many other assets, plant and equipment are tradable goods, and taxes are due on realized gains when they are sold. If the original owner takes up accelerated depreciation on a capital good, that leads to a higher tax bill at resale by reducing the asset's remaining tax basis. This is especially salient since 9/11, as investment stimulus policies have typically accelerated depreciation deductions. This paper studies how these subsidies affect capital reallocation. I show theoretically that tax policies which stimulate new capital formation can impede its subsequent reallocation and provide evidence that this tension is empirically important, with important implications for the design of investment subsidies.

Investment subsidies, which stimulate new capital accumulation (Zwick and Mahon 2017), shape capital reallocation through two opposing forces. First, by stimulating new investment, a subsidy may expand the capital stock, leaving a larger quantity of capital goods to be sold in the future, which would increase reallocation (Lanteri and Rampini 2023). I refer to this as the capital stock expansion channel. However, when a firm sells a depreciable asset, it is taxed on the difference between the sale price and the portion of the original purchase price that has not yet been deducted—the tax basis. Because this tax is realized only when the asset is sold, it can discourage reallocation and lock assets into their current owners (Feldstein, Slemrod, and Yitzhaki 1980; Hassett and Viard 2007; Hines 2016). Policies like bonus depreciation strengthen this force by reducing the tax basis carried by the subsidized asset, raising the effective capital gains tax rate on sale. I refer to this as the lock-in channel.[1] Whether an investment subsidy ultimately increases reallocation therefore cannot be inferred from its effect on new investment alone.

To study these two channels, I develop a new theoretical framework for secondary capital markets in a multi-age vintage model incorporating the tax treatment of capital purchases and sales.[2] In the model, capital lives for four periods. Unconstrained firms are taxed at the corporate rate and produce using new capital and retained capital of ages one through three. Each physical unit of older capital is less efficient than a unit of new capital, so the firm may prefer to sell it and replace it with new capital. Constrained firms cannot invest in new capital and instead purchase used assets, producing with a similar CES technology over vintages. This asymmetry generates a used capital market at each capital age, resulting in prices of used assets adjusting to clear the market. New investment receives a cohort-specific depreciation schedule that can be accelerated through bonus depreciation, while the tax due upon sale depends on the asset's

1. The purchaser of a used asset also receives a new tax basis equal to the price it pays, so depreciation allowances for used purchases affect demand for used capital.

2. The vintage structure is related to Edgerton (2011), Lanteri (2018), and Lanteri and Rampini (2023). The key distinction is that I attach a depreciation schedule to each purchase cohort and jointly model the corporate tax rate, the taxation of resale, and depreciation allowances for new and used purchases.

remaining basis. Purchasers of used assets receive a new basis equal to the purchase price, with their own eligibility for bonus depreciation.

The model's central prediction is that investment subsidies for new capital have ambiguous effects on total reallocation, and that complementarity determines whether lower young sales are offset at older ages. On the one hand, bonus reduces the cost of new investment and expands the stock of capital that can subsequently be sold. This is the capital stock expansion channel. On the other hand, by reducing the asset's remaining basis, bonus reduces the tax benefit of selling young capital and strengthens the incentive to retain it. This is the lock-in channel. Young sales fall when the increase in retention exceeds the increase in investment, but whether total reallocation falls also depends on the extent to which sales at older ages offset this decline. With substitutable vintages, including the standard perfect-substitutes benchmark, total reallocation instead rises whenever investment rises, even if young sales fall, because older sales more than offset the early decline. I show that when new and retained vintages are complementary, additional new investment raises the marginal product of retained capital, reinforcing lock-in. If this retention response is sufficiently strong, later sales do not fully offset the early decline, so bonus creates more capital while causing less of it to be subsequently reallocated. Finally, when new purchases are fully expensed, extending bonus to used purchases releases lock-in and, with complementary vintages, shifts transactions toward younger assets. Indeed, fully symmetric full expensing for new and used equipment restores the efficient allocation. This extends the result from Abel (2007) to secondary markets for capital.

I test the predictions of the model using the universe of financed used equipment transactions in Texas from 2001 through 2019. These data, also known as Uniform Commercial Code (UCC) filings, allow me to study the quantity of physical transactions and have previously been used in other settings to study used asset transactions (Edgerton 2011; Ma, Murfin, and Pratt 2022). I construct a parser to identify each transaction's equipment type, manufacturer, and model year, which allows me to trace each capital vintage's reallocation given its tax treatment. With that, I study two policy experiments. First, I compare subsequent transactions involving equipment from bonus and nonbonus purchase vintages, when bonus generally applied only to new capital. Second, I examine what happened to reallocation following the 2017 Tax Cuts and Jobs Act, which extended bonus eligibility to used asset purchases. These comparisons test whether the subsidy discourages the subsequent circulation of the capital it creates and whether extending it to used purchases releases lock-in.

I use the UCC filings to compare how bonus depreciation, a common investment subsidy, affected the likelihood of bonus vintages transacting. Comparing machines within manufacturers at the same asset age, I find that bonus vintages transact approximately 17.5 percent less than equipment from nonbonus vintages. This means that, for example, a four-year-old 2008 Chevrolet

truck in 2012—which received bonus depreciation—would transact roughly 17.5 percent less than a four-year-old 2007 Chevrolet truck in 2011. Since equipment in my sample ordinarily has no remaining tax basis after age seven, differences in remaining original deductions become less relevant as capital ages. Consistent with this, bonus vintages transact substantially less between ages two and seven, with smaller differences among older equipment. This shortfall in bonus transactions does not appear to experience any catch-up in later years, indicating that the effect on depressed reallocation is persistent. Indeed, these patterns are consistent with the model's prediction that complementarity across vintages can allow lock-in to dominate the capital stock expansion channel, reducing total reallocation rather than simply delaying young sales.

The 2017 Tax Cuts and Jobs Act provides a complementary test by extending bonus depreciation to qualifying used asset purchases, making both new and used equipment eligible for full expensing. I find that, following the reform, the age distribution of transactions shifts toward younger assets: the share involving equipment aged one through seven rises from approximately 51.7 percent in 2012–2017 to 61.2 percent in 2018–2019. This corresponds exactly with the model's prediction. Moreover, relative to nonbonus vintages, transactions involving preexisting bonus vintages rise by approximately 18.6 percent after TCJA. This recovery is larger for machines that would still have had more tax basis remaining under ordinary depreciation, linking the response to the deductions brought forward by the original subsidy. Moreover, I find, using an event study approach, that the timing is consistent with a causal interpretation of TCJA releasing lock-in. Together with the pre-TCJA evidence, the findings suggest that the effect of investment subsidies on reallocation depends on their treatment of used as well as new capital.

The model and evidence from UCC filings in Texas are broadly supported by additional data. In particular, I find that firms which are more exposed to bonus depreciation in Compustat are less likely to sell their capital even when they invest more in new capital. Moreover, using the Statistics of Income, I show that bonus take-up rises less with the statutory bonus rate in industries where capital is more tradable. This implies that the tax mechanism is salient to firms when choosing whether to accept the subsidy; if capital had no secondary market and thus was untradable after purchasing it, then tradability would have no effect on bonus take-up. Finally, using industry-level data from the Census Bureau's Business Dynamics Statistics, I find that industries more exposed to bonus experience larger subsequent declines in entry, with larger declines in industries where capital is more readily traded. These results are consistent with the model's prediction that investment subsidies can affect entry through secondary markets for capital goods.

In sum, I find that subsidies for new investment can create more capital while causing less of it to be subsequently reallocated. This result is quantitatively and economically significant. Aggregated upward, back-of-the-envelope calculations suggest that lower-bonus or no-bonus counterfactuals would have increased reallocation by roughly $14–$29 billion per year from 2012–

2017. These findings stand in contrast to the standard argument that these policies expand the capital stock and, consequently, the supply of used assets to young and financially constrained firms (Smith 2009; Lanteri and Rampini 2023).

Related Literature. This paper relates to three main strands of literature.

First, my paper builds on the literature on secondary markets for capital and capital reallocation (Eisfeldt and Rampini 2006). In my model, constrained firms purchase used capital from unconstrained firms. This structure is motivated by Eisfeldt and Rampini (2007), who show that financially constrained firms rely disproportionately on used capital, and Ma, Murfin, and Pratt (2022), who show that startups purchase older machines and that local used capital availability shapes startup hiring and subsequent investment. Darmouni and Sutherland (2024) show that when it is difficult to acquire new capital, many firms turn toward purchasing used assets, and this increases competition with new firms. This increase in demand endogenously raises the price of used assets. With that in mind, I follow Lanteri (2018), in endogenizing the price of used assets to clear the market, and show that this equilibrium feedback propagates into investment, retention, and transactions across capital ages. The closest theoretical paper is Lanteri and Rampini (2023). They show that subsidizing new investment expands the future supply of old capital, which generates the capital stock expansion channel and provides an efficiency rationale for investment subsidies. While I retain this force, I add an opposing lock-in channel generated by how investment subsidies affect the tax treatment of future resale.[3]

Second, my paper contributes to the literature on how taxation affects reallocation of durable assets, including housing and financial capital. For example, on the seller side, realization-based taxation can discourage sales because tax on any gain becomes due only when an asset is sold. This lock-in effect has been documented for corporate stock (Feldstein, Slemrod, and Yitzhaki 1980; Ivković, Poterba, and Weisbenner 2005), corporate dispositions of property, plant, and equipment (Desai and Gentry 2004), and depreciable business assets more generally (Hines 2016).[4] In the model, the wedge generated by the lock-in effect is jointly determined by the corporate tax rate and the asset's remaining basis, so an investment subsidy can create a subsequent incentive to retain the subsidized asset. On the buyer side, a taxable transfer establishes a new depreciable basis, and the resulting deductions can encourage trade. Prior work shows that this force can generate tax-motivated transfers of real estate and structures (Hendershott and Ling 1984; Gordon, Hines, and Summers 1987; Hassett and Viard 2007). I incorporate both sides of the transaction: the seller carries the basis created by the asset's original depreciation schedule, while

3. Baley, Blanco, and Oviedo (2025) study how corporate taxes interact with fixed adjustment costs and an exogenous purchase–resale price wedge.

4. Related work on housing finds that transaction taxes distort prices and reduce trade (Kopczuk and Munroe 2015; Best and Kleven 2018).

the buyer receives a new basis equal to the purchase price, whose deductions may themselves be accelerated. A smaller empirical literature shows that tax policy toward new capital can affect contemporaneous used equipment prices and sales (Smith 2009; Edgerton 2011; Basu, Kim, and Singh 2024). I ask what happens one step later, when the tax-favored machine itself becomes old and can return to the market. Because it carries the depreciation schedule from its purchase vintage, the subsidy that encouraged its initial purchase may discourage its subsequent sale. I place these seller- and buyer-side forces in an equilibrium market in which used prices, new investment, and retention across capital ages are endogenous and test the model's vintage-specific predictions using individual equipment transactions, examining both original bonus treatment and the subsequent extension of eligibility to used purchases.

Finally, this paper contributes to an empirical literature on the effects of business tax policy on investment and entry. Following Hall and Jorgenson (1967), this literature typically studies the effects of tax policy through its effect on the user cost of capital. For example, House and Shapiro (2008) show that temporary bonus depreciation creates strong incentives to move investment into the subsidy window, while Kitchen and Knittel (2011), Zwick and Mahon (2017), and Ohrn (2019) provide empirical evidence that accelerated depreciation raises investment. This new investment response is evident in my own model, but it is the beginning of the analysis. Because accelerated depreciation also reduces the tax basis of the capital being purchased, it changes the incentive to sell that capital later. I therefore follow the same reforms from new investment into subsequent transactions of bonus vintages and their recovery after the Tax Cuts and Jobs Act extended bonus eligibility to used purchases. Whereas the investment literature asks how strongly tax policy stimulates capital formation, I show that this response does not reveal whether the capital it creates is subsequently reallocated. Moreover, Sapollnik and Swonder (2025) and Fairlie et al. (2026) show, at a different level of aggregation than I use, that corporate tax cuts increase business entry. In supplementary analysis, I study a different policy margin and show that, at the industry level, investment subsidies reduce entry through the secondary market for capital goods.

2 Used Capital Markets and their Tax Treatment

The secondary market for physical capital is large. Setting aside markets for residential and nonresidential structures, the used equipment market alone is roughly 10–15 percent as large as the market for new equipment, meaning that 2–3 percent of the equipment stock churns every year. In dollar terms, that translates to well over $100 billion annually.[5] These markets play a key role

5. See Appendix Figures A.1a and A.1b for these magnitudes over time.

in easing the financial constraints of young firms. Indeed, evidence from Ma, Murfin, and Pratt (2022) shows that buyers of used assets skew young and financially constrained, while sellers tend to be larger. These mutually beneficial trades allow entrants access to used capital, while permitting larger corporations, which Appendix Figure A.2 shows are typically in trucking, agriculture, and construction, to finance new equipment purchases. Consequently, used asset markets facilitate reallocation of capital, which both eases financial constraints and makes investment more reversible.[6]

Tax policy plays a key role in organizing the incentives to buy and sell used capital, thereby affecting both sides of the segmented market for used assets. The remainder of this section discusses that in detail.

Rules for Buying Capital. Historically, the government has subsidized investment through a mixture of tax credits and accelerated depreciation policies. The latter has been dominant since the Tax Reform Act of 1986. Under the Modified Accelerated Cost Recovery System (MACRS), capital is depreciated over a prescribed recovery period that typically understates its economic useful life. Since firms discount the future at a positive rate (Gormsen and Huber 2022), one dollar of tax depreciation spread over several years has less than one dollar of value in present value terms. To see that concretely, suppose a non-negative fraction d_t of a dollar of investment can be deducted in each period t, where t measures time since the purchasing firm places the asset in service, with $\sum_{t=0}^{T} d_t = 1$ and some deductions occurring after the purchase period. Then, with a discount rate $r > 0$, the present value of those deductions is

$$z = \sum_{t=0}^{T} \frac{d_t}{(1+r)^t} < 1.$$

Under ordinary MACRS, the same depreciation schedule generally applies to used and new asset purchases symmetrically. For a used asset purchase, the buyer's tax basis is reset at the purchase price, regardless of the seller's remaining basis. Used assets, however, are physically older and so have typically depreciated economically, so they usually command a lower price and may even be scrapped before the scheduled recovery period ends. If the asset is held through its scheduled recovery period, the associated tax savings have present value τz, where τ is the business income tax rate. If an investment tax credit c applies, the tax-adjusted price of a dollar of investment is therefore $1 - c - \tau z$.

Bonus depreciation, which has been studied extensively with respect to new capital (House and Shapiro 2008; Kitchen and Knittel 2011; Zwick and Mahon 2017), allows firms to deduct

6. Kermani and Ma (2022) show that asset resale values tend to be higher in agriculture and construction, which indicates a higher degree of reversibility in those markets.

a percentage θ of investment immediately, with the remainder following the standard MACRS schedule. The modified present value of depreciation allowances is

$$\widetilde{z} = \theta + (1 - \theta)z.$$

The associated tax savings are $\tau\widetilde{z}$, so a higher bonus rate lowers the tax-adjusted purchase price. As a result, many studies, like those referenced above, have found significant effects of bonus on new investment.

Historically, bonus has applied differentially to new and used equipment purchases. Until the 2017 Tax Cuts and Jobs Act, bonus only applied to new investment. Bonus was first introduced at 30% in 2002, raised to 50% in 2003, allowed to lapse in 2005, reinstated at 50% in 2008, temporarily raised to 100% from late 2010 through 2011, and raised again to 100% for several years following the 2017 Tax Cuts and Jobs Act (TCJA). The TCJA extended bonus depreciation to qualifying used property acquired after September 27, 2017, making used equipment eligible for bonus depreciation for the first time.

As firms claim deductions over time for an asset, the portion of its purchase price not yet deducted declines. The remaining amount is the asset's tax basis. Below, I discuss how the remaining tax basis figures prominently into tax rules around selling used assets.

Rules for Selling Used Capital. The statutory rules governing sales of used capital have been relatively stable since the Tax Reform Act of 1986 (TRA86), but the tax due when an asset is sold has varied with depreciation policy and the business income tax rate. Since TRA86, the basic principle has been that firms pay tax on the difference between the sale price and the asset's remaining tax basis. To see that, suppose an asset purchased in period s is sold in period t. If q_s is the purchase price and d_i is the fraction of that price deducted in period i, its adjusted basis at sale is

$$B_t = q_s \left(1 - \sum_{i=s}^{t} d_i\right),$$

where d_t includes any allowable sale-year depreciation. If the firm sells the asset for q_t, its after-tax sale proceeds are

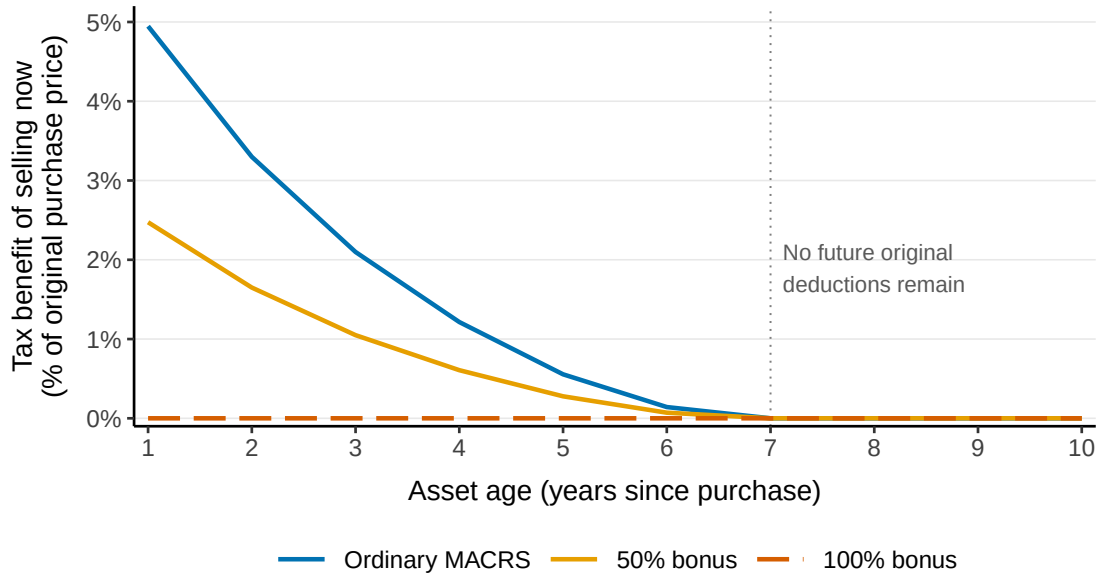
$$q_t - \tau(q_t - B_t) = (1 - \tau)q_t + \tau B_t.$$

A gain attributable to depreciation deductions is known as depreciation recapture.[7] This expression describes the tax due conditional on sale. Holding the sale price fixed, the liability is larger

7. Strictly, §1245 recapture applies only to the portion of the gain attributable to depreciation deductions, i.e., $\min(q_t - B_t,\ q_s - B_t)$ when the sale generates a gain. Any gain above the original purchase price, $q_t - q_s$, is generally subject to §1231 treatment. For used equipment sold below its original purchase price, the entire gain is recapture. For corporations, the distinction is immaterial in either case because both are taxed at rate τ.

when deductions have reduced B_t or when the business income tax rate is higher and the sale generates a gain. Bonus depreciation increases the tax due upon a subsequent sale by reducing the asset's basis more quickly. Under full expensing, $B_t = 0$, so the entire sale price is included in taxable income.

Figure 1: Bonus Depreciation and the Tax Benefit of Selling Capital Early



Notes: The figure plots $\tau \sum_{j\geq 1}(1-\beta^j)D_{t+j}$ as a share of the original purchase price for a \$100 seven-year MACRS asset under the half-year convention. The tax rate is $\tau = 0.35$, the discount rate is $r = 0.10$, and $\beta = 1/(1+r)$. The deductions D_{t+j} would be claimed in future periods if the asset were retained. Sale proceeds remain taxable under full expensing.

While the decision about whether and when to sell an asset depends on its productivity, whether a replacement exists, and other economic factors, a narrow tax tradeoff exists between selling in the present and retaining into the future. If the asset is retained, its remaining basis is recovered through the remaining depreciation deductions. If it is sold, the remaining basis offsets taxable sale proceeds immediately. Since firms discount future deductions, selling brings forward the recovery of basis and raises the present value of the associated tax savings. The value of doing that has a precise expression. Let D_{t+j} denote the deduction the firm would claim j periods later if it retained the asset, and let $\beta = 1/(1+r)$. The tax benefit from accelerating those deductions through a sale is

$$\underbrace{\tau \sum_{j\geq 1} D_{t+j}}_{\text{selling}} - \underbrace{\tau \sum_{j\geq 1} \beta^j D_{t+j}}_{\text{retaining}} = \tau \sum_{j\geq 1}(1-\beta^j)D_{t+j}.$$

When full expensing makes all future D_{t+j} zero, the tax benefit of selling earlier goes away. Fig-

ure 1 makes this clear. The figure plots $\tau \sum_{j\geq 1}(1-\beta^j)D_{t+j}/q_s$ by age for an example seven-year asset under a 10% discount rate, which is the tax benefit of selling now. The tax benefit of selling early declines as the asset ages and is lower under higher bonus. With full expensing, there is no tax benefit to selling early. Consequently, bonus encourages investment—as is widely recognized in public finance—but the same policy also reduces the tax incentive to sell capital earlier.[8] Nevertheless, the decision about whether and when to sell depends on a variety of economic factors in equilibrium, which the next section takes seriously in a vintage model of capital.

3 A Model of Capital Investment and Sales

This section studies how firm choices over investment in new capital and sales of used capital vary in equilibrium when tax policy changes. I use a stylized vintage capital model to separate two competing margins. On the one hand, a literature on the effects of investment subsidies would suggest that investment subsidies would increase reallocation by increasing the stock of new assets (Lanteri and Rampini 2023). However, theoretical and empirical studies of the effects of higher capital gains tax rates on housing (Kopczuk and Munroe 2015) and financial assets (Ivković, Poterba, and Weisbenner 2005) would suggest that the higher effective tax rates following bonus would generate lock-in, leading to lower reallocation. The model I develop below delineates the conditions under which each margin may dominate, and shows that it is possible for investment subsidies to simultaneously lead to a larger stock of new capital *and* lower reallocation.

3.1 Environment

I study secondary markets for capital goods in discrete time $t = 0, 1, \ldots$ using a vintage model. There are two types of firms. *Unconstrained firms* invest in a new vintage v at date v. The vintage lives for four periods, from v through $v+3$, and is scrapped at $v+4$. From $v+1$ onward, some of the vintage may be purchased by *constrained firms*, which cannot invest in new capital. Superscripts u and c denote unconstrained and constrained firms, respectively.[9] In practical terms, the three used ages represent relatively young equipment with substantial ordinary tax life remaining, equipment approaching the end of its tax recovery period, and fully depreciated equipment that remains productive. As I describe in detail below, tax treatment differs between purchases of

8. Until 2018, sellers could also defer recapture by exchanging equipment for like-kind property under §1031. TCJA changed several of these rules simultaneously: it increased bonus depreciation for new capital, extended bonus depreciation to used property, reduced the corporate tax rate from 35 to 21 percent, and repealed like-kind treatment for equipment.

9. This is motivated by evidence from Ma, Murfin, and Pratt (2022), which shows that used asset markets tend to be segmented: financially constrained and young firms tend to buy used capital, while unconstrained firms tend to supply it.

new and used capital, but the firms share a discount factor $\beta \in (0,1)$ and produce with similar CES technologies over vintages, as in Jovanovic and Yatsenko (2012). The price of new capital is exogenous, while the prices of used vintages are determined in equilibrium, with both firm types taking prices as given. The remainder of this subsection describes each firm type's decision problem in detail, along with how tax policy affects their optimal decisions.

New Capital Purchasers

There is a unit mass of identical unconstrained firms that produce an output good, which serves as the numeraire. Each firm produces output with production function

$$Y_t^u = \left[\omega_0 I_t^\rho + \sum_{a=1}^{3} \omega_a \left(\phi_a \widetilde{K}_{a,t} \right)^\rho \right]^{\nu/\rho}, \tag{1}$$

where the production function is CES over capital vintages, governed by a returns to scale parameter $\nu \in (0,1)$, an elasticity of substitution between capital vintages $\sigma = \frac{1}{1-\rho} > 0$, and weights $\omega_a > 0$ satisfying $\sum_{a=0}^{3} \omega_a = 1$.[10] At $\rho = 0$, I use the continuous Cobb–Douglas limit.

At the beginning of each period t, each firm decides how much to invest in a new vintage, I_t. I assume that new investments are elastically supplied at a price of one and are immediately available for production. The efficiency of capital of age a is ϕ_a, with $1 = \phi_0 \geq \phi_1 \geq \phi_2 \geq \phi_3 > 0$. An investment I_v made at date v becomes age-one capital at date $v+1$. At that time, the firm chooses how much to sell, $S_{1,v+1}$, at price $q_{1,v+1}$, leaving retained capital $\widetilde{K}_{1,v+1} = I_v - S_{1,v+1}$. Here, the first subscript denotes age and the second denotes calendar time. This process of choosing how much to sell and retain continues at ages two and three, until the capital is scrapped at $v+4$. The following diagram illustrates this.

Tax Policy. The firm's tax treatment has three relevant components. First, operating income is taxed at rate $\tau \in [0,1)$. I assume that firms can use all deductions at rate τ and that gains and losses from asset sales are taxed symmetrically.

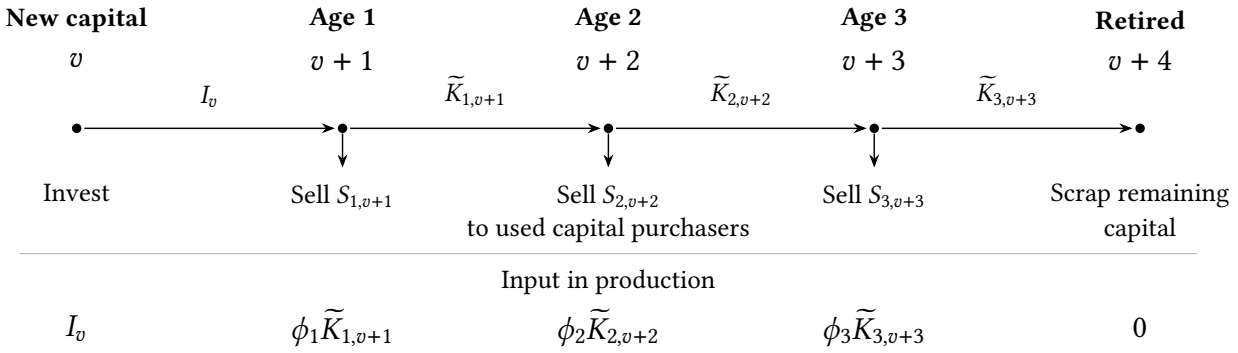
Second, the firm faces a tax depreciation schedule. Its ordinary schedule, which is the stylized model counterpart to MACRS, is

$$\bar{d}_0, \quad \bar{d}_1, \quad \bar{d}_2 \geq 0, \qquad \bar{d}_0 + \bar{d}_1 + \bar{d}_2 = 1.$$

If the firm purchases a unit of capital in period t, a fraction $\bar{d}_0$ would be deducted immediately,

10. This formulation resembles a stylized version of the neoclassical benchmark with ν taking the role of the capital share. In the perfect-substitutes limit, $\rho \to 1$, the firm's production can be written as $Y^u = K^\nu$, where $K = \omega_0 I + \sum_{a=1}^{3} \omega_a \phi_a \widetilde{K}_a$ is a linear aggregate of new and retained capital (Fisher 1965).

Figure 2: Investment, sales, and retention for a vintage purchased at date v.



$\bar{d}_1$ in the following period, and the remainder two periods after purchase. Consequently, tax deductions end two periods before scrappage, which reflects the empirical reality that the useful life of an asset typically exceeds its tax life. Under ordinary depreciation, age-one capital still has future deductions available, age-two capital is in its final deduction period, and age-three capital is fully tax-depreciated but remains productive. I am interested in how bonus depreciation, which brings those deductions forward in time, affects the firm's decisions about investment and sales of capital. I model bonus depreciation at rate $\theta^N \in [0,1]$ as

$$d_0(\theta^N) = \theta^N + (1-\theta^N)\bar{d}_0, \qquad d_j(\theta^N) = (1-\theta^N)\bar{d}_j, \quad j = 1,2. \tag{2}$$

When $\theta^N = 0$, the firm faces the ordinary depreciation schedule and when $\theta^N = 1$, all new investment is immediately tax-deductible. Intermediate values pull a proportional amount of depreciation forward to the present. Conditional on retaining capital through its deduction dates, the present value of depreciation tax savings per unit is $\tau\left[d_0(\theta^N) + \beta d_1(\theta^N) + \beta^2 d_2(\theta^N)\right]$.

Third, I assume that the firm is taxed on asset sales on the price net of tax basis at rate τ. Suppose a machine of age $a \in \{1,2,3\}$ is sold in period t. Its remaining basis immediately before sale is

$$B_1(\theta^N) = d_1(\theta^N) + d_2(\theta^N), \qquad B_2(\theta^N) = d_2(\theta^N), \qquad B_3(\theta^N) = 0.$$

Mirroring the treatment of recapture in the tax code, the seller's after-tax proceeds per unit of capital sold are

$$(1-\tau)q_{a,t} + \tau B_a(\theta^N).$$

The sale ends the firm's depreciation deductions on that capital. Conversely, if age-one capital is retained rather than sold, the firm deducts $d_1(\theta^N)$ in period t, leaving $d_2(\theta^N)$ to be deducted in period $t+1$ if the firm still holds the capital. Thus, the tax benefit of accelerating the remaining deduction by selling at age one is $\tau(1-\beta)d_2(\theta^N)$.

Remark (Depreciation). In this model, a vintage's age-price profile determines its economic depreciation profile. As discussed later on, these prices are determined in equilibrium, which makes depreciation endogenous. Importantly, the efficiency profile influences these prices, but that *physical* process is distinct from economic depreciation. Tax depreciation is a separate policy object, specified independently of the physical efficiency profile, but it also influences equilibrium economic depreciation. This distinction between physical efficiency, retirement, policy, and economic depreciation mirrors the treatment in Feldstein and Rothschild (1974).

Optimal Investment and Sales. The firm chooses investment in new capital and sales of used capital, taking prices as given. The state consists of the inherited capital stocks $K_{1,t}, K_{2,t}, K_{3,t}$, and retained capital satisfies $\widetilde{K}_{a,t} = K_{a,t} - S_{a,t}$. Under permanent tax policy and stationary prices $q_{a,t} = q_a$, the firm's Bellman equation is

$$V^u(K_{1,t}, K_{2,t}, K_{3,t}) = \max_{\substack{I_t \geq 0 \\ 0 \leq S_{a,t} \leq K_{a,t},\ a=1,2,3}} \Big\{ (1-\tau) Y_t^u - \big(1 - \tau d_0(\theta^N)\big) I_t + \sum_{a=1}^{3} \big[(1-\tau) q_a + \tau B_a(\theta^N) \big] S_{a,t} + \tau d_1(\theta^N) \widetilde{K}_{1,t} + \tau d_2(\theta^N) \widetilde{K}_{2,t} + \beta V^u\big(I_t, \widetilde{K}_{1,t}, \widetilde{K}_{2,t}\big) \Big\}, \tag{3}$$

where output is given by the production function above. The continuation value reflects the aging of investment and retained capital.

At an interior stationary allocation, write I for investment and $\widetilde{K}_a$ for the quantity of age-a capital retained by the firm, with $0 < \widetilde{K}_3 < \widetilde{K}_2 < \widetilde{K}_1 < I$. The optimality conditions are

$$\frac{\partial Y^u}{\partial I} = 1 + \frac{\tau(1-\beta)}{1-\tau} \big[d_1(\theta^N) + d_2(\theta^N) \big] - \beta q_1, \tag{4}$$

$$\frac{\partial Y^u}{\partial \widetilde{K}_1} = q_1 - \beta q_2 + \frac{\tau(1-\beta)}{1-\tau} d_2(\theta^N), \tag{5}$$

$$\frac{\partial Y^u}{\partial \widetilde{K}_2} = q_2 - \beta q_3, \tag{6}$$

$$\frac{\partial Y^u}{\partial \widetilde{K}_3} = q_3. \tag{7}$$

Equation (4) governs investment in new capital, while the remaining conditions govern the decisions to sell or retain used capital. The optimality condition for new investment states that its marginal product is equal to its marginal cost. While the marginal cost involves two standard terms—the unit price of new investment and the cost of delayed tax deductions—it also involves a term βq_1. This captures the value of being able to sell capital in the following period, which encourages additional investment today. The optimality conditions (5)–(7) relate the marginal product of retaining capital to its marginal cost: foregone sale proceeds net of discounted fu-

ture resale value. When future deductions remain, selling also accelerates their recovery. This additional tax benefit appears only in the age-one condition.

Studying the optimality conditions individually reveals a tension. When bonus goes up, $d_1(\theta^N)$ and $d_2(\theta^N)$ decline. Holding the other capital inputs and prices fixed, the reduction in $d_1(\theta^N) + d_2(\theta^N)$ reduces the marginal cost of new investment, which stimulates more capital creation. But the reduction in $d_2(\theta^N)$ also reduces the benefit of accelerating deductions by selling age-one capital. This lowers the marginal cost of retention and generates a lock-in effect. As I discuss later, the strength of these responses depends on substitution across vintages. When new and retained capital are complements, additional new investment raises the marginal value of retained capital and reinforces retention. When they are substitutes, additional investment lowers the marginal value of retained capital and weakens the retention incentive. The following sections close the model with demand for used capital and illustrate when each force is stronger in equilibrium.

Used Asset Purchasers

On the demand side of the used capital market, I assume there is a unit mass of constrained firms that produce the output good with production function

$$Y_t^c = A^c \left[\sum_{a=1}^{3} \eta_a \left(\phi_a K_{a,t}^c \right)^\rho \right]^{\nu/\rho}, \tag{8}$$

where $K_{a,t}^c$ denotes holdings of age-a capital available for production, $A^c > 0$ is a productivity parameter, and the weights satisfy $\eta_a > 0$ and $\sum_{a=1}^{3} \eta_a = 1$. The parameters ν and ρ and the efficiency profile ϕ_a are the same as for unconstrained firms.

I make the substantive assumption that constrained firms cannot invest in new capital goods and do not resell any used goods they purchase.[11] Instead, they are restricted to the following conditions. At the beginning of each period t, each firm chooses how much capital of each age $a \in \{1, 2, 3\}$ to purchase, $U_{a,t}$, at price $q_{a,t}$. These purchases are immediately available for production, and the firm holds each unit until it is scrapped upon reaching age four. Consequently, the firms hold capital of each vintage consisting of both current purchases and capital acquired in earlier periods:

$$K_{1,t}^c = U_{1,t}, \quad K_{2,t}^c = K_{1,t-1}^c + U_{2,t}, \qquad K_{3,t}^c = K_{2,t-1}^c + U_{3,t}. \tag{9}$$

11. Allowing these firms to purchase new capital would introduce an additional margin of substitution between new and used assets. Allowing resale would make their purchase decisions depend on future resale values and tax basis. The resale restriction also rules out repeated transfers that restart depreciation allowances, as studied by Gordon, Hines, and Summers (1987).

Any profits earned by these firms are taxed at τ. Following the tax code, a unit of used capital purchased at price $q_{a,t}$ receives a new tax basis equal to that price, which restarts at the ordinary depreciation schedule. Those purchases receive a bonus depreciation rate $\theta^U \in [0,1]$, which may differ from the bonus rate on new purchases. Importantly, I assume that any basis remaining when the asset is scrapped is deducted at retirement, so an age-three purchase recovers any remaining basis in the following period. Under the hold-until-retirement assumption, the tax benefits at the purchase date can be summarized by the present value of deductions per dollar purchased:

$$\widetilde{z}_1^c = \widetilde{z}_2^c = d_0(\theta^U) + \beta d_1(\theta^U) + \beta^2 d_2(\theta^U), \qquad \widetilde{z}_3^c = d_0(\theta^U) + \beta\big[d_1(\theta^U) + d_2(\theta^U)\big].$$

Consequently, the after-tax acquisition cost per unit of age-a capital is $q_{a,t}\big(1-\tau\widetilde{z}_a^c\big)$. Note that this is the same representation as appears in, for example, Zwick and Mahon (2017), and many other papers in public finance on the tax treatment of investment good purchases.

Optimal Purchases. Taking prices as given, firms choose how much used capital to purchase. For them, the state consists of $K_{1,t-1}^c$ and $K_{2,t-1}^c$, which enter the current period as age-two and age-three capital. Under permanent tax policy and stationary prices $q_{a,t} = q_a$, the firm's Bellman equation, with depreciation benefits incorporated into acquisition costs and fixed tax benefits from past purchases omitted, is

$$\begin{aligned} V^c(K_{1,t-1}^c, K_{2,t-1}^c) = \max_{\substack{U_{a,t}\geq 0 \\ a=1,2,3}} \Big\{ &(1-\tau)Y_t^c - \sum_{a=1}^{3} q_a\big(1-\tau\widetilde{z}_a^c\big)U_{a,t} \\ &+ \beta V^c(K_{1,t}^c, K_{2,t}^c)\Big\}, \end{aligned} \tag{10}$$

where current holdings satisfy the accumulation equations above. The continuation value reflects the aging of these holdings. At an interior stationary allocation, write K_a^c for capital holdings and U_a for purchases. The optimality conditions are

$$q_1\big(1-\tau\widetilde{z}_1^c\big) = (1-\tau)\left[\frac{\partial Y^c}{\partial K_1^c} + \beta\frac{\partial Y^c}{\partial K_2^c} + \beta^2\frac{\partial Y^c}{\partial K_3^c}\right], \tag{11}$$

$$q_2\big(1-\tau\widetilde{z}_2^c\big) = (1-\tau)\left[\frac{\partial Y^c}{\partial K_2^c} + \beta\frac{\partial Y^c}{\partial K_3^c}\right], \tag{12}$$

$$q_3\big(1-\tau\widetilde{z}_3^c\big) = (1-\tau)\frac{\partial Y^c}{\partial K_3^c}. \tag{13}$$

Each condition equates the purchase price net of depreciation tax savings to the discounted marginal output generated over the asset's remaining life. An age-one purchase provides three periods of production, an age-two purchase two, and an age-three purchase one. Bonus on used

capital brings deductions forward and reduces acquisition costs at given prices, just like in a standard model along the lines of Hall and Jorgenson (1967).

3.2 Stationary Competitive Equilibrium

To obtain transparent analytical comparative statics, I consider constant policies and interior stationary equilibria.

Definition 1 (Stationary competitive equilibrium). Given constant policy parameters $(\tau, \theta^N, \theta^U)$, $\bar{d}_0, \bar{d}_1, \bar{d}_2$, a stationary competitive equilibrium consists of a price vector (q_1, q_2, q_3), value functions V^u and V^c and their associated decision rules, and stationary capital stocks and allocations such that:

(i) Given prices, the value functions solve (3) and (10), respectively, and the associated decision rules attain the maxima at every feasible state.

(ii) The stationary allocations are generated by these decision rules, and the firms' inherited capital stocks remain unchanged under their respective laws of motion.

(iii) The used capital market clears at each age:

$$S_a = U_a, \qquad a = 1, 2, 3.$$

Appendix B.1 shows that, under conditions I maintain throughout the rest of this section, a unique interior stationary competitive equilibrium exists with positive trading at every age, and the equilibrium varies smoothly with policy. Trade at all ages reflects differences in valuations from two sources: constrained firms need to buy used assets to produce since they cannot invest in new capital, and the two types of firms place different relative weights on younger and older vintages in their CES aggregators.

3.3 The Effect of Investment Subsidies on Used Asset Markets

Having characterized the stationary equilibrium, I now consider how variation in tax policy affects the markets for new and used capital. I start with bonus depreciation for new capital, then turn to bonus depreciation for used capital.

Bonus depreciation for new capital. Consider an increase in bonus depreciation for new investment, θ^N. In a stationary competitive equilibrium, how does it affect capital goods markets at each age? When θ^N rises, new investment becomes cheaper by bringing tax deductions forward in time. As long as investment rises, that expands the stock of assets that firms can sell, potentially increasing supply at subsequent ages. However, the policy also makes it more expensive for

the firm to sell young used capital at $v+1$, the period immediately following its original purchase, since its tax basis is lower, increasing the tax bill at a given resale price. This creates a tension between capital stock expansion and lock-in. I study how this tension manifests in two steps. First, I examine the equilibrium response of age-one sales. Here, the tension is evident from the equilibrium relationship between reallocation and investment, $U_1 = I - \widetilde{K}_1$. If $d\widetilde{K}_1/d\theta^N > dI/d\theta^N$, then lock-in dominates for age-one capital. Later, I examine the effect on *total* reallocation, including sales at ages two and three, where selling no longer accelerates a future deduction on the original purchase. The following proposition characterizes when lock-in dominates capital stock expansion for age-one sales in the first step.

Proposition 1 (Bonus Depreciation and Young Reallocation)**.** *Suppose $\rho \leq \nu$, $\tau > 0$, and $\bar{d}_2 > 0$. Around an interior stationary competitive equilibrium, an increase in θ^N, holding θ^U fixed, raises investment in new capital. Young reallocation, U_1, falls if and only if*

$$-\frac{\partial}{\partial I}\underbrace{\left(\frac{\partial Y^u}{\partial I}+\sum_{a=1}^{3}\beta^a\frac{\partial Y^u}{\partial \widetilde{K}_a}\right)}_{\text{discounted marginal products}}\Bigg|_{U_1} > \left[-\frac{\partial}{\partial I}\underbrace{\left(\frac{\partial Y^u}{\partial \widetilde{K}_1}-q_1+\beta q_2\right)}_{\substack{\text{net benefit of retaining}\\ \text{young capital}}}\Bigg|_{U_1}\right]\times\underbrace{\left(1+\beta+\frac{\bar{d}_1}{\bar{d}_2}\right)}_{\text{tax incentive ratio}}. \tag{14}$$

Reversing the inequality gives an increase in young reallocation; equality gives no first-order change.

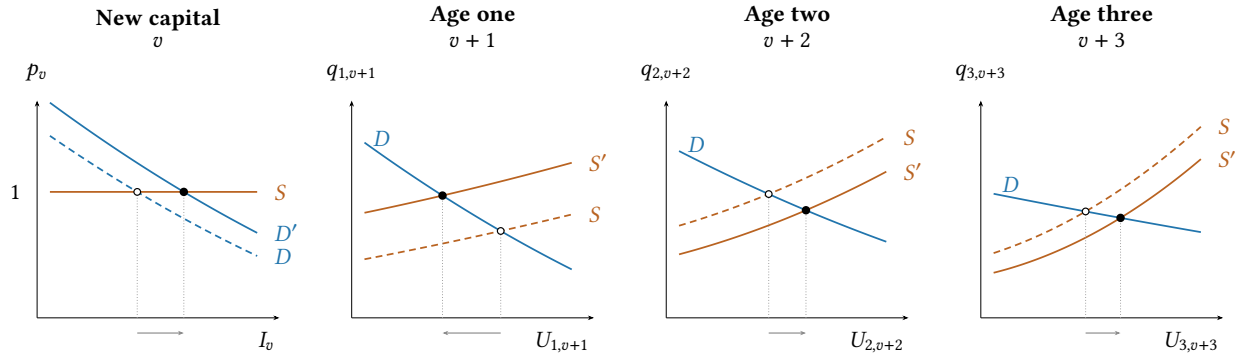
Proof. *See Appendix B.2.*

The inequality in (14), which governs whether lock-in dominates capital expansion for young used capital sales, is determined by two forces: technology and tax policy.[12] Two of the underbraces come from technological terms. The underbrace on the left-hand side collects the discounted marginal products of capital over its lifetime. How these change depends on returns to scale, captured by ν, substitution across vintages, and the capital mix. If they decline quickly as firms invest more, then bonus produces a smaller expansion in the stock of machines that later become available for sale. All else equal, that makes a decline in young reallocation more likely. The middle underbrace captures how the net benefit of keeping a machine, its marginal product less the current resale price plus its discounted future resale value, changes when investment rises. When vintages are complementary, new investment raises the marginal product of older capital, which lowers the incentive to reallocate existing capital. Next, tax policy is captured by the tax incentive ratio, which weights the technological terms. When bonus rises, deductions

12. The outer derivatives with respect to I hold U_1 fixed. Older holdings and all used capital prices adjust to satisfy the older-market equilibrium conditions.

associated with $\bar{d}_1$ come forward by one period and those associated with $\bar{d}_2$ by two periods at purchase, encouraging investment through both deductions. However, when a larger share of delayed deductions falls in $\bar{d}_2$, the ratio $\bar{d}_1/\bar{d}_2$ is lower. All else equal, bonus then reduces the tax benefit from selling young proportionately more relative to its investment incentive, making it more likely that the lock-in effect dominates.[13]

Figure 3: Bonus Depreciation Across Vintage Capital Markets.



Proposition 1 establishes that new investment can rise while reallocation can fall at age-one. The first two panels of Figure 3 illustrate this case, showing an outward shift in the demand for new assets at age-zero, with a retraction in supply at age-one. However, we have not yet characterized the effects of increasing bonus for new assets on age-two and age-three sales. These markets differ from the age-one market because selling no longer accelerates a future deduction on the original purchase. It could be that supply increases at age-two and age-three, illustrating a partial or full catch-up effect, or supply could contract in both markets. The right two panels illustrate the case in which supply expands for older assets, showing some catch-up in asset sales. In this case, Proposition 1 is simply a timing result: total reallocation may still rise even if it contracts for younger, more efficient assets. The following proposition characterizes when older sales offset the decline in young sales and when total reallocation falls, linking investment in new assets to its downstream effects on other asset markets.

Proposition 2 (Bonus Depreciation and Total Reallocation)**.** *Suppose* $\tau > 0$ *and* $\bar{d}_2 > 0$*. Let* $U = U_1 + U_2 + U_3$*. Around an interior stationary competitive equilibrium, holding* θ^U *fixed:*

(i) If $\rho \geq \nu$*, total reallocation rises whenever investment rises. At* $\rho = \nu$*, bonus raises both investment and total reallocation.*

13. This points to a potentially important difference between tax regimes. For instance, the ordinary depreciation schedule enacted by the Economic Recovery Tax Act of 1981—the Accelerated Cost Recovery System (ACRS)—generally accelerated deductions more quickly than the ordinary depreciation schedule enacted by the Tax Reform Act of 1986, the Modified Accelerated Cost Recovery System (MACRS). As a result, Proposition 1 suggests that, all else equal, lock-in would be stronger under MACRS than ACRS.

(ii) If $\rho < \nu$, an increase in θ^N raises investment and reduces total reallocation if and only if

$$\underbrace{\left(1 + \left.\frac{\partial U_2}{\partial U_1}\right|_I + \left.\frac{\partial U_3}{\partial U_1}\right|_I\right)\left(-\frac{dU_1}{d\theta^N}\right)}_{\textit{young lock-in, net of changes in older sales}} > \underbrace{\left.\frac{\partial U}{\partial I}\right|_{U_1}\frac{dI}{d\theta^N}}_{\textit{capital expansion}}. \tag{15}$$

Proof. *See Appendix B.3.*

The earlier proposition concerns whether lock-in happens for age-one capital, which is the only age at which there is a direct trade-off induced by bonus depreciation on the original purchase. That was determined by the sign of $\frac{dI}{d\theta^N} - \frac{d\widetilde{K}_1}{d\theta^N}$. This proposition examines whether young lock-in reduces total reallocation over the lifespan of bonus-treated capital.[14] Since total reallocation is given by $U = U_1 + U_2 + U_3 = I - \widetilde{K}_3$, the change in reallocation is given by the sign of $\frac{dI}{d\theta^N} - \frac{d\widetilde{K}_3}{d\theta^N}$. The proposition establishes that this is largely determined by technology and, in particular, the sign of $\rho - \nu$. When $\rho > \nu$, vintages are substitutable; when $\rho < \nu$, they are complements.

First, consider the case in which $\rho > \nu$. The limiting version of this case, in which $\rho \to 1$, yields the familiar Solow (1960) benchmark in which vintages are perfectly substitutable, and it also serves as the standard capital aggregator in macroeconomics and public finance. To see how these markets are linked, first hold investment fixed and consider a reduction in age-one sales, leaving constrained firms with less young capital. Since vintages are substitutable, more age-one capital lowers the marginal product of capital at ages two and three for unconstrained firms, increasing their supply on the used asset market. Since vintages are also substitutes for constrained firms, a reduction in the supply of young capital makes the marginal product of older capital higher, raising their demand for capital at ages two and three. After prices adjust, combined sales at ages two and three increase by more than the reduction in young sales when investment is fixed. Now, allowing investment to adjust, the proposition establishes that, with substitute vintages, total reallocation rises whenever investment rises, regardless of whether young sales rise or fall.

Next, suppose vintages are neither substitutes nor complements, i.e., $\rho = \nu$. Here, the marginal product of each vintage depends on its own quantity and is independent of the quantities of other vintages. Holding investment fixed, higher bonus still encourages unconstrained firms to retain more young capital, reducing age-one sales. However, there are now no cross-vintage effects for either side of the used asset market. Since $K_2^c = U_1 + U_2$ and $\Delta K_2^c = 0$ holding investment fixed, constrained firms compensate for the contraction in the age-one market by setting $-\Delta U_1 = \Delta U_2$.

14. In (15), the partial derivatives with respect to U_1 hold I fixed, while $\partial U / \partial I$ holds U_1 fixed. In both comparisons, older holdings and all used capital prices adjust to satisfy the older-market equilibrium conditions.

Age-three sales would not change when investment is held fixed. When investment adjusts, bonus increases the total number of machines, which causes total reallocation to rise.

Now, suppose that vintages are complementary, i.e., $\rho < \nu$. Holding investment fixed, greater young retention now raises the marginal product of older capital for unconstrained firms, while acquiring less young capital lowers the marginal product of older capital for constrained firms. This reinforces retention at older ages, but once investment adjusts, capital expansion can still outweigh that effect and cause total reallocation to rise. Whether or not that is true depends on the inequality in (15). It is more likely to hold when vintages are more complementary, which is the opposite of how the capital aggregator is typically modeled.

Bonus depreciation for used capital. While policymakers typically only vary bonus for new capital, they can also vary it for used capital. Indeed, up until the 2017 Tax Cuts and Jobs Act, bonus only applied to new capital purchases. This section characterizes why that asymmetry matters, examining how variation in bonus for used capital θ^U affects the age distribution of used asset purchases, holding fixed $\theta^N = 1$. The following proposition characterizes exactly that.

Proposition 3 (Used Bonus and Lock-In Release)**.** *Under the maintained equilibrium conditions, suppose $\tau > 0$ and $\bar{d}_2 > 0$. Let $U = U_1 + U_2 + U_3$, and hold $\theta^N = 1$.*

(i) If $\rho \leq \nu$, an increase in θ^U satisfies

$$\frac{d}{d\theta^U}\left(\frac{\widetilde{K}_a}{I}\right) < 0, \qquad a = 1, 2, 3,$$
$$\frac{d}{d\theta^U}\left(\frac{U_1}{U}\right) > 0.$$

(ii) For every $\rho < 1$, an increase in θ^U satisfies

$$\frac{d}{d\theta^U}\left(\frac{U_1 + U_2}{U}\right) > 0.$$

(iii) At $\theta^U = 1$, equilibrium allocations and gross prices coincide with those in the no-tax equilibrium.

Proof. *See Appendix B.4.*

When there is full expensing for used and new equipment, no tax distortions remain on either the secondary market for capital goods, or on the primary market. This yields the same alloca-

tions and prices as would be obtained if $\tau = 0$.[15] This is important because full expensing of new purchases alone does not eliminate distortions in the timing of capital sales. Increasing θ^U to full expensing removes that distortion entirely, bringing the deductions of constrained firms forward in time and raising demand for used capital. With complementary vintages, this raises the equilibrium resale price of young capital, encouraging unconstrained firms to retain a smaller fraction of the vintage. This channel is reinforced by complementarity. At a given level of investment, transferring more young capital lowers the marginal product of older vintages for unconstrained firms and raises it for constrained firms, encouraging further transfers. As a result, firms retain a smaller fraction of each investment vintage, and transactions of used assets shift toward younger assets.

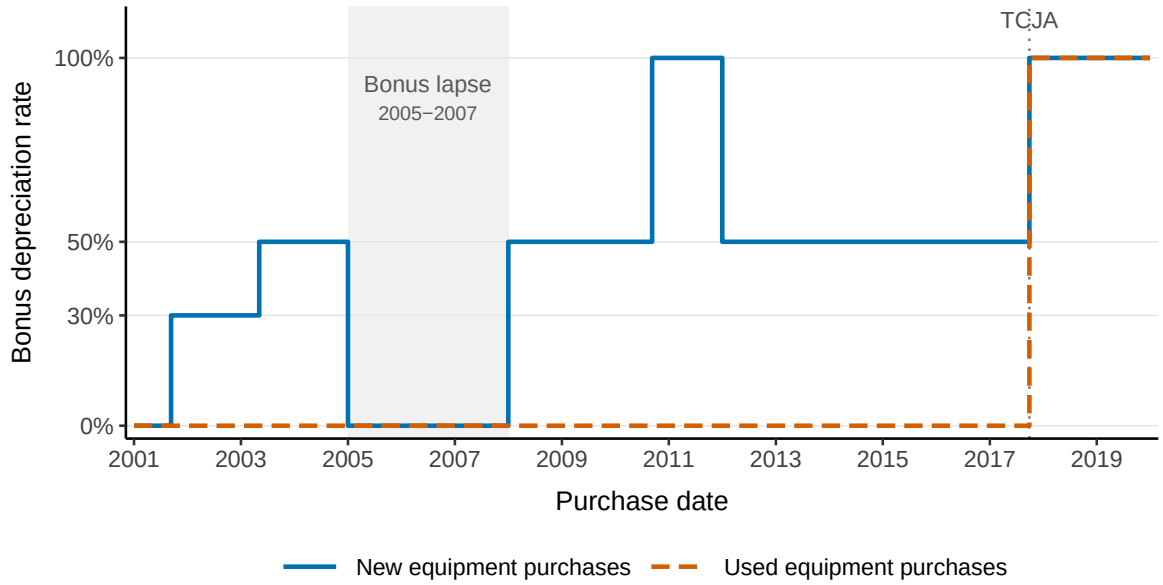
Joint full expensing restores neutrality regardless of technology. When vintages are substitutes rather than complements, the adjustment in capital markets may occur through both age-one and age-two capital. When they are substitutes, selling intermediate-age capital raises the marginal product of younger capital for unconstrained firms, which dampens the mechanism in the previous paragraph. Nevertheless, the combined share of age-one and age-two transactions increases, shifting transactions away from assets that would already have exhausted their original basis under ordinary depreciation toward assets that would still be within that recovery period.

3.4 Empirical Predictions

The model yields empirical predictions for both capital sales and the age distribution of transactions. On the one hand, holding θ^U fixed, raising θ^N may raise new investment while reducing sales of young assets that would still have future original deductions under ordinary depreciation. If sales at older ages do not fully offset this decline, total reallocation falls despite higher investment, a response that requires complementarity and is inconsistent with the standard perfect substitutes benchmark. Thus, evidence of lock-in is consistent with complementarity among vintages. On the other hand, when new purchases are fully expensed and vintages are complementary or additive, granting bonus to used asset purchases should shift the age distribution of used asset sales toward younger assets and release lock-in. Both predictions are relevant for policy, which substantially varied bonus for new assets throughout the 2000s and 2010s, and granted it to used assets following the 2017 Tax Cuts and Jobs Act. Figure 4 shows this variation over time. In Sections 5 and 6, I leverage this variation to test the predictions of the model.

15. In a setting without resale and no other frictions, as in the standard neoclassical model, it is sufficient to remove all distortions by simply giving 100% bonus to new capital (Abel 2007). This proposition shows that it is also necessary to extend full expensing to used asset purchases as well.

Figure 4: Bonus Depreciation for New and Used Equipment



Notes: Statutory bonus depreciation rates for qualifying equipment purchases, shown by policy effective date. The 30-percent allowance enacted in 2002 applied retroactively to qualifying purchases after September 10, 2001. Used equipment became eligible after September 27, 2017.

4 Data

I test the model's predictions using Uniform Commercial Code (UCC) filings in Texas. They provide asset-level records of equipment entering secured lending relationships, allowing me to measure financed used asset transactions by vintage, type, and age. These transactions are the empirical counterpart to used asset purchases U_a in the model, which equal asset sales S_a in equilibrium.

Many used asset transactions involve financing, and when a business uses equipment as collateral, the lender typically files a financing statement with the state government to document its claim. These UCC filings are a standard feature of commercial lending in the United States and have been used in recent work on equipment markets, capital reallocation, and small business lending (Edgerton 2012; Ma, Murfin, and Pratt 2022; Basu, Kim, and Singh 2024). I use the universe of UCC filings in Texas, obtained from Jensen (2024). The records provide broad coverage of equipment entering secured lending relationships, although they exclude unfinanced purchases. I treat the appearance of a positive-age machine in an itemized filing as a financed used asset transaction. This linkage is not exact because a UCC filing records a security interest rather than a bill of sale, and refinancing may generate a filing without a contemporaneous ownership transfer.

In each filing, I observe the borrower, the secured party, and the collateral in free-form text.

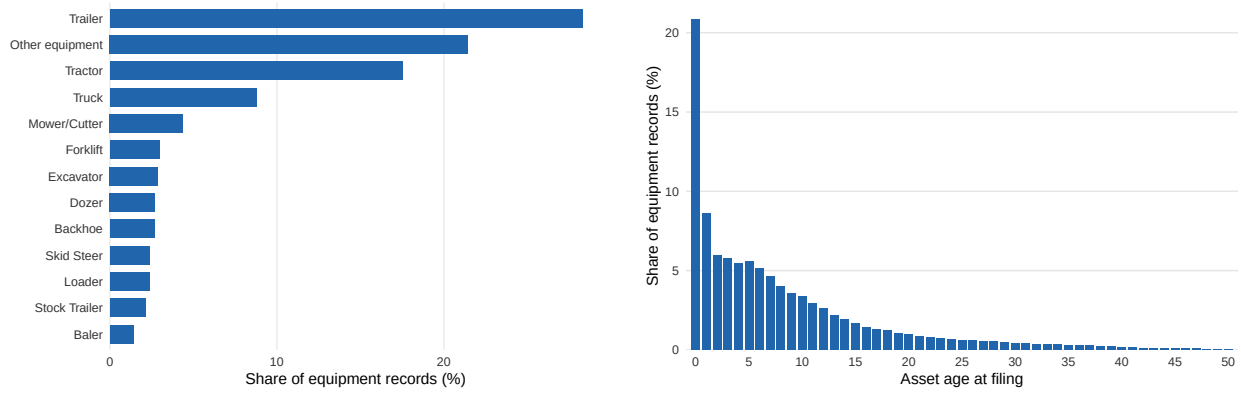
Some descriptions only contain a blanket claim on the assets of the borrower, but many filings itemize specific pieces of equipment and report their type, manufacturer, model year, and serial number. These descriptions are sufficient to identify individual machines associated with each financed transaction and recover their characteristics. When a filing lists multiple machines, I count each one as a separate observation. For each observation, I can construct transaction quantities by equipment type and vintage.[16] I date the age of each machine using the filing event on which it appears and comparing that to the model year recovered from the description. The description also allows me to figure out the equipment type. Many filings contain a serial number or vehicle identification number; when the latter is valid, it can be used to recover the model year. Age is then filing year less the model year. Several types of records contain blanket liens, legal and financial boilerplate, and records that cannot be reconstructed unambiguously; I exclude all such filings. Appendix C.1 describes the construction in detail.

I am able to link each transaction to the tax system using each observation's equipment type, model year, and asset age. The equipment type allows a linkage to the baseline ordinary MACRS depreciation schedule. For example, construction equipment and trucks are generally five-year property, while agricultural equipment is generally seven-year property, resulting in some variation across assets. Next, since I treat model year as a proxy for the original purchase vintage, that allows me to identify if a particular machine was exposed to bonus when it was originally sold to the first owner. In that case, its effective tax rate on sale would differ from an otherwise equivalent asset initially bought in a year without bonus. Third, the asset age determines how much the asset has progressed through the applicable tax depreciation schedule since its initial purchase. Together, these fields allow me to construct the predicted statutory basis remaining when the equipment appears in the UCC data, which is the empirical counterpart to the model's remaining-basis margin. As an example, a three-year-old tractor belonging to the 2007 non-bonus vintage would have followed an ordinary MACRS depreciation schedule, while a three-year-old machine appearing in 2011 would belong to the 2008 bonus vintage. The machines may be the same age but they would have different predicted remaining basis because only the latter was exposed to bonus depreciation.

I retain equipment assigned to five- or seven-year property with an identified equipment type, model year, and valid age, excluding confirmed leases, administrative events, and identified inventory filings. The resulting sample contains 373,046 equipment items from 2001 through 2019. Using age zero to classify new equipment and positive age to classify used equipment, the sample contains 77,811 new and 295,235 used equipment items. Appendix Tables C.1, C.2, and C.3 report the sample construction, summary statistics, and largest equipment categories.

16. UCC filings do not report transaction prices, loan amounts, or loan terms. I therefore observe the quantity and characteristics of equipment involved in financed transactions but not the prices at which they sell.

Figure 5: Equipment Composition and Age



(a) Equipment composition

(b) Asset age

Notes: Both panels use the full sample of 373,046 equipment records from 2001–2019, including age zero. Confirmed leases, administrative events, and identified inventory filings are excluded. Equipment categories outside the largest twelve are grouped as other equipment. Asset age is filing year less model year.

Figure 5 describes the equipment composition and age distribution of the sample. Panel 5a shows that trailers, tractors, and trucks together account for approximately 55 percent of equipment records, with the remaining observations spanning other agricultural, construction, and commercial equipment. Panel 5b shows that approximately 21 percent of records concern age-zero equipment. Among positive-age records, the median age is seven years, so the sample includes both young machines and equipment well beyond ordinary tax recovery.

5 Bonus Depreciation and the Lock-in Effect

I compare transactions involving equipment whose model year assigns it to a bonus purchase vintage—which I call *bonus vintages*—with equipment assigned to nonbonus purchase vintages. The model highlights that the comparison between the two is *ex ante* ambiguous. On the one hand, bonus encourages firms to create more capital, expanding the cohort that may eventually reach the secondary market. On the other hand, the resulting machines face higher effective tax rates on sale, which strengthens the retention incentive. The analysis here asks which channel dominates in the subsequent circulation of bonus-treated vintages.

To study which channel dominates, I aggregate the data into cells organized by equipment type, MACRS recovery class, basis schedule, filing year, and asset age. I group these cells by manufacturer because manufacturers may differ in the durability of the goods they produce, the extensiveness of their secondary-market networks, and other characteristics. For example, John Deere, which is one of the largest manufacturers of tractors in the data, has a wide network of

secondary market dealers. My preferred specification therefore compares vintages within manufacturers, holding equipment type, MACRS recovery class, and basis schedule fixed. Let m index groups defined by manufacturer, equipment type, MACRS recovery class, and basis schedule, and let $N_{m,t,a}$ denote the number of transactions in group m, filed in year t, at age a. I use model year as a proxy for the machine's original purchase vintage, $v = t - a$, and estimate

$$\mathbb{E}\left[N_{m,t,a} \mid \alpha_{m,a}, \gamma_t\right] = \exp\left[\alpha_{m,a} + \gamma_t + \beta\,\text{BonusVintage}_{t-a}\right], \tag{16}$$

where BonusVintage_v equals one if purchase vintage v was eligible for a positive statutory bonus rate and zero otherwise. The fixed effect $\alpha_{m,a}$ allows each group to have its own transaction profile over the asset lifecycle and γ_t absorbs shocks common to all transactions in a filing year. Thus, the preferred specification compares machines within the same group at the same age that appear in different filing years and therefore belong to different purchase vintages. I estimate (16) by PPML over filing years 2001–2017 and cluster standard errors by purchase vintage, which is the level at which bonus eligibility changes.

Identification comes from changes in the statutory bonus schedule across vintages. The identifying assumption is that, absent their initial differential tax treatment, bonus and nonbonus vintages within these groups would have otherwise generated comparable transaction counts at the same age, after accounting for filing-year shocks. Table 1 reports the main results.

Column (1) reports a broad comparison within equipment types and tax schedules. This, for example, considers whether bonus-vintage trucks transact less than non-bonus trucks. Across equipment categories, bonus-vintage equipment appears approximately 26 percent less often in subsequent used asset transactions than nonbonus-vintage equipment. Column (2) restricts the sample to observations for which I identify a manufacturer and yields a similar estimate. Column (3) then compares transactions within manufacturers—Chevrolet trucks with other Chevrolet trucks—holding equipment type, MACRS recovery class, and basis schedule fixed. This attenuates the implied difference to approximately 15 percent. Finally, column (4) allows each group to have its own asset-age profile, comparing four-year-old Chevrolet trucks from bonus vintages with four-year-old Chevrolet trucks from nonbonus vintages appearing in different filing years. For example, a four-year-old 2004 Chevrolet truck appearing in a 2008 filing belongs to a bonus vintage, while a four-year-old 2007 Chevrolet truck appearing in a 2011 filing does not. This is my preferred specification and implies that bonus-vintage equipment appears in approximately 17.5 percent fewer subsequent transactions. Columns (5) and (6) restrict the sample to cohorts for which I observe initial cohort size. Within this sample, controlling for initial cohort size reduces the estimated difference from 23.1 to 18.4 percent. Thus, the negative relationship persists after accounting for measured differences in the initial stock of machines. While the results

attenuate when moving from broad categories to the within-manufacturer design, the differences remain economically and statistically significant, consistent with lock-in dominating capital stock expansion.

Table 1: Bonus Depreciation and Used Equipment

	Full sample	Maker observed	Group FE	Group × age FE	Size observed	Size controlled
	(1)	(2)	(3)	(4)	(5)	(6)
Bonus vintage	−0.300***	−0.276***	−0.160**	−0.193**	−0.262***	−0.203***
	(0.077)	(0.083)	(0.063)	(0.078)	(0.066)	(0.041)
Implied difference (%)	−25.9	−24.1	−14.8	−17.5	−23.1	−18.4
Type × class × schedule FE	X	X				
Equipment-group FE			X			
Equipment group × age FE				X	X	X
Common age FE	X	X	X	Absorbed	Absorbed	Absorbed
Filing-year FE	X	X	X	X	X	X
Log initial cohort size						X
Equipment mentions	249,090	185,770	185,770	185,770	92,218	92,218
Estimation cells	21,370	17,843	62,679	62,679	20,924	20,924
Equipment groups	—	3,088	3,088	3,088	1,055	1,055
Vintage clusters	66	66	66	66	16	16

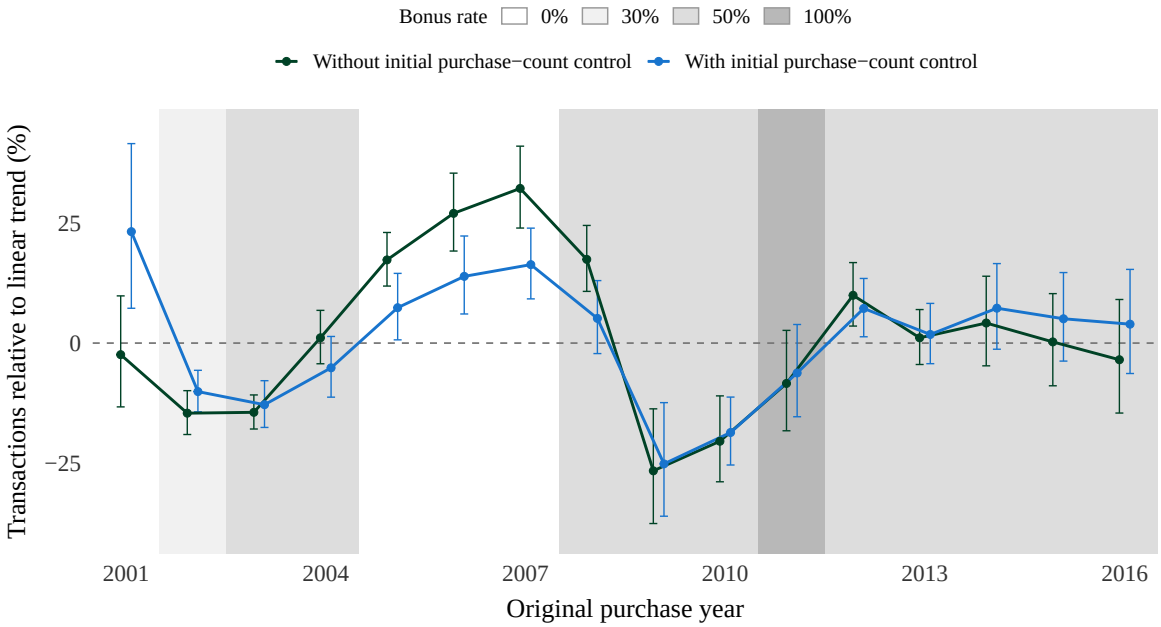
Notes: PPML on observed positive-age cells in the preferred sample, excluding confirmed lease and administrative events and narrow inventory rosters. Bonus vintage indicates any positive bonus at the model-year purchase date. Equipment group is manufacturer × type × recovery class × depreciation schedule. Columns (1)–(2) aggregate across manufacturers. Filing years are 2001–2017 in columns (1)–(4). Columns (5)–(6) use identical cells with observed age-zero cohort size, 2002–2017. Standard errors in parentheses cluster by purchase vintage; no zero completion. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ using t_{G-1}.

I account for several concerns about measurement. For example, UCC filings proxy for ownership transfers, so repeated filings involving the same machine may overstate transactions. Appendix Table D.1 shows that the results are similar when I restrict the sample to initial filings or count only the first observed appearance of each machine with a given borrower. Additionally, I show that excluding age-one equipment, which may include new purchases, also leaves the estimate largely unchanged. Across these checks, bonus-vintage equipment appears in approximately 16.5–17.7 percent fewer subsequent transactions.

Next, I replace the bonus-vintage indicator from the table with separate coefficients for each

original purchase year. This is important for testing whether the measured differences in transactions align with changes in policy. Figure 6 plots these coefficients relative to their linear purchase-year trend, with and without controlling for initial cohort size, using the same observations in both specifications. The 2005-07 lapse in bonus depreciation yields a large positive hump compared to surrounding coefficients in both specifications. Indeed, equipment from these nonbonus vintages appears in approximately 40% more subsequent transactions than equipment from the surrounding 2002–2004 and 2008–2010 bonus vintages. Controlling for initial cohort size reduces this difference to approximately 27%, but the pattern remains pronounced.[17]

Figure 6: Used Equipment Transactions by Original Purchase Year



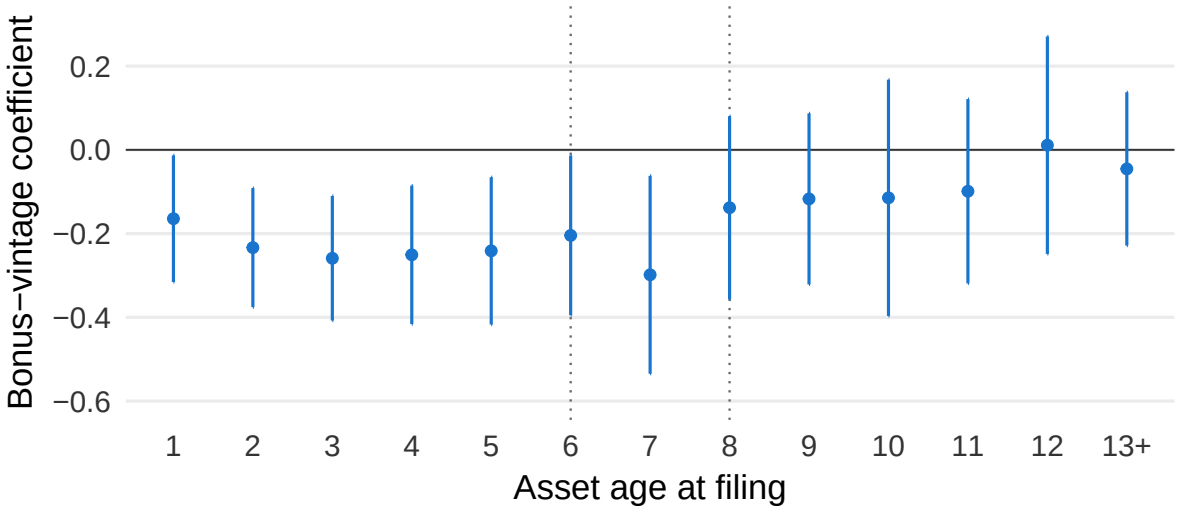
Notes: Both PPML profiles use identical observations from filing years 2002–2017 and purchase vintages 2001–2016, with equipment-group-by-exact-age and filing-year fixed effects. The controlled specification adds the log age-zero transaction count for the same group and vintage. Both profiles remove their equal-weight level and linear purchase-year trend. Shading indicates the statutory bonus rate. Bars report pointwise 95% confidence intervals, clustering by manufacturer.

Next, I investigate whether the model's main mechanism holds by examining differences in coefficients by age, focusing on the specification in Column 4 of the main table. Compared to nonbonus vintages, the direct tax penalty on selling a bonus vintage early should be most evident

17. Appendix Table D.4 shows that much of the identifying variation comes from the 2005-07 bonus lapse and is robust to controlling for initial cohort size and including years with no transactions for cohorts observed at age zero. Appendix Table D.3 reports similar results when comparing 50-percent bonus with no bonus on identical observations. Moreover, Appendix Table D.2 shows that allowing for gradual changes across purchase vintages by adding quadratic and cubic trends in model year leaves the estimated relationship similar in magnitude and statistically significant.

at early ages and then decline and eventually disappear as the asset approaches the end of its ordinary recovery period. I examine this in Figure 7 by allowing the bonus-vintage coefficient to vary with asset age. The estimated differences are negative among younger equipment and attenuate among older equipment. The pattern is consistent with the model. However, the point estimates do not decline monotonically: the age seven coefficient is more negative than age six. Appendix Table D.5 reports the formal tests; the difference between the mean coefficients at ages two through seven and at ages eight and older has a p-value of 0.055. This pattern survives a number of checks in the appendix.[18]

Figure 7: The Bonus-Vintage Transaction Gap by Asset Age



Notes: PPML estimates use positive-age equipment filings from 2001–2017, with equipment-group-by-exact-age and filing-year fixed effects. Ages 13 and older share one bonus coefficient; fixed effects retain exact age. Dotted lines mark ages six and eight. Bars report 95% confidence intervals using purchase-vintage clustering and t_{65} inference.

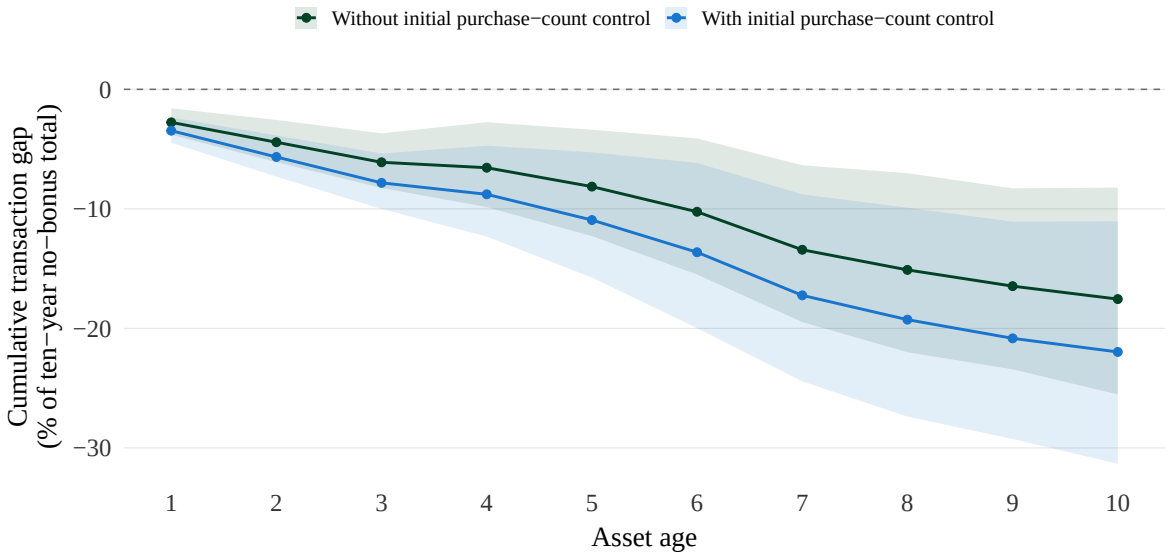
The model distinguishes a delay in sales from a reduction in total reallocation. Proposition 2 shows that, when vintages are substitutes, total reallocation rises whenever investment rises, even if young sales fall, because older sales more than offset the early decline. With complementary vintages, that recovery need not occur. The cumulative analysis examines whether the early shortfall is recovered as bonus-treated vintages age.

While Figure 7 shows the point estimates becoming smaller at older ages, that does not necessarily mean that there is catch-up later in the capital lifecycle. That would require bonus vintages to generate enough additional transactions at older ages to offset the earlier deficit. To test this,

18. Appendix Figure D.1 shows a similar profile when I restrict the sample to initial filings or count only the first observed appearance of a machine with a given borrower. Additionally, Appendix Figure D.2 and Appendix Table D.6 show that the younger-age deficit remains after restricting the sample to equipment types assigned five-year status directly from their type.

Figure 8 follows the same 2001–2007 purchase cohorts through ages one to ten and accumulates the estimated transaction differences. Both curves express these cumulative differences as a share of the same predicted no-bonus ten-year transaction total. Without controlling for initial acquisition counts, the estimated cumulative shortfall reaches approximately 17.5% by age ten. Controlling for those counts yields a shortfall of approximately 22%. I find that subsequent transactions do not fully offset the early shortfall through age ten, which is consistent with complementary vintages. Appendix Figure D.3 shows that the cumulative shortfall also remains when I restrict the data to initial filings or count only the first observed appearance of each machine with a given borrower.

Figure 8: Cumulative Transactions Through Age Ten



Notes: PPML follows purchase vintages 2001–2007 through ages 1–10, using identical cohorts and cells, including supported zeros. Specifications include equipment-group-by-age and filing-year fixed effects, with and without the log age-zero transaction count. Both curves use common baseline age weights and the same predicted no-bonus ten-year total from the controlled model. Bands report conditional pointwise 95% intervals using seven purchase-vintage clusters and t_6 draws, holding weights and fitted fixed effects fixed.

In sum, the results are highly supportive of the idea that bonus depreciation may cause reallocation to decline. At first glance, that conclusion appears to differ from those drawn by Basu, Kim, and Singh (2024), who find that bonus depreciation increases investment in used assets and lowers their prices. However, we differ in one key way. Whereas I trace whether bonus vintages transact more or less than non-bonus vintages, they track whether bonus induces higher used capital purchases, regardless of vintage. They interpret this through a replacement margin, in which firms may replace their old machines with subsidized new ones. As I discuss later,

Compustat firms increase capital expenditures while reporting lower capital sale proceeds.[19]

Altogether, the results are consistent with the model's prediction that lock-in can dominate capital stock expansion. In that case, subsidizing new investment does not necessarily increase reallocation, which is consistent with complementarity among vintages rather than the standard perfect-substitutes benchmark. In the next section, I examine whether the 2017 Tax Cuts and Jobs Act, which extended bonus eligibility to used equipment, released the lock-in generated by bonus depreciation.

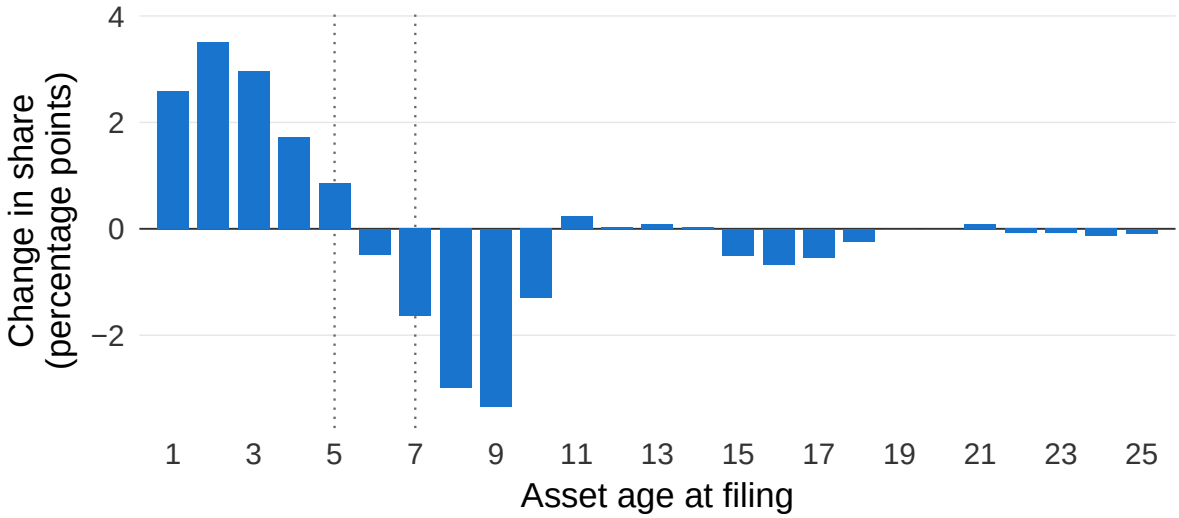
6 The Tax Cuts and Jobs Act and the Reversal of Lock-In

Until the 2017 Tax Cuts and Jobs Act, bonus depreciation generally applied only to purchases of new equipment. The reform extended bonus depreciation to qualifying used asset purchases, making both new and used equipment purchases have 100% bonus depreciation. By bringing used purchasers' deductions forward, the reform increases their willingness to pay for used equipment, which can raise resale prices and encourage existing owners to sell rather than retain it. The model predicts that extending bonus to used purchases can release lock-in and shift transactions toward younger assets. I examine whether this is accompanied by greater subsequent circulation of pre-TCJA bonus vintages. At first glance, the reform appears effective in the aggregate data. Figure A.1 shows a sharp spike in used asset purchases relative to new investment, as well as the share of the equipment capital churning. Moreover, Figures C.1a and C.1b both show an increase in transaction counts and a decrease in median age of capital in the UCC data. The remainder of this section examines whether these changes are accompanied by a narrowing of the earlier transaction shortfall of pre-TCJA bonus vintages.

Figure 9 compares the age distribution of used equipment transactions in 2012–2017 with that in 2018–2019, using all purchase vintages and manufacturers. Among equipment aged one through twenty-five, the share of transactions involving assets aged one through seven rises from approximately 51.7 percent to 61.2 percent. The increase is concentrated at ages one through five, broadly overlapping with the younger ages at which bonus vintages previously transacted less. Thus, the age shift is consistent with the model's lock-in-release mechanism.

19. A closely related replacement mechanism appears in automobile markets: Berger et al. (2026) estimate that pandemic transfers raised both new and used car sales and use a quantitative model in which new car purchases generate trade-ins that expand used car supply.

Figure 9: TCJA and the Age Distribution of Used Equipment Transactions



Notes: The figure reports changes in transaction shares between 2012–2017 and 2018–2019, in percentage points. Both periods include equipment aged 1–25, using all purchase vintages and manufacturers. Shares sum to one within each period. Dotted lines mark ages five and seven.

Next, I examine whether the earlier transaction shortfall of bonus vintages narrows after TCJA. Both bonus and nonbonus equipment became eligible for bonus when purchased used, but original bonus had reduced the tax basis of bonus vintages more quickly, strengthening their owners' incentive to retain them. A larger recovery for bonus vintages would therefore be indicative of lock-in release. Consequently, I compare post-reform changes in transactions involving bonus and nonbonus equipment already in existence before TCJA, restricting the sample to model-year vintages through 2016. This focuses the comparison on the subsequent circulation of existing equipment rather than machines newly created after the reform. I aggregate transactions into cells defined by equipment type, MACRS recovery class, depreciation schedule, filing year, and asset age, including equipment with and without an identified manufacturer. Let g index equipment groups defined by type, recovery class, and depreciation schedule, and let $N_{g,t,a}$ denote the number of transactions in group g, filed in year t, at age a. I estimate

$$\mathbb{E}[N_{g,t,a}] = \exp\left[\alpha_g + \gamma_{e(g),t} + \mu_a + \beta\,\text{BonusVintage}_{t-a} + \delta\,\text{BonusVintage}_{t-a}\text{Post}_t\right], \qquad (17)$$

where $e(g)$ denotes the equipment type associated with group g, and Post_t equals one in 2018 and 2019. Equipment-group fixed effects account for persistent differences in transaction counts across groups, while equipment-type-by-year fixed effects absorb annual shocks specific to each equipment type. Age fixed effects account for the common transaction profile over the asset lifecycle. The coefficient β captures the pre-TCJA transaction differential between bonus and nonbonus vintages, while δ measures its change after the reform. I estimate the equation by

PPML over filing years 2001–2019 and cluster standard errors by original purchase vintage. The identifying assumption is that, absent the reform, transaction counts for bonus and nonbonus vintages would not have changed differentially after 2017, conditional on the equipment-group, equipment-type-by-year, and age controls.

Table 2: TCJA and Used Equipment Transactions

	Pooled	Binary	Continuous
	(1)	(2)	(3)
Bonus vintage (pre-TCJA)	−0.317***	−0.302***	−0.305***
	(0.083)	(0.080)	(0.081)
Bonus vintage × Post-TCJA	0.170***	0.091	0.084*
	(0.043)	(0.056)	(0.042)
Post × Positive basis gap		0.242***	
		(0.071)	
Post × Basis gap (10 pp)			0.141***
			(0.022)
Pre-reform basis-gap control		X	X
Equipment group FE	X	X	X
Equipment type × year FE	X	X	X
Asset-age FE	X	X	X
Cohort-size control	No	No	No
Equipment mentions	286,478	286,478	286,478
Estimation cells	24,354	24,354	24,354
Vintage clusters	66	66	66

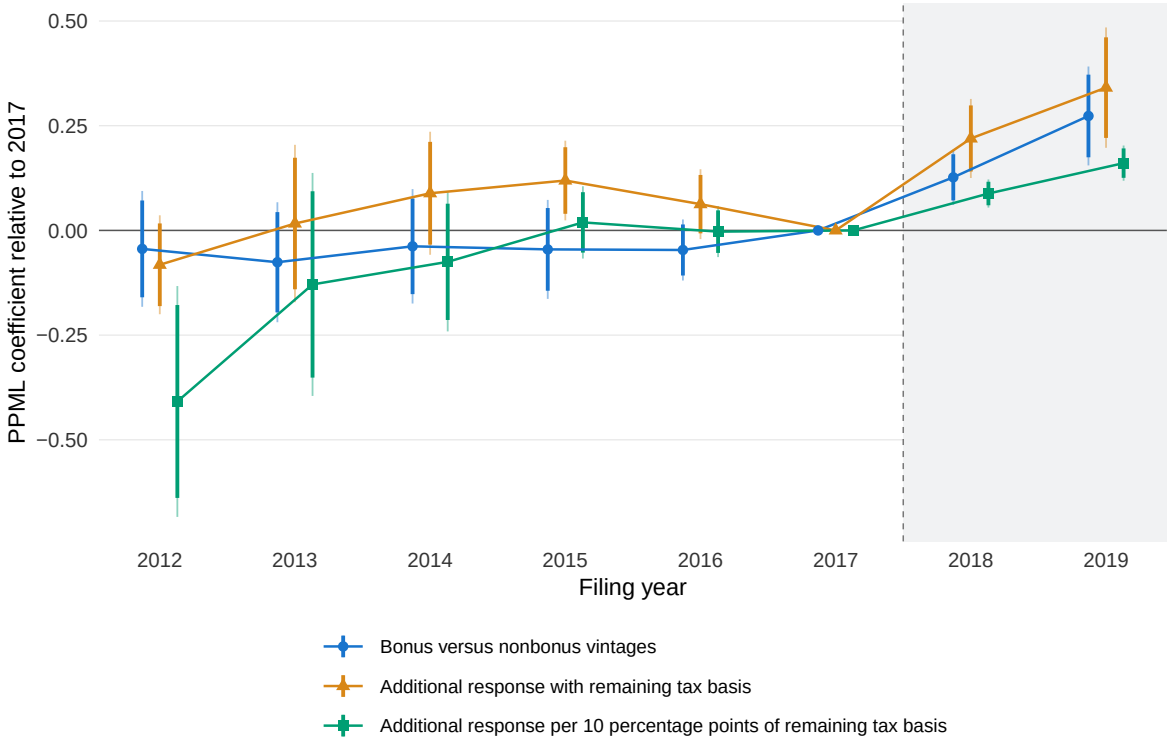
Notes: PPML, 2001–2019; post: 2018–2019. The sample uses preexisting model-year vintages through 2016 and excludes confirmed leases, administrative events, and flagged inventory records. The basis gap is the bonus-induced reduction in original basis after the 2017 deduction; positive gap exceeds 1% of cost. Columns (2)–(3) include exposure main effects. Standard errors clustered by original vintage are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ using t_{65}.

Table 2 shows that the earlier transaction shortfall of bonus vintages narrows after TCJA.

Relative to pre-TCJA, column (1) shows that bonus vintages transacted approximately 18.6 percent more relative to nonbonus vintages. This coefficient averages across equipment at different stages of ordinary tax recovery. The lock-in mechanism suggests that the response should differ depending on how much tax basis remains, with a larger effect—if lock-in is released—for machines with more remaining basis. As a first step, I test whether that is true in column (2). The predicted basis gap is the difference between remaining basis under ordinary depreciation and under the original bonus schedule, evaluated at end-2017. The basis gap coefficient interacts the post-TCJA indicator with an indicator for whether this gap exceeds one percent of original purchase cost for bonus-vintage assets. I find that the recovery is substantially larger for this group and absorbs all of the statistical significance from the bonus vintage × Post indicator. As a second step, I test whether the release of lock-in varies with the amount of tax basis remaining, not just whether *any* remains. I find that the estimated response rises by 0.141 log points for every ten-percentage-point increase in remaining basis, indicating a larger recovery where original bonus had reduced remaining basis more.

Next, I examine whether the pooled estimates in the table follow timing consistent with a causal interpretation. Figure 10 estimates annual versions of all three specifications, using 2017 as the reference year. For the overall bonus–nonbonus comparison, the estimates for 2012–2016 are modestly negative relative to 2017, but none is statistically distinguishable from zero. Following TCJA, transactions involving bonus vintages rise relative to nonbonus vintages by approximately 13.5 percent in 2018 and 31.4 percent in 2019, both measured relative to 2017. The recovery is also larger for equipment with remaining tax basis and increases with the amount remaining. In particular, the additional response per ten percentage points of remaining basis rises to 0.088 log points in 2018 and 0.161 in 2019. The remaining-basis comparisons show more variation before the reform, particularly early in the sample, so their preceding paths are not uniformly flat. Nevertheless, the increase after TCJA across all three comparisons is consistent with the reform causally reducing lock-in.

Figure 10: TCJA and the Recovery in Transactions of Bonus Vintages



Notes: Annual PPML estimates corresponding to columns (1)–(3) of Table 2, using 2001–2019 filings and model-year vintages through 2016. Coefficients are relative to 2017. Thick and thin bars show 90% and 95% confidence intervals, clustering by purchase vintage with t_{65} critical values. Points are offset horizontally for readability.

In the previous section, we observed that the transaction shortfall of bonus vintages narrows with age, which may suggest here that the apparent recovery following TCJA reflects normal catch-up. I address this by estimating a bonus-specific age profile using observations through 2016 and holding it fixed in the TCJA regressions. Appendix Table D.7 reports that the pooled recovery falls from 18.6 percent to 15.4 percent, while the annual pattern changes little and the basis-gap responses strengthen, suggesting that historical aging explains little of the post-reform recovery.[20] Separately, one may worry that the recovery reflects an increase in amended UCC statements rather than asset purchases. Restricting the analysis to initial filings, Appendix Table D.8 shows similar estimates, indicating that amendments do not drive the recovery.[21] Finally, the basis-gap relationship could depend on uncertain depreciation assignments because some equipment's recovery schedule depends on uses I cannot observe. For example, a trailer may be used on a farm or in a factory, and its depreciation schedule would depend on how it is used. Appendix

20. Appendix Figure D.4 compares the annual estimates with and without the historical age adjustment.

21. Estimates attenuate substantially when identifiers are required, before repeated machine–borrower appearances are removed. Some age-one equipment may be new, potentially affecting the historical benchmark. Excluding these pre-reform observations leaves the estimates similar (Appendix Table D.9).

Table D.10 shows that both basis-gap responses remain positive and statistically significant under alternative schedules on unchanged observations, indicating that the relationship survives these alternative tax assignments.

The Tax Cuts and Jobs Act extended symmetric 100% bonus depreciation to used capital, and the changes in transactions are consistent with the model's prediction of lock-in release. This, together with the evidence on bonus, strongly suggests that the capital aggregator must feature complementarity between vintages. We observe a shift in the age distribution of asset transactions toward those of younger vintages. Together with the pre-TCJA evidence, these findings suggest that the effect of investment subsidies on reallocation depends on their treatment of used as well as new capital.

In sum, both the bonus and TCJA results are consistent with complementarity among vintages. Before TCJA, bonus raises investment while reducing the subsequent circulation of bonus vintages. With complementary vintages, additional new investment raises the marginal product of older capital, encouraging firms to retain more rather than sell it. Complementarity also helps explain the release of lock-in following TCJA. When firms sell more young capital, the marginal product of their older capital falls, while used purchasers' demand for that older capital rises, encouraging further sales. Thus, complementarity can reinforce both the initial lock-in and its release when bonus extends to used purchases.

7 Additional Evidence from other Data Sources

The appendix and supplement provide additional evidence on the model's implications using three additional data sources. Using these, I examine whether firms increase investment while reducing capital sales, whether bonus take-up depends on how readily an industry's capital can be resold, and whether reduced reallocation is accompanied by lower firm entry. Together, all three shed further light on the mechanisms underlying the model and empirics.

First, I use Compustat. In Compustat, firms record the dollar value of their sales of used capital. Following an identification approach from Zwick and Mahon (2017), I show that firms in industries which are more exposed to bonus depreciation sell less of their existing capital while significantly expanding their capital expenditures. With industry-specific quadratic trends, a one-percentage-point increase in the present value of depreciation deductions is associated with approximately 1 percent higher investment and 5 percent lower sale proceeds, as reported in columns (2) and (6) of Appendix Table F.1. While the Compustat data cannot separately identify prices or quantities, nor can it track vintages, the pattern of reduced capital sales with expanding investment is consistent with the model's predictions about capital lock-in and the UCC evidence around bonus.

Next, I use Compustat to examine lock-in release around the 2017 Tax Cuts and Jobs Act using two separate strategies. First, I compare foreign firms to U.S. firms using a difference-in-differences approach. In Appendix Table F.3, I find that, relative to foreign firms, U.S. firms report approximately 22 percent higher capital sale proceeds after the reform. Second, I use the same foreign firms to construct synthetic controls for each domestic firm. I find a similar pattern in Appendix Figure F.2: used capital sales are closely matched before 2018, but diverge sharply afterward, with U.S. firms significantly increasing their used capital sales relative to their synthetic matches. Thus, overall, I find a similar pattern with Compustat as I do in the evidence from Texas. Bonus seems to lock in used capital and reduce reallocation, consistent with complementarity between vintages, but the TCJA evidence is consistent with lock-in release following full expensing for both sides of the market.

Second, I use the Statistics of Income to track whether industry-level bonus take-up is correlated with how tradable an industry's capital is. Bonus take-up is voluntary and roughly half of firms actually use it (Kitchen and Knittel 2011). In principle, the more tradable capital is, as measured by Kermani and Ma (2022), the lower an incentive a firm would have to claim bonus because the tax benefit of accelerating deductions at purchase is smaller when the firm expects to recover its remaining basis through an early sale. I find that this holds in the data. Appendix Table E.1 shows that bonus take-up responds less to increases in the statutory bonus rate in industries where assets are more readily resold, and that holds when I control for profitability and other determinants of firms' ability to use deductions. However, excluding agriculture, which has particularly deep secondary markets for used assets (Kermani and Ma 2022), leaves a similar estimate that is no longer statistically significant.

Finally, I examine the model's implications for downstream outcomes like firm entry using evidence from the Business Dynamics Statistics. Ma, Murfin, and Pratt (2022) indicate that used capital is more important for young firms, which may also be entrants. When the cost of used capital is lower, as would happen from an increase in its supply, then entry should rise. I show that this holds in an extension of the baseline model to entry. Then, using the Census Bureau's Business Dynamics Statistics, I show in Supplemental Figure S1 that industries more exposed to bonus experience larger subsequent declines in entry, with larger declines in industries where capital is more readily traded using the industry-level measure of tradability from Kermani and Ma (2022).[22] This provides a third supporting piece of evidence consistent with lock-in reducing

22. Basu, Kim, and Singh (2024) find that bonus increases the number of small establishments. The key difference is that, whereas I estimate entry responses by *firms* across industries, they compare the number of small establishments across industries within counties, using county-by-year fixed effects. While these are prima facie contradictory, they need not be. Their design absorbs equilibrium changes common to industries within each county, leaving a relative local response. My own results absorb nationally common changes, but can retain local adjustments because industries have different geographic concentrations. Thus, each has its own missing intercept problem, and the two coefficients need not have the same sign.

reallocation and entry. However, I find that the results are considerably noisier following the Tax Cuts and Jobs Act, and I cannot establish that entry rose following its enactment.

8 Aggregating the Effects of Bonus on Reallocation

The empirical results largely conform to the theoretical model and suggest that bonus depreciation for new capital reduces capital reallocation. From 2012 through most of 2017, bonus depreciation was a constant 50 percent. Over the same time period, average annual reallocation, as measured by aggregate gross sale proceeds for Property A on IRS Form 4797, was roughly $104 billion. I use that figure as a baseline to gauge how much extra reallocation my estimates imply there may have been under lower-bonus counterfactuals from 2012–17 via three different back-of-the-envelope calculations. All three should be interpreted as back-of-the-envelope accounting exercises rather than estimates of the full general equilibrium response.

The first exercise is a deliberately conservative calculation in the spirit of Zwick and Mahon (2017). I assign every industry the bonus increment received by the least-exposed quintile of industries while preserving its ordinary MACRS depreciation schedule. Applying the Compustat capital sales coefficient from Column 1 of Table F.1, using the 2012–16 Compustat weights, implies that this would increase reallocation by $13.7 billion relative to observed policy, with a 95% confidence interval spanning $7.8 billion to $21 billion.

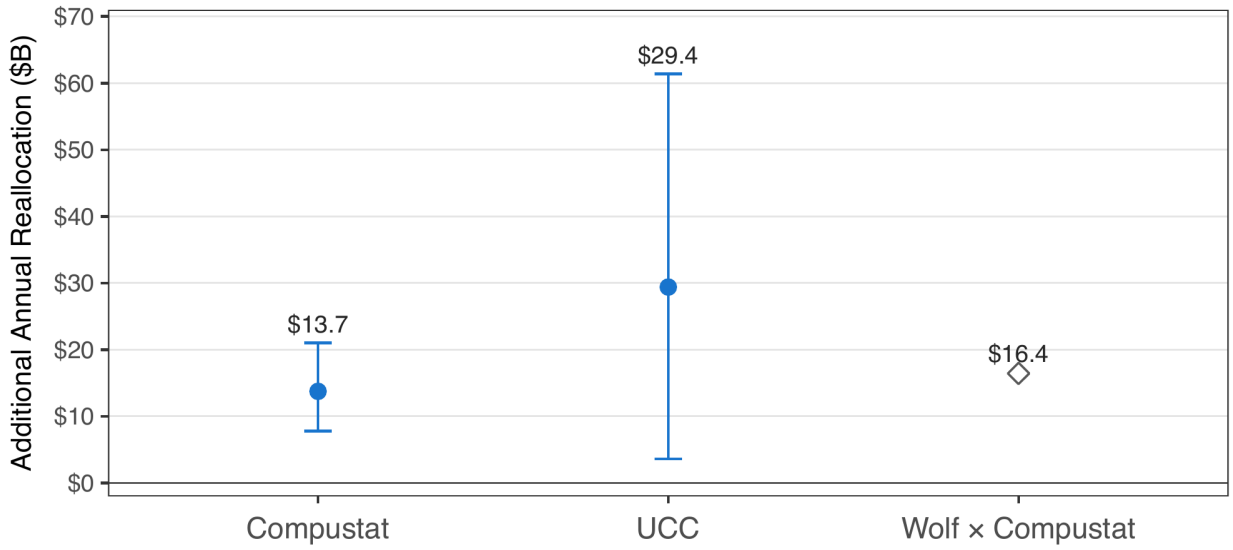
As a next estimate, I use the direct comparison in transactions between vintages receiving 50-percent bonus and zero-bonus vintages from Column 1 of Table D.4. That specification implies that zero-bonus vintages transact about 28.3 percent more than vintages with 50 percent bonus. Applied proportionately to the full SOI baseline and holding the average value of used asset transactions fixed, this implies that reallocation would rise by roughly $29.4 billion, with a 95% confidence interval spanning $3.6 billion to $61.4 billion.

Finally, I compute a third point of comparison using Koby and Wolf (2020) and Wolf (2023). The former supplies a partial equilibrium path of the investment response to bonus depreciation, while the latter uses demand equivalence to estimate the corresponding general equilibrium investment path in his macroeconomic environment.[23] After rescaling the investment response to a 50-percentage-point change in bonus depreciation, I multiply the first-year average of that general equilibrium path by a lagged-sales/current-investment response ratio estimated on a common Compustat sample, following the timing in Appendix Table F.2, to estimate the response of

23. The partial equilibrium investment and output responses determine the excess-demand path created by the investment subsidy. Wolf (2023) chooses a weighted combination of empirically identified demand shocks that generates the same path and adds their investment response to the partial equilibrium response to obtain the general equilibrium path.

reallocation to bonus. The resulting estimated $16.4 billion in additional reallocation should be interpreted cautiously for four reasons. First, the sales-to-investment response ratio is estimated from differences in industry exposure and held fixed even though, in general equilibrium, both capital sales and investment respond to the used price. Second, the investment coefficient is imprecisely estimated, making the response ratio highly uncertain, so I do not report an interval. Third, the calculation uses the first-year response to an investment-subsidy shock rather than the response to the persistent bonus rate observed during 2012–17. Finally, the mapping from partial to general equilibrium relies on a particular macroeconomic closure and invariance conditions in Wolf (2023) which need not hold in this setting. Hence, it should largely be interpreted as a sensitivity exercise.

Figure 11: Economic Magnitude of the Reallocation Response



Notes: Dollar values use average annual Form 4797 Part III Property A gross sale proceeds from 2012 through 2017. Confidence intervals propagate coefficient uncertainty only. No interval is reported for the Wolf × Compustat calculation because the sales-to-investment ratio is imprecisely estimated.

Across the three approaches, the point estimates are on the same order of magnitude and imply roughly $14–$29 billion in additional reallocation per year under lower-bonus or no-bonus counterfactuals relative to observed policy from 2012–17. While these estimates are not the general equilibrium effect of reducing bonus, they do point toward an economically meaningful effect of bonus depreciation on reallocation.

9 Conclusion

This paper studies how the tax system shapes capital reallocation. I show that tax policies which stimulate new capital formation may simultaneously impede its subsequent reallocation. That ambiguity arises from the actual implementation of investment subsidies through the tax code. Accelerated depreciation lowers the after-tax cost of purchasing new capital, generating a capital stock expansion channel, but it also leaves the subsidized asset with less tax basis when it later reaches the secondary market, generating a lock-in channel that strengthens the incentive to retain it. In general equilibrium, lock-in dominates when capital vintages are sufficiently complementary in production. And, as long as that is true, granting bonus to used asset purchases should reduce that lock-in effect.

Using the universe of financed equipment purchases in Texas, I find that lock-in dominates. I find that equipment from bonus vintages subsequently transacts nearly 20 percent less than equipment from nonbonus vintages. Moreover, I find that the transaction gap is concentrated at ages two through seven and attenuates among older assets, when differences in remaining tax basis are less relevant. The transaction shortfall narrows following the Tax Cuts and Jobs Act. Aggregated upward, my estimates imply approximately $13–$30 billion of additional reallocation per year under lower-bonus or no-bonus counterfactuals from 2012–2017.

In sum, investment subsidies can create more capital while causing less of that capital to be subsequently reallocated. This matters because tax policy is a potentially important instrument to reduce the misallocation of capital and may ease the financial constraints of young entrants through a larger supply of used assets. Nevertheless, fully answering exactly how much the reallocation channel matters would require establishing the aggregate effects of the policy, characterizing optimal policy, and quantifying its consequences for welfare.

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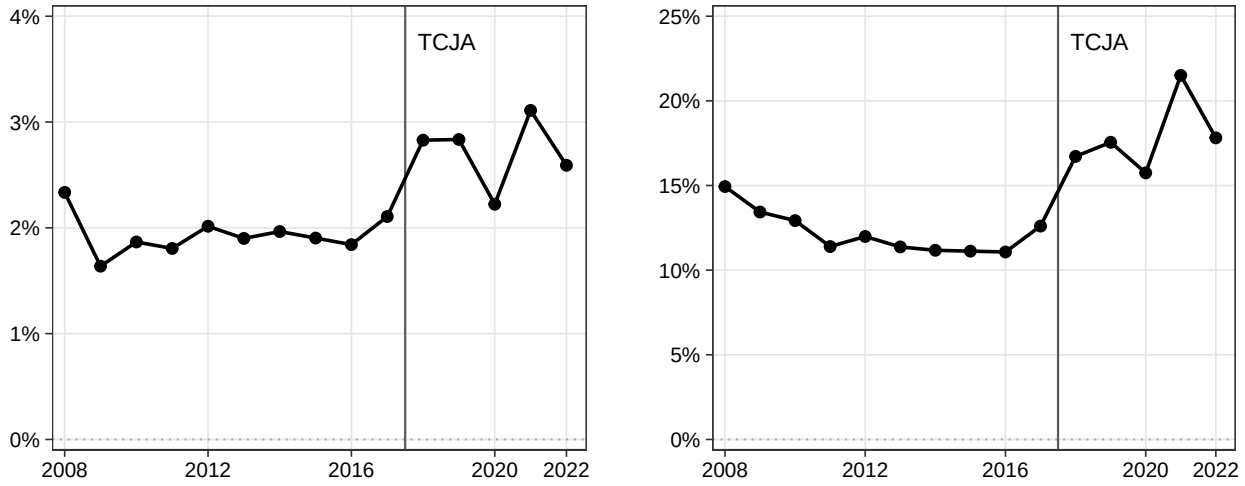
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A Institutional Detail

Figure A.1: Aggregate Used Equipment Sales

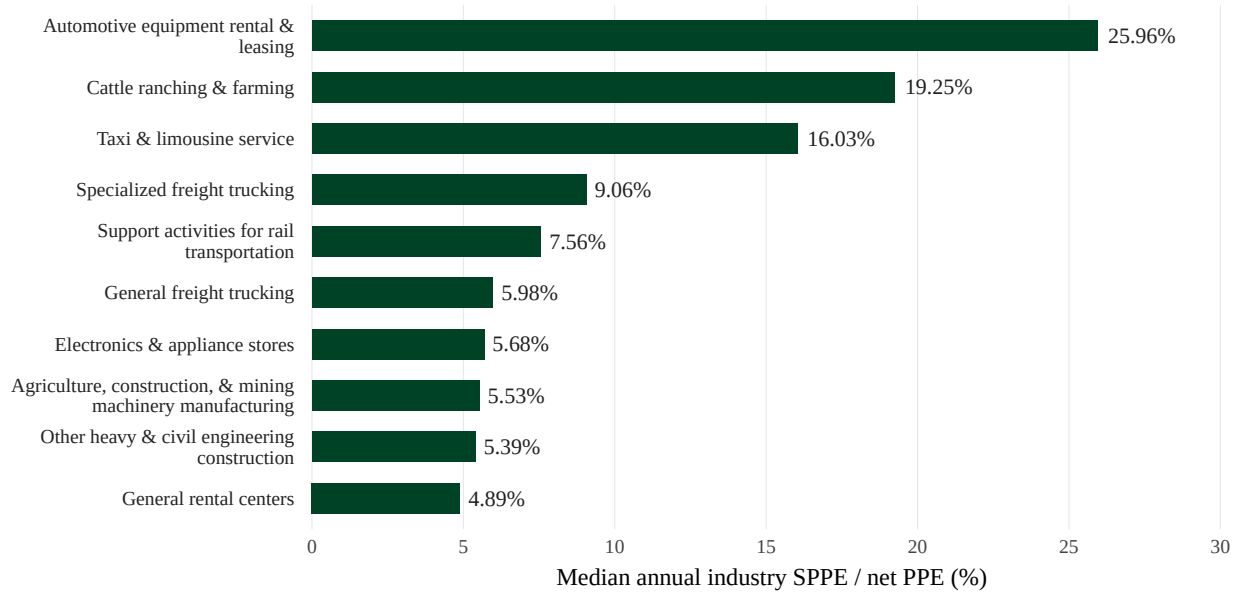


(a) Relative to equipment capital stock

(b) Relative to new equipment investment

Notes: Used equipment sales are aggregate Section 1245 dispositions from Form 4797 in SOI Publication 5108. Equipment capital stock and new investment are from BEA Fixed Asset Tables 4.1 and 4.7, respectively, line 18. The 2015 ratios are linearly interpolated. The vertical line marks TCJA.

Figure A.2: Asset Dispositions by Industry



Notes: The figure shows the ten four-digit NAICS industries with the highest median annual ratios of asset-sale proceeds to net property, plant, and equipment among U.S. nonfinancial Compustat firms, 1995–2016. Annual ratios are calculated using industry totals, and the figure reports their median over time.

B Theoretical Derivations

B.1 Stationary Competitive Equilibrium

Maintain the production functions

$$Y^u = \left[\omega_0 I^\rho + \sum_{a=1}^{3} \omega_a (\phi_a \widetilde{K}_a)^\rho\right]^{\nu/\rho},$$

$$Y^c = A^c \left[\sum_{a=1}^{3} \eta_a (\phi_a K_a^c)^\rho\right]^{\nu/\rho},$$

with their normalized Cobb–Douglas limits at $\rho = 0$. Both production functions are defined on nonnegative inputs by continuous extension. For $\rho \leq 0$, output is zero whenever any capital input is zero.

Both firm types have unit mass and discount factor $\beta \in (0,1)$. Assume $0 < \nu < 1$, $\rho < 1$, $A^c > 0$, positive weights summing to one within each technology, and

$$1 \geq \phi_1 \geq \phi_2 \geq \phi_3 > 0.$$

Policies are constant, with $\tau \in [0,1)$, $\theta^N, \theta^U \in [0,1]$, and

$$\bar{d}_j \geq 0, \qquad \bar{d}_0 + \bar{d}_1 + \bar{d}_2 = 1, \qquad \bar{d}_2 > 0.$$

The constrained firm's deduction values are

$$\widetilde{z}_1^c = \widetilde{z}_2^c = \theta^U + (1-\theta^U)(\bar{d}_0 + \beta \bar{d}_1 + \beta^2 \bar{d}_2),$$

$$\widetilde{z}_3^c = \theta^U + (1-\theta^U)[\bar{d}_0 + \beta(\bar{d}_1 + \bar{d}_2)].$$

Write

$$m = \frac{1-\tau}{1-\tau \widetilde{z}_1^c}, \qquad n = \frac{1-\tau}{1-\tau \widetilde{z}_3^c}.$$

Combining the unconstrained firm's investment and retention conditions gives

$$c_N = 1 + \frac{\tau(1-\beta)(1-\theta^N)}{1-\tau}[\bar{d}_1 + (1+\beta)\bar{d}_2],$$

$$\lambda = \frac{\tau(1-\beta)(1-\theta^N)\bar{d}_2}{(1-\tau)c_N}.$$

The deduction schedules imply

$$0 < m \leq n \leq 1, \qquad 1 \leq \frac{n}{m} < 1+\beta, \qquad 0 \leq \lambda < \frac{1}{1+\beta}.$$

Lemma 1 (Local stationary equilibrium)**.** *Suppose an interior stationary competitive equilibrium*

exists and

$$\sigma \geq \frac{\tau(1-\beta)(1-\theta^U)\beta(1+\beta)\bar{d}_2}{(1-\nu)\left[1-\tau+\tau(1-\beta)(1-\theta^U)\{\bar{d}_1+(1+\beta+\beta^2)\bar{d}_2\}\right]}, \qquad \sigma = \frac{1}{1-\rho}. \tag{B.1}$$

Then, for every sufficiently small admissible policy change, stationary equilibrium allocations and prices have a unique nearby continuation. This continuation is continuously differentiable in policy and remains interior.

Proof. Constrained firms' optimality conditions determine prices:

$$\begin{aligned} q_1 &= m\left(\partial_{K_1^c} Y^c + \beta\partial_{K_2^c} Y^c + \beta^2\partial_{K_3^c} Y^c\right), \\ q_2 &= m\left(\partial_{K_2^c} Y^c + \beta\partial_{K_3^c} Y^c\right), \\ q_3 &= n\partial_{K_3^c} Y^c. \end{aligned} \tag{B.2}$$

Stationary accumulation and market clearing give

$$K_a^c = I - \widetilde{K}_a.$$

Substituting the price conditions into the unconstrained firm's conditions gives

$$\begin{aligned} \partial_I Y^u + \sum_{a=1}^{3} \beta^a \partial_{\widetilde{K}_a} Y^u &= c_N, \\ \partial_{\widetilde{K}_1} Y^u - m\partial_{K_1^c} Y^c &= \lambda c_N, \\ \partial_{\widetilde{K}_2} Y^u - m\partial_{K_2^c} Y^c + \beta(n-m)\partial_{K_3^c} Y^c &= 0, \\ \partial_{\widetilde{K}_3} Y^u - n\partial_{K_3^c} Y^c &= 0. \end{aligned} \tag{B.3}$$

Normalize holdings by investment:

$$c_a = \frac{K_a^c}{I} = 1 - \frac{\widetilde{K}_a}{I}.$$

Interiority implies

$$0 < c_1 < c_2 < c_3 < 1.$$

Until investment is recovered below, evaluate outputs and marginal products at $I = 1$, $\widetilde{K}_a = 1 - c_a$, and $K_a^c = c_a$, taking physical-input derivatives before evaluation.

Homogeneity removes investment's scale from the retention conditions. In particular, the normalized young-capital condition is

$$\partial_{\widetilde{K}_1} Y^u - m\partial_{K_1^c} Y^c = \lambda\left(\partial_I Y^u + \sum_{a=1}^{3} \beta^a \partial_{\widetilde{K}_a} Y^u\right). \tag{B.4}$$

The ratio of the two firms' common CES marginal-product factors, including m, is

$$\mathcal{H}(c) = m(A^c)^{\rho/\nu}\left(\frac{Y^c}{Y^u}\right)^{(\nu-\rho)/\nu}. \tag{B.5}$$

At $\rho = 0$, this expression is mY^c/Y^u.

Temporarily treat that ratio as $H > 0$. The normalized retention conditions become

$$\omega_3\phi_3^\rho(1-c_3)^{\rho-1} = \frac{n}{m}H\eta_3\phi_3^\rho c_3^{\rho-1}, \tag{B.6}$$

$$\omega_2\phi_2^\rho(1-c_2)^{\rho-1} + \beta(1-m/n)\omega_3\phi_3^\rho(1-c_3)^{\rho-1} = H\eta_2\phi_2^\rho c_2^{\rho-1}, \tag{B.7}$$

$$\begin{aligned}(1-\beta\lambda)\omega_1\phi_1^\rho(1-c_1)^{\rho-1} - H\eta_1\phi_1^\rho c_1^{\rho-1} \\ = \lambda\left[\omega_0 + \sum_{a=2}^{3}\beta^a\omega_a\phi_a^\rho(1-c_a)^{\rho-1}\right].\end{aligned} \tag{B.8}$$

These equations determine c_3, c_2, c_1 successively. Each residual has a strictly positive derivative with respect to its own c_a. For the young equation, that derivative is

$$(1-\rho)\left[(1-\beta\lambda)\omega_1\phi_1^\rho(1-c_1)^{\rho-2} + H\eta_1\phi_1^\rho c_1^{\rho-2}\right] > 0.$$

The older equations have analogous positive own derivatives. Successive applications of the implicit-function theorem define a unique smooth $c(H)$ near the maintained equilibrium. Its strict ordering persists in a sufficiently small neighborhood.

Consistency of the marginal-product factors requires

$$H = \mathcal{H}(c(H)).$$

Define

$$G(H) = \log H - \log\mathcal{H}(c(H)). \tag{B.9}$$

The supporting CES lemma below establishes $G'(H) > 0$ at the maintained equilibrium. The implicit-function theorem gives a unique nearby solution for H under sufficiently small policy changes.

Homogeneity then recovers investment:

$$I = \left[\frac{\partial_I Y^u + \sum_{a=1}^{3}\beta^a\partial_{\widetilde{K}_a}Y^u}{c_N}\right]^{1/(1-\nu)}.$$

The marginal products in this expression are evaluated at unit investment. Recover physical holdings and transactions as

$$\widetilde{K}_a = I(1-c_a), \qquad K_a^c = Ic_a,$$

$$S_1 = U_1 = Ic_1, \qquad S_2 = U_2 = I(c_2-c_1), \qquad S_3 = U_3 = I(c_3-c_2).$$

The constrained firm's price conditions recover prices. These quantities are smooth in policy, and

strict positivity persists locally.

It remains to verify that the nearby stationary choices solve the firms' dynamic problems. Fix the recovered prices and use a star for the stationary allocation. Concavity gives

$$Y_t^u - Y^{u*} \le (\partial_I Y^{u*})(I_t - I^*) + \sum_{a=1}^{3} (\partial_{\widetilde{K}_a} Y^{u*})(\widetilde{K}_{a,t} - \widetilde{K}_a^*),$$

$$Y_t^c - Y^{c*} \le \sum_{a=1}^{3} (\partial_{K_a^c} Y^{c*})(K_{a,t}^c - K_a^{c*}).$$

Multiply by $(1-\tau)\beta^t$, sum through T, and substitute the laws of motion and first-order conditions. The linear terms cancel across dates.

Let Π_T^i denote discounted profits through T, using capitalized deduction costs for constrained firms. For any feasible path from the stationary inherited stocks,

$$\Pi_T^u - \Pi_T^{u*} \le \beta^{T+1} \sum_{a=1}^{3} [(1-\tau)q_a + \tau B_a] K_a^*, \tag{B.10}$$

where

$$(K_1^*, K_2^*, K_3^*) = (I^*, \widetilde{K}_1^*, \widetilde{K}_2^*),$$

and

$$\Pi_T^c - \Pi_T^{c*} \le \beta^{T+1} \left[q_2(1 - \tau\widetilde{z}_2^c) K_1^{c*} + q_3(1 - \tau\widetilde{z}_3^c) K_2^{c*} \right].$$

Here

$$B_1 = (1-\theta^N)(\bar{d}_1 + \bar{d}_2), \qquad B_2 = (1-\theta^N)\bar{d}_2, \qquad B_3 = 0.$$

These bounds drop the nonnegative concavity gaps and the competing path's nonnegative terminal stock values. Their right-hand sides converge to zero, so the stationary choices maximize discounted profits.

The same calculation supplies Bellman solutions at every finite inherited state. Each concavity gap grows at least linearly when inputs become large: its supporting plane has positive coefficients, while output has degree $\nu < 1$. Apart from terms fixed by the inherited stocks and prices, finite-horizon profits equal minus the discounted concavity gaps and a nonnegative terminal stock value. If discounted gaps are infinite, profits tend to $-\infty$. If discounted gaps are finite, the linear bound implies finite discounted inputs and vanishing discounted terminal stocks. Maximizing profits is therefore equivalent to minimizing discounted concavity gaps.

A feasible path with finite discounted gaps exists because inherited capital has finite life. A minimizing sequence has bounded inputs at every date. A diagonal subsequence and Fatou's lemma give an attaining feasible path. The Bellman principle supplies finite value functions and maximizing decision rules.

The nearby stationary allocations therefore constitute competitive equilibria. Every nearby interior equilibrium satisfies the same reduced conditions, whose local solution for allocations and prices is unique. □

Lemma 2 (CES curvature)**.** *At the maintained interior stationary competitive equilibrium satisfy-*

ing (B.1),

$$G'(H) > 0.$$

Proof. Evaluate outputs and marginal products at unit investment. Write

$$y_a = (1-\rho)Hc_a'(H).$$

Differentiating (B.6) gives

$$y_3 = c_3(1-c_3).$$

Substituting this into the derivative of (B.7) gives

$$\frac{y_2}{c_2(1-c_2)} = \frac{\partial_{\widetilde{K}_2} Y^u + \beta(1-m/n)(1-c_3)\partial_{\widetilde{K}_3} Y^u}{\partial_{\widetilde{K}_2} Y^u + \beta(1-m/n)(1-c_2)\partial_{\widetilde{K}_3} Y^u} \leq 1.$$

Both numerator and denominator are positive, so $y_2 > 0$.

Differentiating (B.8) and applying these older-age bounds gives $0 < y_1 \leq \bar{y}_1$, where

$$\bar{y}_1 = c_1(1-c_1)\frac{m\partial_{K_1^c} Y^c + \lambda \sum_{a=2}^3 \beta^a c_a \partial_{\widetilde{K}_a} Y^u}{m\partial_{K_1^c} Y^c + \lambda c_1 \left(\partial_I Y^u + \sum_{a=2}^3 \beta^a \partial_{\widetilde{K}_a} Y^u\right)}.$$

Set

$$\bar{y}_a = c_a(1-c_a), \qquad a = 2, 3.$$

The normalized young condition and Euler's identities imply

$$\begin{aligned} &\frac{\sum_{a=1}^3 \bar{y}_a \partial_{\widetilde{K}_a} Y^u}{\nu Y^u} \\ &\quad + \frac{m\bar{y}_1 \partial_{K_1^c} Y^c + \sum_{a=2}^3 \bar{y}_a \partial_{\widetilde{K}_a} Y^u}{m c_1 \partial_{K_1^c} Y^c + \sum_{a=2}^3 c_a \partial_{\widetilde{K}_a} Y^u} < 1. \end{aligned} \tag{B.11}$$

To verify this inequality, multiply the difference between one and its left-hand side by the two positive denominators. Substitute (B.4) and the expression for $\bar{y}_1$. The coefficient bounds used in the expansion follow from $\lambda < 1/(1+\beta)$:

$$0 < 1 - \lambda\beta(1-\beta^{a-1}) \leq 1,$$

$$\beta^a[1+\lambda(1-\beta)] \leq \beta, \qquad \beta^a \leq \beta^2, \qquad a = 2, 3.$$

The resulting expression is bounded below by

$$\begin{aligned} &(\partial_I Y^u)\left[m c_1^2 \partial_{K_1^c} Y^c + \sum_{a=2}^3 c_a^2 \partial_{\widetilde{K}_a} Y^u\right] \\ &\quad + m(\partial_{K_1^c} Y^c) \sum_{a=2}^3 (c_a - c_1)^2 \partial_{\widetilde{K}_a} Y^u \\ &\quad + (\partial_{\widetilde{K}_2} Y^u)(\partial_{\widetilde{K}_3} Y^u)(c_3 - c_2)^2 > 0. \end{aligned} \tag{B.12}$$

This proves (B.11).

The older allocation conditions give

$$m\partial_{K_2^c} Y^c = \partial_{\widetilde{K}_2} Y^u + \beta(1 - m/n)\partial_{\widetilde{K}_3} Y^u,$$

$$m\partial_{K_3^c} Y^c = \frac{m}{n}\partial_{\widetilde{K}_3} Y^u.$$

Consequently, Euler's identity gives

$$mvY^c \geq \frac{m}{n}\left[mc_1\partial_{K_1^c} Y^c + \sum_{a=2}^{3} c_a \partial_{\widetilde{K}_a} Y^u\right]. \tag{B.13}$$

Also,

$$\frac{\bar{y}_2 \partial_{\widetilde{K}_3} Y^u}{mc_1\partial_{K_1^c} Y^c + \sum_{a=2}^{3} c_a \partial_{\widetilde{K}_a} Y^u} < \frac{c_2(1 - c_2)}{c_3} < 1. \tag{B.14}$$

Using these bounds, the constrained-firm term

$$\frac{\sum_{a=1}^{3} y_a \partial_{K_a^c} Y^c}{vY^c}$$

is bounded above by n/m times the second fraction in (B.11), plus at most $\beta(n/m - 1)$. Both fractions in (B.11) are positive and sum to less than one. Since $y_a \leq \bar{y}_a$, it follows that

$$\sum_{a=1}^{3} y_a \left[\frac{\partial_{\widetilde{K}_a} Y^u}{vY^u} + \frac{\partial_{K_a^c} Y^c}{vY^c}\right] < 1 + (1 + \beta)\left(\frac{n}{m} - 1\right). \tag{B.15}$$

Differentiating the definition of G gives

$$HG'(H) = 1 - \frac{\nu - \rho}{1 - \rho}\sum_{a=1}^{3} y_a \left[\frac{\partial_{\widetilde{K}_a} Y^u}{vY^u} + \frac{\partial_{K_a^c} Y^c}{vY^c}\right]. \tag{B.16}$$

If $\rho \geq \nu$, this is positive immediately. If $\rho < \nu$, the deduction schedules imply

$$\frac{n}{m} - 1 = \frac{\tau(1 - \beta)(1 - \theta^U)\beta\bar{d}_2}{1 - \tau + \tau(1 - \beta)(1 - \theta^U)(\bar{d}_1 + \bar{d}_2)}.$$

Condition (B.1) is equivalent to

$$\frac{\nu - \rho}{1 - \rho}\left[1 + (1 + \beta)\left(\frac{n}{m} - 1\right)\right] \leq 1.$$

Combining this with the strict bound in (B.15) gives $G'(H) > 0$, including when (B.1) holds at equality. □

B.2 Proof of Proposition 1

Proof. I first establish how constrained firms' holdings per unit of investment change. This ordering will establish that investment rises.

Retain the SCE notation $c_a = K_a^c/I$, $\mathcal{H}$, G, and $y_a = (1-\rho)Hc_a'(H)$. Initially evaluate output and marginal products at unit investment, as in the SCE appendix.

In the comparison calculation from Lemma 2, replace $\bar{y}_1$ by $c_1(1-c_1)$. The expressions for ages two and three are unchanged. This gives

$$\frac{\sum_{a=1}^{3} c_a(1-c_a)\partial_{\widetilde{K}_a} Y^u}{vY^u} + \frac{mc_1(1-c_1)\partial_{K_1^c} Y^c + \sum_{a=2}^{3} c_a(1-c_a)\partial_{\widetilde{K}_a} Y^u}{mc_1\partial_{K_1^c} Y^c + \sum_{a=2}^{3} c_a\partial_{\widetilde{K}_a} Y^u} < 1. \tag{B.17}$$

To verify the inequality, multiply the gap between one and the left-hand side by its two positive denominators. The result is the positive sum of squares in (B.12) plus

$$\left(\partial_{\widetilde{K}_1} Y^u - m\partial_{K_1^c} Y^c\right)(1-c_1)\sum_{a=2}^{3} c_a(c_a-c_1)\partial_{\widetilde{K}_a} Y^u.$$

This additional term is nonnegative: the normalized young-retention equation makes its first factor nonnegative, and $c_1 < c_2 < c_3$.

The constrained firm's actual deduction costs differ across ages. The older equilibrium conditions give

$$m\partial_{K_2^c} Y^c = \partial_{\widetilde{K}_2} Y^u + \beta\left(1-\frac{m}{n}\right)\partial_{\widetilde{K}_3} Y^u, \qquad m\partial_{K_3^c} Y^c = \frac{m}{n}\partial_{\widetilde{K}_3} Y^u.$$

Using these identities and Euler's identity,

$$\begin{aligned} mvY^c &= mc_1\partial_{K_1^c} Y^c + c_2\partial_{\widetilde{K}_2} Y^u \\ &\quad + \left[\frac{m}{n}c_3 + \beta\left(1-\frac{m}{n}\right)c_2\right]\partial_{\widetilde{K}_3} Y^u \\ &\geq \frac{m}{n}\left[mc_1\partial_{K_1^c} Y^c + \sum_{a=2}^{3} c_a\partial_{\widetilde{K}_a} Y^u\right]. \end{aligned}$$

The corresponding numerator satisfies

$$\begin{aligned} m\sum_{a=1}^{3} c_a(1-c_a)\partial_{K_a^c} Y^c &\leq mc_1(1-c_1)\partial_{K_1^c} Y^c \\ &\quad + \sum_{a=2}^{3} c_a(1-c_a)\partial_{\widetilde{K}_a} Y^u \\ &\quad + \beta\left(1-\frac{m}{n}\right)c_2(1-c_2)\partial_{\widetilde{K}_3} Y^u. \end{aligned}$$

Consequently, the constrained firm's actual fraction is at most n/m times the second fraction in

(B.17), plus $\beta(n/m-1)$ times

$$\frac{c_2(1-c_2)\partial_{\widetilde{K}_3}Y^u}{mc_1\partial_{K_1^c}Y^c+\sum_{a=2}^3 c_a\partial_{\widetilde{K}_a}Y^u}<\frac{c_2(1-c_2)}{c_3}<1.$$

The second fraction in (B.17) is also strictly between zero and one. Adding the unconstrained firm's fraction and applying (B.17) gives

$$\sum_{a=1}^3 c_a(1-c_a)\left[\frac{\partial_{\widetilde{K}_a}Y^u}{\nu Y^u}+\frac{\partial_{K_a^c}Y^c}{\nu Y^c}\right]<1+(1+\beta)\left(\frac{n}{m}-1\right). \tag{B.18}$$

Since

$$\partial_{c_a}\log\mathcal{H}=(\nu-\rho)\left[\frac{\partial_{\widetilde{K}_a}Y^u}{\nu Y^u}+\frac{\partial_{K_a^c}Y^c}{\nu Y^c}\right]\geq 0,$$

and the maintained elasticity restriction implies

$$\frac{\nu-\rho}{1-\rho}\left[1+(1+\beta)\left(\frac{n}{m}-1\right)\right]\leq 1,$$

equation (B.18) yields

$$\sum_{a=1}^3 c_a(1-c_a)\partial_{c_a}\log\mathcal{H}<1-\rho. \tag{B.19}$$

Lemma 2 gives $0<y_a\leq c_a(1-c_a)$ for $a=2,3$. Hence

$$(1-\rho)-\sum_{a=2}^3 y_a\partial_{c_a}\log\mathcal{H}>c_1(1-c_1)\partial_{c_1}\log\mathcal{H}\geq 0. \tag{B.20}$$

At fixed H, the older triangular equations do not depend on λ. The young equation has a residual that increases in c_1 and decreases in λ, so $\partial_\lambda c_1|_H>0$. Differentiating $G(H)=0$, including the dependence of the young equation on λ, gives

$$\frac{dc_1}{d\lambda}=\partial_\lambda c_1|_H\,\frac{(1-\rho)-\sum_{a=2}^3 y_a\partial_{c_a}\log\mathcal{H}}{(1-\rho)HG'(H)}>0.$$

The numerator is positive by (B.20), and $G'(H)>0$ follows from Lemma 2.

Along the older equations,

$$(1-\rho)\,dc_a=y_a\,d\log H,\qquad a=2,3.$$

Combining these equations with $H=\mathcal{H}(c)$ gives

$$\frac{dc_a}{dc_1}=\frac{y_a\partial_{c_1}\log\mathcal{H}}{(1-\rho)-\sum_{b=2}^3 y_b\partial_{c_b}\log\mathcal{H}},\qquad a=2,3.$$

The older derivative formulas also imply

$$\frac{y_3}{c_3} = 1 - c_3 < \frac{y_2}{c_2} \leq 1 - c_2 < 1 - c_1.$$

Together with (B.20), these inequalities show that the proportional response of c_1 exceeds those of c_2 and c_3. Finally,

$$\frac{d\lambda}{d\theta^N} = -\frac{\tau(1-\beta)\bar{d}_2}{(1-\tau)c_N^2} < 0.$$

It follows that

$$-\frac{d\log c_1}{d\theta^N} > -\frac{d\log c_2}{d\theta^N} \geq -\frac{d\log c_3}{d\theta^N} \geq 0. \tag{B.21}$$

At $\rho = \nu$, the older fractions remain unchanged because $\partial_{c_1} \log \mathcal{H} = 0$.

Return now to actual investment and holdings. Suppose, for a contradiction, that $dI/d\theta^N \leq 0$. Since $K_a^c = Ic_a$, equation (B.21) implies

$$\frac{d\log K_1^c}{d\theta^N} < \frac{d\log K_2^c}{d\theta^N} \leq \frac{d\log K_3^c}{d\theta^N} \leq 0. \tag{B.22}$$

The constrained firm's young-purchase condition is

$$q_1 = m\sum_{a=1}^{3} \beta^{a-1}\partial_{K_a^c} Y^c,$$

where m is fixed. CES differentiation gives

$$\frac{d\log q_1}{d\theta^N} = (\nu-\rho)\frac{\sum_{a=1}^3 K_a^c(\partial_{K_a^c}Y^c)\, d\log K_a^c/d\theta^N}{\nu Y^c} - (1-\rho)\frac{\sum_{a=1}^3 \beta^{a-1}(\partial_{K_a^c}Y^c)\, d\log K_a^c/d\theta^N}{\sum_{a=1}^3 \beta^{a-1}\partial_{K_a^c}Y^c}.$$

The first weighted average is at least the second. To verify this, cross-multiply their difference. Its numerator is

$$\sum_{1\leq a<b\leq 3} (\partial_{K_a^c}Y^c)(\partial_{K_b^c}Y^c)\left(\beta^{a-1}K_b^c - \beta^{b-1}K_a^c\right) \times \left(\frac{d\log K_b^c}{d\theta^N} - \frac{d\log K_a^c}{d\theta^N}\right) \geq 0.$$

Every term is nonnegative by (B.22), $K_a^c < K_b^c$ for $a < b$, and $\beta \in (0,1)$. Thus

$$\frac{d\log q_1}{d\theta^N} \geq -\frac{1-\nu}{\nu Y^c}\sum_{a=1}^{3} K_a^c(\partial_{K_a^c}Y^c)\frac{d\log K_a^c}{d\theta^N} > 0. \tag{B.23}$$

The strict inequality follows because young holdings strictly decline.

For the unconstrained firm, CES differentiation gives

$$\frac{d\log(\partial_I Y^u)}{d\theta^N} = -(1-\nu)\frac{d\log I}{d\theta^N} - \frac{(\nu-\rho)I}{\nu Y^u}\sum_{a=1}^{3}(\partial_{\widetilde{K}_a} Y^u)\frac{dc_a}{d\theta^N} \geq 0.$$

But differentiating its investment condition requires

$$\frac{d(\partial_I Y^u)}{d\theta^N} + \beta\frac{dq_1}{d\theta^N} = -\frac{\tau(1-\beta)}{1-\tau}(\bar{d}_1 + \bar{d}_2) < 0.$$

This contradicts (B.23) and the nonnegative change in the marginal product of investment. Hence

$$\frac{dI}{d\theta^N} > 0.$$

To characterize young sales, solve the older equations conditional on (I, U_1), with $\widetilde{K}_1 = I - U_1$. Equation (B.20) ensures that this local solution exists. The two remaining equilibrium equations are

$$\partial_I Y^u + \sum_{a=1}^{3}\beta^a \partial_{\widetilde{K}_a} Y^u = c_N,$$
$$\partial_{\widetilde{K}_1} Y^u - q_1 + \beta q_2 = \lambda c_N.$$

All allocation derivatives below allow older holdings and prices to adjust through the older equilibrium conditions.

Both left-hand sides are homogeneous of degree $\nu - 1$. Their ratio depends only on $c_1 = U_1/I$ and equals λ in equilibrium. Since $dc_1/d\lambda > 0$,

$$\frac{\partial}{\partial U_1}\left[\frac{\partial_{\widetilde{K}_1} Y^u - q_1 + \beta q_2}{\partial_I Y^u + \sum_{a=1}^{3}\beta^a \partial_{\widetilde{K}_a} Y^u}\right]\Bigg|_I = \frac{1}{I}\left(\frac{dc_1}{d\lambda}\right)^{-1} > 0.$$

Applying Euler's identity to the two left-hand sides gives their Jacobian determinant with respect to (I, U_1):

$$-\frac{(1-\nu)c_N^2}{I^2}\left(\frac{dc_1}{d\lambda}\right)^{-1} < 0.$$

The policy derivatives of the right-hand sides are

$$\frac{dc_N}{d\theta^N} = -\frac{\tau(1-\beta)}{1-\tau}\left[\bar{d}_1 + (1+\beta)\bar{d}_2\right], \qquad \frac{d(\lambda c_N)}{d\theta^N} = -\frac{\tau(1-\beta)}{1-\tau}\bar{d}_2.$$

Differentiating the two equilibrium equations and applying Cramer's rule expresses $dU_1/d\theta^N$ as

the following numerator divided by the negative determinant:

$$\frac{\tau(1-\beta)}{1-\tau}\Bigg\{\left[\bar{d}_1+(1+\beta)\bar{d}_2\right]\frac{\partial}{\partial I}\left(\partial_{\widetilde{K}_1}Y^u-q_1+\beta q_2\right)\Big|_{U_1} - \bar{d}_2\frac{\partial}{\partial I}\left(\partial_I Y^u+\sum_{a=1}^{3}\beta^a\partial_{\widetilde{K}_a}Y^u\right)\Big|_{U_1}\Bigg\}.$$

Young reallocation falls precisely when this numerator is positive. Dividing by $\bar{d}_2 > 0$ and rearranging gives

$$-\frac{\partial}{\partial I}\left(\partial_I Y^u+\sum_{a=1}^{3}\beta^a\partial_{\widetilde{K}_a}Y^u\right)\Big|_{U_1} > \left(1+\beta+\frac{\bar{d}_1}{\bar{d}_2}\right)\left[-\frac{\partial}{\partial I}\left(\partial_{\widetilde{K}_1}Y^u-q_1+\beta q_2\right)\Big|_{U_1}\right].$$

Reversing the inequality gives an increase in young reallocation; equality gives no first-order change. □

B.3 Proof of Proposition 2

Proof. Maintain $\tau > 0$, as in Proposition 1. Hold θ^U fixed and retain the notation of the SCE appendix and the proof of Proposition 1. Stationary accumulation and market clearing imply

$$U_1 = Ic_1, \qquad U_2 = I(c_2-c_1), \qquad U_3 = I(c_3-c_2).$$

Thus total reallocation satisfies

$$U = U_1+U_2+U_3 = Ic_3 = I-\widetilde{K}_3.$$

After substituting the constrained firm's price conditions, the older equilibrium conditions are

$$\partial_{\widetilde{K}_2}Y^u - m\partial_{K_2^c}Y^c+\beta(n-m)\partial_{K_3^c}Y^c = 0,$$
$$\partial_{\widetilde{K}_3}Y^u - n\partial_{K_3^c}Y^c = 0.$$

These conditions contain no direct dependence on θ^N. They are homogeneous of degree $\nu-1$, so their normalized equations are independent of investment's scale.

To obtain those responses, evaluate marginal products at unit investment. Along the older triangular equations,

$$(1-\rho)\,dc_a = y_a\,d\log H, \qquad a = 2,3.$$

Differentiating $H = \mathcal{H}(c)$ and substituting these expressions gives

$$\left[(1-\rho)-\sum_{a=2}^{3}y_a\partial_{c_a}\log\mathcal{H}\right]d\log H = (1-\rho)\partial_{c_1}\log\mathcal{H}\,dc_1.$$

Hence

$$\frac{dc_a}{dc_1} = \frac{y_a \partial_{c_1} \log \mathcal{H}}{(1-\rho) - \sum_{b=2}^{3} y_b \partial_{c_b} \log \mathcal{H}}, \qquad a = 2, 3. \tag{B.24}$$

The CES derivative is

$$\partial_{c_a} \log \mathcal{H} = (\nu - \rho) \left[\frac{\partial_{\widetilde{K}_a} Y^u}{\nu Y^u} + \frac{\partial_{K_a^c} Y^c}{\nu Y^c} \right]. \tag{B.25}$$

For $\rho \geq \nu$, these derivatives are nonpositive, making the denominator in (B.24) positive. For $\rho < \nu$, the curvature bound established in the proof of Proposition 1 gives

$$(1-\rho) - \sum_{a=2}^{3} y_a \partial_{c_a} \log \mathcal{H} > c_1(1-c_1)\partial_{c_1} \log \mathcal{H} > 0. \tag{B.26}$$

The older equations are thus locally solvable conditional on c_1 in every case, determining c_2 and c_3 as functions of c_1. Since $y_3 = c_3(1-c_3) > 0$, equations (B.24)–(B.25) show that dc_3/dc_1 is negative for substitutes, zero for additive vintages, and positive for complements.

First consider $\rho > \nu$. The differentiation of the young equation and $G(H) = 0$, used in the proof of Proposition 1, gives

$$\frac{dc_1}{d\lambda} = \partial_\lambda c_1|_H \frac{(1-\rho) - \sum_{a=2}^{3} y_a \partial_{c_a} \log \mathcal{H}}{(1-\rho) H G'(H)} > 0.$$

Here $\partial_\lambda c_1|_H > 0$ follows from the young triangular equation, the numerator is positive by (B.25), and $G'(H) > 0$ follows from the SCE lemma. Since

$$\frac{d\lambda}{d\theta^N} = -\frac{\tau(1-\beta)\bar{d}_2}{(1-\tau)c_N^2} < 0,$$

the young fraction satisfies $dc_1/d\theta^N < 0$. Equation (B.24) then gives

$$\frac{dc_3}{d\theta^N} = \frac{dc_3}{dc_1}\frac{dc_1}{d\theta^N} > 0.$$

Differentiating $U = Ic_3$,

$$\frac{dU}{d\theta^N} = c_3 \frac{dI}{d\theta^N} + I \frac{dc_3}{d\theta^N}.$$

Both terms are positive whenever investment rises. This proves the substitutes result.

At $\rho = \nu$, equation (B.25) is zero, so equation (B.24) gives

$$\frac{dc_2}{dc_1} = \frac{dc_3}{dc_1} = 0.$$

The older fractions remain unchanged when θ^N changes. Proposition 1 establishes $dI/d\theta^N > 0$, yielding

$$\frac{dU}{d\theta^N} = c_3 \frac{dI}{d\theta^N} > 0.$$

Now suppose $\rho < \nu$. To express total reallocation in terms of investment and young sales, use $c_1 = U_1/I$. After the older markets clear,

$$U = I\, c_3(U_1/I).$$

Differentiating with respect to each argument gives

$$\left.\frac{\partial U}{\partial U_1}\right|_I = \frac{dc_3}{dc_1} > 0, \qquad \left.\frac{\partial U}{\partial I}\right|_{U_1} = c_3 - c_1\frac{dc_3}{dc_1}. \tag{B.27}$$

The second derivative is also positive. Indeed, equations (B.24) and (B.26) imply

$$0 < c_1\frac{dc_3}{dc_1} < \frac{y_3}{1-c_1} = c_3\frac{1-c_3}{1-c_1} < c_3,$$

where the final inequality uses $c_1 < c_3$. Thus

$$\left.\frac{\partial U}{\partial I}\right|_{U_1} > 0. \tag{B.28}$$

The individual sales identities give

$$\left.\frac{\partial U_2}{\partial U_1}\right|_I = \frac{dc_2}{dc_1} - 1, \qquad \left.\frac{\partial U_3}{\partial U_1}\right|_I = \frac{dc_3}{dc_1} - \frac{dc_2}{dc_1}.$$

Adding them yields

$$1 + \left.\frac{\partial U_2}{\partial U_1}\right|_I + \left.\frac{\partial U_3}{\partial U_1}\right|_I = \frac{dc_3}{dc_1}. \tag{B.29}$$

All these partial derivatives incorporate the adjustment of older holdings and prices through the older equilibrium conditions.

Since those conditions have no direct dependence on θ^N, the total policy response is

$$\begin{aligned}\frac{dU}{d\theta^N} = {} & \left.\frac{\partial U}{\partial I}\right|_{U_1}\frac{dI}{d\theta^N} \\ & + \left(1 + \left.\frac{\partial U_2}{\partial U_1}\right|_I + \left.\frac{\partial U_3}{\partial U_1}\right|_I\right)\frac{dU_1}{d\theta^N}.\end{aligned}$$

Proposition 1 establishes that investment rises. Equations (B.27)–(B.29) establish that both coefficients in this decomposition are positive. Consequently, total reallocation falls if and only if

$$\left(1 + \left.\frac{\partial U_2}{\partial U_1}\right|_I + \left.\frac{\partial U_3}{\partial U_1}\right|_I\right)\left(-\frac{dU_1}{d\theta^N}\right) > \left.\frac{\partial U}{\partial I}\right|_{U_1}\frac{dI}{d\theta^N}.$$

Reversing the inequality gives an increase in total reallocation; equality gives no first-order change. □

B.4 Proof of Proposition 3

Proof. Hold $\theta^N = 1$ and retain the notation of the SCE appendix. Stationary accumulation and market clearing give

$$\frac{\widetilde{K}_a}{I} = 1 - c_a, \qquad \frac{U_1}{U} = \frac{c_1}{c_3}, \qquad \frac{U_1 + U_2}{U} = \frac{c_2}{c_3}.$$

It suffices to establish the corresponding responses of the normalized holdings and their ratios.

First, the constrained firm's deduction multipliers satisfy

$$\frac{d\log n}{d\theta^U} = \frac{\tau(1-\beta)(\bar{d}_1 + \bar{d}_2)}{1 - \tau\widetilde{z}_3^c} > 0,$$

and

$$\frac{d\log m}{d\theta^U} - \frac{d\log n}{d\theta^U} = \frac{\tau(1-\tau)\beta(1-\beta)\bar{d}_2}{(1-\tau\widetilde{z}_1^c)(1-\tau\widetilde{z}_3^c)} > 0.$$

Thus both multipliers rise, with a larger proportional increase in m.

Evaluate output and marginal products at unit investment. Because $\mathcal{H}(c)$ is m times a function of the normalized holdings, differentiating $H = \mathcal{H}(c)$ gives

$$\frac{d\log(H/m)}{d\theta^U} = \sum_{a=1}^{3} \partial_{c_a} \log \mathcal{H} \, \frac{dc_a}{d\theta^U}, \tag{B.30}$$

where

$$\partial_{c_a} \log \mathcal{H} = (\nu - \rho)\left[\frac{\partial_{\widetilde{K}_a} Y^u}{\nu Y^u} + \frac{\partial_{K_a^c} Y^c}{\nu Y^c}\right].$$

Since $\theta^N = 1$, we have $\lambda = 0$, $y_1 = c_1(1-c_1)$, and $y_3 = c_3(1-c_3)$. Differentiating the young and age-three triangular equations yields

$$(1-\rho)\frac{dc_1}{d\theta^U} = y_1\left[\frac{d\log(H/m)}{d\theta^U} + \frac{d\log m}{d\theta^U}\right], \tag{B.31}$$

$$(1-\rho)\frac{dc_3}{d\theta^U} = y_3\left[\frac{d\log(H/m)}{d\theta^U} + \frac{d\log n}{d\theta^U}\right]. \tag{B.32}$$

The age-two triangular equation also contains the coefficient $\beta(1 - m/n)$. Its derivative is

$$\frac{d}{d\theta^U}\left[\beta\left(1 - \frac{m}{n}\right)\right] = -\beta\frac{m}{n}\left(\frac{d\log m}{d\theta^U} - \frac{d\log n}{d\theta^U}\right).$$

Including this term, substituting the age-three response, and collecting terms gives

$$
\begin{aligned}
&(1-\rho)\frac{\partial_{\widetilde{K}_2} Y^u + \beta(1-m/n)(1-c_2)\partial_{\widetilde{K}_3} Y^u}{c_2(1-c_2)}\frac{dc_2}{d\theta^U} \\
&\quad = \left[\partial_{\widetilde{K}_2} Y^u + \beta(1-m/n)(1-c_3)\partial_{\widetilde{K}_3} Y^u\right]\left[\frac{d\log(H/m)}{d\theta^U} + \frac{d\log n}{d\theta^U}\right] \\
&\qquad + \left[\partial_{\widetilde{K}_2} Y^u + \beta\partial_{\widetilde{K}_3} Y^u\right]\left(\frac{d\log m}{d\theta^U} - \frac{d\log n}{d\theta^U}\right).
\end{aligned}
$$

Using the expression for y_2 in the CES lemma, this becomes

$$
\begin{aligned}
(1-\rho)\frac{dc_2}{d\theta^U} = y_2\Bigg[&\frac{d\log(H/m)}{d\theta^U} + \frac{d\log n}{d\theta^U} \\
&+ \left(\frac{d\log m}{d\theta^U} - \frac{d\log n}{d\theta^U}\right)\frac{\partial_{\widetilde{K}_2} Y^u + \beta\partial_{\widetilde{K}_3} Y^u}{\partial_{\widetilde{K}_2} Y^u + \beta(1-m/n)(1-c_3)\partial_{\widetilde{K}_3} Y^u}\Bigg].
\end{aligned}
\tag{B.33}
$$

Here $y_2 > 0$ is the older derivative from the CES lemma.

Suppose first that $\rho \le \nu$. Then $\partial_{c_a}\log\mathcal{H} \ge 0$. Subtracting $y_a\, d\log(H/m)/d\theta^U$ from each response equation leaves a strictly positive direct policy term. Multiplying by $\partial_{c_a}\log\mathcal{H}$, summing, and using (B.30) gives

$$
\begin{aligned}
&\left[(1-\rho) - \sum_{a=1}^{3} y_a \partial_{c_a}\log\mathcal{H}\right]\frac{d\log(H/m)}{d\theta^U} \\
&\qquad = \sum_{a=1}^{3}\partial_{c_a}\log\mathcal{H}\left[(1-\rho)\frac{dc_a}{d\theta^U} - y_a\frac{d\log(H/m)}{d\theta^U}\right] \ge 0.
\end{aligned}
$$

The CES lemma establishes

$$
(1-\rho) - \sum_{a=1}^{3} y_a \partial_{c_a}\log\mathcal{H} = (1-\rho)HG'(H) > 0.
$$

Consequently,

$$
\frac{d\log(H/m)}{d\theta^U} \ge 0. \tag{B.34}
$$

Equations (B.31)–(B.33) now imply $dc_a/d\theta^U > 0$ for every age. Hence

$$
\frac{d}{d\theta^U}\left(\frac{\widetilde{K}_a}{I}\right) = -\frac{dc_a}{d\theta^U} < 0, \qquad a = 1, 2, 3.
$$

To establish the young transaction-share result, subtract the logarithmic versions of (B.31)

and (B.32):

$$(1-\rho)\frac{d}{d\theta^U}\log\left(\frac{c_1}{c_3}\right) = (c_3-c_1)\left[\frac{d\log(H/m)}{d\theta^U}+\frac{d\log n}{d\theta^U}\right] + (1-c_1)\left(\frac{d\log m}{d\theta^U}-\frac{d\log n}{d\theta^U}\right) > 0.$$

The inequality follows from (B.34), the strict ordering of holdings, and the deduction-multiplier derivatives. Since $U_1/U = c_1/c_3 > 0$, this proves part (i).

For part (ii), the CES expression for y_2 gives

$$\frac{y_2}{c_2} - (1-c_3) = \frac{(c_3-c_2)\partial_{\widetilde{K}_2} Y^u}{\partial_{\widetilde{K}_2} Y^u + \beta(1-m/n)(1-c_2)\partial_{\widetilde{K}_3} Y^u} > 0. \tag{B.35}$$

We first show that $dc_2/d\theta^U > 0$ also holds when $\rho > \nu$. The ratio in (B.33) satisfies

$$\frac{\partial_{\widetilde{K}_2} Y^u + \beta\partial_{\widetilde{K}_3} Y^u}{\partial_{\widetilde{K}_2} Y^u + \beta(1-m/n)(1-c_3)\partial_{\widetilde{K}_3} Y^u} > 1.$$

Thus the direct policy term inside the brackets in (B.33) is strictly larger than $d\log m/d\theta^U$.

Suppose $dc_2/d\theta^U \le 0$. Equation (B.33) would then imply

$$\frac{d\log(H/m)}{d\theta^U}+\frac{d\log m}{d\theta^U} < 0, \qquad \frac{d\log(H/m)}{d\theta^U}+\frac{d\log n}{d\theta^U} < 0.$$

Equations (B.31)–(B.32) would give $dc_1/d\theta^U < 0$ and $dc_3/d\theta^U < 0$. But when $\rho > \nu$, all the derivatives $\partial_{c_a}\log\mathcal{H}$ are strictly negative. Equation (B.30) would imply

$$\frac{d\log(H/m)}{d\theta^U} > 0,$$

contradicting the preceding inequalities. Hence $dc_2/d\theta^U > 0$ for substitutes as well.

Now consider the two possible signs of the bracket in (B.32). If

$$\frac{d\log(H/m)}{d\theta^U}+\frac{d\log n}{d\theta^U} \ge 0,$$

equations (B.32)–(B.33) and (B.35) imply

$$\begin{aligned}(1-\rho)\frac{d\log c_2}{d\theta^U} &> \frac{y_2}{c_2}\left[\frac{d\log(H/m)}{d\theta^U}+\frac{d\log n}{d\theta^U}\right]\\ &\ge (1-c_3)\left[\frac{d\log(H/m)}{d\theta^U}+\frac{d\log n}{d\theta^U}\right]\\ &= (1-\rho)\frac{d\log c_3}{d\theta^U}.\end{aligned}$$

If that bracket is negative, equation (B.32) gives $dc_3/d\theta^U < 0$, while $dc_2/d\theta^U > 0$. In either case,

$$\frac{d}{d\theta^U}\left(\frac{c_2}{c_3}\right) > 0.$$

Since $(U_1 + U_2)/U = c_2/c_3$, this proves part (ii) for every $\rho < 1$.

Finally, let $\theta^U = 1$. With both new and used purchases fully expensed, the unconstrained firm's period payoff is

$$(1-\tau)\left[Y_t^u - I_t + \sum_{a=1}^{3} q_{a,t} S_{a,t}\right],$$

and the constrained firm's period payoff is

$$(1-\tau)\left[Y_t^c - \sum_{a=1}^{3} q_{a,t} U_{a,t}\right].$$

At the same gross prices, these are $1-\tau$ times their respective no-tax payoffs. The feasible choices, accumulation equations, and discount factor are unchanged. Multiplication by the positive constant $1-\tau$ leaves both firms' optimal decisions unchanged, and the market-clearing conditions are identical. Thus the full-expensing and no-tax economies have the same stationary equilibrium allocations and gross prices. □

C UCC Filings

C.1 Parser Details

This appendix describes how I transform raw Texas UCC financing statements into the analysis sample. The raw data are fixed-width text files originally from the Texas Secretary of State's office, but are provided by Jensen (2024). Each row in a text file begins with a one-digit label indicating the type of information it contains. There are seven substantive labels distinguishing filing metadata, debtors, secured parties, amendments, and collateral descriptions. A final summary record reports totals across the bulk-file chunks, allowing me to verify that the complete set has been parsed correctly. I proceed in several steps to construct the data. The resulting observations are equipment mentions in financing statements, rather than verified ownership transfers.

Reconstructing Collateral Descriptions and Filing Dates

The first step is to recover a complete collateral description and the correct observation date for each filing event. This is challenging because a description may be divided across multiple rows, and it is possible that the equipment either appears on an original financing statement or in a later amendment.

As a first step, I verify that every row has the expected format and that the aggregate number of rows of each kind agrees with the totals supplied by the Texas Secretary of State in the final summary record.

Each file divides collateral descriptions into consecutive 80-character pieces which may span multiple rows. Each row contains the original filing number, an event number, a sequence number, and the 80-character piece. I group rows with the same filing and event numbers, which ensures that they belong to the same filing event, and join their text fields end to end without inserting spaces. After reconstructing the full description, I remove padding and standardize white space. Doing it this way (rather than padding ex ante) preserves model and serial numbers that cross an 80-character boundary. The sequence numbering indicates the order in which the descriptions should be concatenated; I require these within a group to be consecutive.

Finally, I label each description with the filing event in which it appears. If the filing is indicated to be an original financing statement, I use the initial filing date. I recover that date from the master filing record; when it is missing, I infer it from a valid expiration date by subtracting five years. Sometimes financing statements are amended. In that case, I link the description using the original filing number and the state's UCC3 event number and use the date of the event. I retain the original filing year separately, but define the observation year as the year in which the equipment appears. Asset age is the observation year minus the model year. In the paper, I refer to the observation year as the filing year.

Blanket Liens and Itemized Equipment

The analysis requires collateral descriptions that identify particular pieces of equipment. Some filing events instead contain blanket liens, which give the lender a security interest in all or broad categories of the borrower's property without identifying any specific machine. References to "equipment" or "inventory" in these descriptions do not establish that a particular asset was involved in the transaction.

I identify blanket language using the complete collateral description associated with each filing event. Examples include "all personal property," "all inventory," "all assets," "general intangibles," "accounts receivable," and "now owned or hereafter acquired." These phrases do not by themselves generate equipment observations. However, a description may combine blanket language with a list of identifiable machines. I retain those machines when their equipment characteristics can be recovered from the itemized portion of the description, rather than excluding the entire event.

Separating Multiple Pieces of Equipment

The unit of observation is an equipment mention, but a single collateral description may list several machines. I identify the text belonging to each machine before extracting its characteristics. Numbered-item prefixes, semicolons, structured table rows, and new combinations of manufacturer, model year, and serial number help identify boundaries. Punctuation alone is not sufficient when it would divide the characteristics of one machine across several fragments.

I also join fragments that describe the same machine, including descriptions continued in attachments or split across columns. Shared headings are applied only to the machines they clearly govern. In a list containing several model years or serial numbers, I associate each with its own machine rather than attaching the first year or manufacturer to every item. Repeated descriptions with the same usable identifier, equipment type, and model year are counted once within an event.

Fragments containing only legal language, such as "together with all accessions," "products and proceeds," or "purchase money security interest," do not contribute equipment observations. The same applies to financing terms, payment schedules, dollar amounts, and fragments without sufficient machine-specific information. These checks are performed before imposing the equipment-type, model-year, and recovery-class restrictions.

Extracting Equipment Characteristics

For each potential equipment description that survives the preceding filters, I extract the characteristics needed to date the equipment, assign its tax treatment, compare similar machines, and track repeat appearances. The five principal fields are equipment type, manufacturer, model year, serial number or VIN, and any explicit indication that the equipment is new or used.

Equipment type. Equipment type is needed to assign a MACRS recovery class and to compare similar assets. I match each machine's description against an ordered dictionary covering construction and agricultural machinery, trucks and trailers, material-handling and forestry equipment, and office and information-technology equipment. More specific descriptions are matched before broader ones, so that, for example, "wheel loader" is not classified simply as "loader" and "gooseneck trailer" is not classified simply as "trailer." I also use manufacturer-specific model conventions when a description contains a model number but no equipment-type term. These matches require support from the identified manufacturer and model, rather than a generic numerical range that could describe several types of equipment.

Manufacturer. Manufacturer identity allows me to compare equipment produced by the same company and helps identify equipment type from manufacturer-specific model numbers. I match descriptions against a dictionary of manufacturers using patterns that accommodate common abbreviations and misspellings. The dictionary includes major construction and agricultural equipment manufacturers, truck manufacturers, and trailer and specialty-equipment producers. For example, the patterns recognize variants such as "MASSEY FERGERSON," "ARTIC CAT," and "NEWHOL." I require the manufacturer to refer to the particular machine, rather than another item in the same filing. In the baseline pre-TCJA specifications, I compare equipment within groups defined by manufacturer, equipment type, recovery class, and basis schedule.

Model year. Model year proxies for the original purchase vintage and, together with the filing year, determines asset age. I retain years supported by the description of the particular machine, including explicitly labeled model years and years paired with its manufacturer, model, or serial number. I exclude legal and filing dates, years embedded in model or serial numbers, and numbers describing capacities or quantities. Unresolved year ranges and conflicting machine years are left unclassified.

A valid 17-character VIN provides an additional source of model-year information. I first verify its check digit and then decode the character identifying the model year. Because that character repeats every 30 years, the filing-event year bounds the possible cycles but does not always identify one uniquely. I use the machine's own description and information tied to that exact VIN to resolve the cycle. If more than one feasible year remains, I do not assign a model year from the VIN alone.

Serial number and VIN. Serial numbers allow me to identify repeated appearances of the same machine. I extract identifiers following explicit labels such as S/N, SER, SERIAL, VIN, PIN, or S#, as well as identifiers clearly attached to a particular equipment description. A bare number sign is not sufficient because it may introduce a model number or other numbered text. I separately retain 17-character VINs that pass the NHTSA check-digit test. Masked, collective, or otherwise ambiguous identifiers are not used to match individual machines.

Tracking repeat appearances. Before constructing the analysis data, I replace filing, equipment, and debtor identifiers with keyed hashes. This permits the same entity or machine to be matched across records without retaining the original identifier. I define a machine using either a validated VIN or a usable serial number combined with its manufacturer or equipment type. To reduce false matches, the tracking sample retains only identifiers associated with a single model year, equipment type, recovery class, and depreciation method. Borrower-pair analyses also exclude filings with multiple debtors because a particular machine cannot be assigned unambiguously to one borrower. These identifiers allow me to count a machine only once within a filing event and to identify the first observed appearance of a machine–borrower pair. Such an appearance need not represent a verified sale because a UCC filing may also arise from refinancing or a later collateral event. A usable serial number is not required for inclusion in the full analysis sample.

New/used label. The main analysis classifies age-zero equipment as new and positive-age equipment as used. The initial parser also recognizes explicit NEW and USED labels in collateral descriptions, but the analysis does not require or use those labels to classify transactions. I exclude phrases in which these words do not describe equipment condition, such as "whether new or used," "now owned or hereafter acquired," "used in the operation of," and "as used herein." I also remove variants of the manufacturer name "New Holland" before searching for NEW. When both labels appear, USED takes priority, so that a description such as "1 USED NEW HOLLAND TRACTOR" receives the used label.

Quantity. I record a quantity when the description states one explicitly. Because a number preceding an equipment name may instead be part of its model designation, I do not use the quantity field as an analytical weight. The regressions count parsed equipment mentions rather than summing reported quantities.

Completing Missing Equipment Information

The initial text parser does not recover every item's manufacturer, equipment type, or model year. I supplement these fields using information tied to the identifiers already present in the collateral description. The purpose of this step is to increase coverage without replacing classifications that can be obtained directly from the filing text.

First, I use validated 17-character VINs to obtain manufacturer, vehicle type, body class, model year, and gross vehicle weight from the National Highway Traffic Safety Administration. To make the results reproducible, I save the NHTSA response for each VIN and reuse that fixed set of responses rather than querying the service each time the data are rebuilt. This information

is attached only to the machine associated with that exact VIN, rather than to other equipment listed in the same filing.

Second, when a manufacturer and model number are observed but the equipment type is missing, I use manufacturer-specific model conventions to identify the type. These rules are limited to model patterns associated with a known manufacturer so that a generic model number cannot generate a classification by itself.

Third, the identity of the secured party can provide information about how otherwise ambiguous equipment is used. I identify Farm Credit System lenders and use that information only as a fallback indicator of agricultural use when assigning the MACRS recovery class. Lender identity never overrides an equipment description that identifies the asset as a truck or as construction equipment.

Information recovered directly from the collateral description generally takes priority over these additional sources. The relevant exception is a generic "tractor" or "truck" classification that a validated VIN identifies unambiguously as an over-the-road tractor unit. I correct those classifications because road tractor units receive a different MACRS recovery period and fall outside the analysis sample.

Assigning MACRS Schedules

The purpose of the MACRS classification is to construct the ordinary depreciation schedule associated with the seller's original purchase cohort and predict the tax basis remaining when the equipment appears in a UCC transaction. A complete schedule requires both a recovery period and a depreciation method. I assign these after combining the available information on equipment type, manufacturer, model year, and secured party. Some equipment types map directly to an asset category, while others require assumptions about use. I retain those conditional assignments and record how they were made. Items for which these rules do not assign a five- or seven-year schedule are excluded.

Recovery period. I assign equipment directly when its observed type maps to a MACRS category. Trailers, for example, are five-year property. Over-the-road tractor units are three-year property and fall outside the paper's five- and seven-year analysis sample. I identify road tractor units from explicit collateral descriptions, manufacturers associated with road tractors, or a validated NHTSA truck-tractor body classification.

Some descriptions do not identify the relevant asset category precisely enough. A generic "semi" or "tractor-trailer" may refer either to the three-year tractor unit or to the five-year trailer, so I leave it unclassified. I also leave terminal tractors unclassified when their use is unknown. Generic aircraft and boats are excluded because their recovery periods depend on uses that are not reported in the collateral description.

Farm machinery. The recovery period for farm machinery depends on when it was originally placed in service. I use model year as a proxy for that purchase vintage. Farm machinery from vintages through 2017 is assigned to the seven-year class, except for qualifying machinery from 2009, when a temporary rule assigned it to the five-year class. Farm machinery from vintages beginning in 2018 is assigned to the five-year class under the maintained assumption that its original use began with the owner associated with its model-year cohort.

I identify farm use from equipment categories and, for otherwise ambiguous equipment, from manufacturer and secured-party information. For example, a generic loader produced by an agricultural manufacturer is treated as farm equipment, while a construction-equipment manufacturer supports a nonfarm assignment. Farm Credit System lenders provide a further fallback indicator. These are use proxies rather than observations of the borrower's actual activity. Secured-party identity never overrides an explicit truck or construction-equipment classification.

Depreciation method. I assign recovery periods and depreciation methods separately. Pre-2018 farm-use property follows the 150-percent declining-balance schedule for its assigned five- or seven-year recovery period under the half-year convention. Nonfarm property and farm machinery from vintages beginning in 2018 follow the corresponding 200-percent declining-balance schedule. Each observation retains its recovery period, depreciation method, and the rule used to assign them.

Explicit NEW or USED labels in a UCC description refer to the equipment's condition when it appears in that filing, not necessarily when it was originally purchased. I do not use those labels to determine the seller's original recovery period or depreciation method. For model-year vintages that predate MACRS and farm-use vintages from 1987–1988, the assigned schedules serve as recovery proxies; their predicted remaining basis is zero throughout the observation period.

Constructing Purchase Vintage and Tax Exposure

The purpose of this step is to assign each machine the tax treatment that would have applied when it was originally purchased and to calculate the resulting difference in remaining basis when it later appears in the UCC data. I treat model year as the original purchase year and match it to the federal bonus-depreciation rate in effect for that year. This produces a measure of statutory eligibility: it records whether the machine would have qualified for bonus depreciation, not whether its owner actually claimed the deduction. I do not observe bonus elections, Section 179 deductions, prior owners, or taxable transfers that may have reset the asset's basis.

Let $B_s(k)$ denote the fraction of ordinary basis remaining after k annual deductions under depreciation schedule s. If an asset's purchase vintage received bonus rate θ_v, bonus immediately deducted that share of its basis and applied the ordinary schedule to the remainder. The predicted difference between the basis of a nonbonus asset and an otherwise identical bonus asset is

$$\theta_v B_s(k).$$

I refer to this difference as the bonus-induced basis gap.

The descriptive tables evaluate ordinary basis after the filing-year deduction, setting $k = a + 1$ for an asset of age a. Under this convention, ordinary basis is exhausted at age five for five-year property and age seven for seven-year property. Positive future basis in the tables means that more than one percent of original cost remains under the ordinary schedule; it does not incorporate bonus depreciation or observed tax elections. Future deduction years count the subsequent annual deductions under that schedule.

For the TCJA comparison, I fix each cohort's exposure at the end of 2017. Its bonus-induced basis gap is $\theta_v B_s(2018 - v)$, which evaluates basis after the 2017 deduction, or equivalently at the

beginning of 2018. I distinguish gaps above one percent of original cost from smaller or exhausted gaps and also use the continuous gap measured in units of ten percentage points. Holding this exposure fixed prevents a cohort from changing groups as its remaining basis is subsequently deducted. A statutory tax wedge can be obtained by multiplying a basis gap by a common tax rate. Because the UCC data contain both corporations and pass-through businesses and do not identify each seller's tax status, such a rate is a benchmark rather than an estimate of actual liability.

The UCC data report model year but not the exact date on which a machine was placed in service. I assign one bonus rate to each calendar year using the rate that applied during most of that year. Under this convention, 2001 is coded as having no bonus, while 2003, 2010, and 2017 are coded at 50 percent despite changes in the statutory rate during those years.

The main pre-TCJA specifications use an indicator equal to one for model-year vintages eligible for any positive bonus depreciation. Under the annual convention, 2002 receives a 30-percent rate; 2003–2004, 2008–2010, and 2012–2017 receive 50 percent; and 2011 receives 100 percent. The 2005–2007 lapse vintages receive no bonus. These assignments describe the original purchase cohort, not the tax treatment of the used purchase recorded in the filing.

Constructing the Analysis Sample

The purpose of the final sample restrictions is to retain equipment mentions for which I can identify the purchase vintage, calculate asset age, and assign the ordinary depreciation schedule needed to measure remaining basis. I retain filing events from 2001 through 2019 whose collateral descriptions pass the sequence-number check described above. Each retained equipment mention must have an identified equipment type and model year, an age between zero and 50, and an assigned five- or seven-year MACRS recovery period and depreciation method.

The unit of observation is an equipment mention in a filing event. A filing that lists several machines contributes one observation for each identified machine, and the same machine may contribute multiple observations if it appears in multiple filing events. Stated quantities are not expanded into additional observations. The alternative counting specifications use the hashed identifiers to remove repeat appearances.

I exclude confirmed equipment leases and administrative events that do not record equipment entering a financing relationship. The latter include verified administrative amendments and current releases or deletions of the named collateral. A general reference to leasing or to the lender's rights upon default is not sufficient for exclusion, and a release applying to one machine does not exclude unrelated equipment in the same filing.

I also exclude identified inventory and equipment-register filings, including accounting or appraisal rosters, title or lien registers, and internal asset-stock tables. An ordinary purchase invoice or a list of several machines is not excluded merely because it is lengthy or tabular. Ambiguous rosters remain in the sample, and there is no filing-size cutoff.

The construction yields 398,111 dated equipment mentions before applying the recovery-class restrictions. Of these, 388,809 receive an assigned five- or seven-year schedule. Removing 9,402 confirmed lease or administrative records leaves 379,407 observations, and excluding 6,361 identified inventory records leaves 373,046. Using age zero to classify new equipment and positive age to classify used equipment, the sample contains 77,811 new and 295,235 used equipment mentions. Table C.1 reports these stages. Tables C.2 and C.3 describe the final sample and its largest

equipment categories.

The baseline pre-TCJA regressions use the 185,770 positive-age observations through 2017 with an identified manufacturer. The TCJA regressions use 286,478 positive-age observations from 2001–2019 belonging to model-year vintages through 2016, without requiring an identified manufacturer.

Table C.1: UCC Sample Construction

Step	Equipment mentions
Equipment candidates after parsing and repairs	398,111
Assigned to five- or seven-year property	388,809
After excluding confirmed leases and administrative events	379,407
After excluding identified inventory filings	373,046

Notes: The final sample covers filing years 2001–2019. Counts are equipment mentions, not distinct machines, and reported quantities are not expanded into additional observations. The second row requires an identified equipment type and model year, age between zero and 50, and an assigned five- or seven-year MACRS recovery period and depreciation method. Ambiguous rosters are retained when they do not satisfy the inventory exclusion.

Table C.2: Summary Statistics for the UCC Analysis Sample

Equipment	Filing years	Items	Mean age	Median age	Positive future basis (%)	Future years
New equipment	2001–2017	66,211	0.0	0	100.0	5.6
New equipment	2018–2019	11,600	0.0	0	100.0	5.1
Used equipment	2001–2017	249,090	10.1	7	35.4	1.1
Used equipment	2018–2019	46,145	9.1	6	43.6	1.4

Notes: New equipment denotes age zero; used equipment denotes positive age. Age is filing year less model year. Positive future basis is the percentage with more than one percent of original cost remaining under the assigned ordinary MACRS schedule after the filing-year deduction, without bonus depreciation. Future years is the mean number of subsequent ordinary deduction years, censored at zero. These measures describe the assigned schedule rather than an observed owner's actual tax basis.

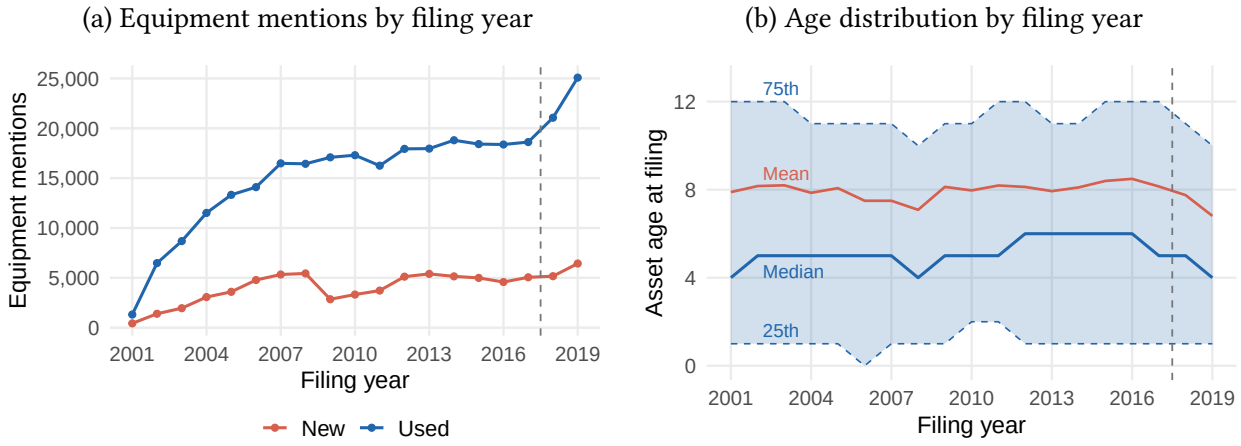
Table C.3: Largest Equipment Categories in the UCC Analysis Sample

Equipment type	MACRS	Method	Items	New (%)	Used age	Basis (%)
Trailer	5	200% DB	103,472	28.7	8.2	55.3
Tractor	7	150% DB	61,577	9.0	15.0	37.2
Truck	5	200% DB	32,703	21.4	9.8	42.6
Mower / Cutter	7	150% DB	13,859	36.0	5.6	82.4
Forklift	5	200% DB	11,149	23.7	8.8	42.6
Excavator	5	200% DB	10,728	15.4	7.7	45.8
Dozer	5	200% DB	10,060	3.6	14.4	16.8
Backhoe	5	200% DB	9,978	9.9	10.8	29.9
Skid Steer	5	200% DB	8,923	14.5	6.6	48.2
Stock Trailer	5	150% DB	7,934	14.0	10.7	35.2
Loader	7	150% DB	7,159	23.8	8.2	66.4
Baler	7	150% DB	4,907	14.4	8.8	54.2
Gooseneck Trailer	5	200% DB	4,272	19.2	10.0	43.4
Welder	7	200% DB	3,868	18.9	12.4	56.3
Wheel Loader	5	200% DB	3,824	18.2	8.3	49.3

Notes: Categories are equipment-type × MACRS-class × depreciation-method cells ranked by the number of equipment mentions. New denotes age zero, and mean used age excludes age-zero observations. Positive basis is the percentage of all mentions in the category with more than one percent of original cost remaining under ordinary depreciation after the filing-year deduction, without bonus depreciation. DB denotes declining balance. Recovery classes and methods are assigned as described in the text.

Figure C.1 documents the sample's coverage over time and the ages of the equipment observed in each filing year. Panel (a) reports equipment mentions separately for age-zero and positive-age equipment. The bulk files do not allow me to distinguish changes in underlying filing activity from changes in data coverage. The regressions include filing-year fixed effects and do not use aggregate changes in the number of records for identification. Panel (b) reports mean and median age together with the interquartile range.

Figure C.1: UCC Sample Coverage and Asset Ages



Notes: Panel (a) reports equipment mentions by filing year. "New" denotes age zero and "used" denotes positive age; both are model-year proxies rather than observed purchase status. Panel (b) reports mean and median age together with the interquartile range. Asset age is filing year minus model year. The vertical line marks 2018, the first filing year following enactment of the TCJA.

D Additional Results and Robustness

D.1 Evidence from Bonus Depreciation

Table D.1: Bonus Depreciation: Transaction Measurement Checks

Records counted	Coefficient	Std. error	Change (%)	Items	Cells
All records	-0.193**	0.078	-17.5	185,770	62,679
Initial filings	-0.184**	0.080	-16.8	167,162	59,615
Records with usable identifiers	-0.191**	0.080	-17.4	101,280	42,494
First observed asset and borrower	-0.181**	0.081	-16.5	94,879	41,545
Exclude age one	-0.195**	0.083	-17.7	166,205	57,697

Notes: PPML estimates use positive-age equipment filings from 2001–2017, with equipment-group × exact-age and filing-year fixed effects. Groups combine manufacturer, equipment type, recovery class, and depreciation schedule. First asset–borrower counts retain the earliest linked filing in the available history, including age-zero records. Usable identifiers require consistent machine and borrower identification and linked dates. Changes are $100[\exp(\widehat{\beta}) - 1]$. Standard errors are clustered by purchase vintage; tests use t_{G-1} (65–66 clusters). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table D.2: Bonus Depreciation and Purchase-Cohort Conditions

Specification	Coefficient	Standard error	Cells	Bonus / nonbonus vintages
Model-year trends				
Baseline	−0.193**	(0.078)	62,679	12 / 54
Quadratic trend in model year	−0.218***	(0.081)	62,679	12 / 54
Quadratic and cubic trends in model year	−0.219***	(0.082)	62,679	12 / 54
Policy comparisons				
Only purchase vintages after 2001	−0.279***	(0.069)	29,071	12 / 3
Exclude the 2005–07 lapse vintages	0.003	(0.080)	53,380	12 / 51
Exclude the 2008–10 bonus vintages	−0.196***	(0.071)	57,005	9 / 54

Notes: Each row estimates PPML with equipment-group × exact-age and filing-year fixed effects, using positive transaction cells in 2001–2017. Equipment groups combine manufacturer, type, recovery class, and depreciation schedule. The first three rows use identical observations. Standard errors cluster by original purchase vintage; tests use t_{G-1}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table D.3: 50-Percent Bonus Versus No Bonus

	(1)	(2)
50-percent bonus vintage	-0.249***	-0.200***
	(0.077)	(0.046)
Log age-zero transaction count		0.186***
		(0.027)
Equipment group × age fixed effects	Yes	Yes
Filing-year fixed effects	Yes	Yes
Initial cohort-size control	No	Yes
Estimation cells	18,527	18,527
Equipment items	82,677	82,677
Equipment groups	1,029	1,029
Purchase-vintage clusters	14	14
Zero-bonus vintages	4	4

Notes: PPML compares 50-percent and zero-bonus vintages, excluding 2002 and 2011, using filing years 2001–2017. Both columns use identical positive-count cells with an observed age-zero count for the same equipment group and vintage. Groups combine manufacturer, equipment type, recovery class, and depreciation schedule. Column (2) adds the log age-zero count. Regressions include equipment-group × exact-age and filing-year fixed effects. Standard errors are clustered by purchase vintage; tests use t_{G-1} (14 clusters, including four zero-bonus vintages). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table D.4: Bonus Depreciation and Used-Equipment Transactions Around the 2005–07 Lapse

	Baseline	Supported sample	Cohort control	Completed cells
	(1)	(2)	(3)	(4)
Bonus relative to 2005–07 lapse	−0.249**	−0.259**	−0.188**	−0.214**
	(0.091)	(0.092)	(0.058)	(0.066)
Exact age-zero cohort support		X	X	X
Log age-zero cohort-size control			X	X
Supported zero cells completed				X
Equipment group × age FE	X	X	X	X
Filing-year FE	X	X	X	X
Equipment mentions	49,538	44,058	44,058	43,498
Cells	13,053	9,631	9,631	14,799
Equipment groups	849	645	645	494
Vintage clusters	8	8	8	8

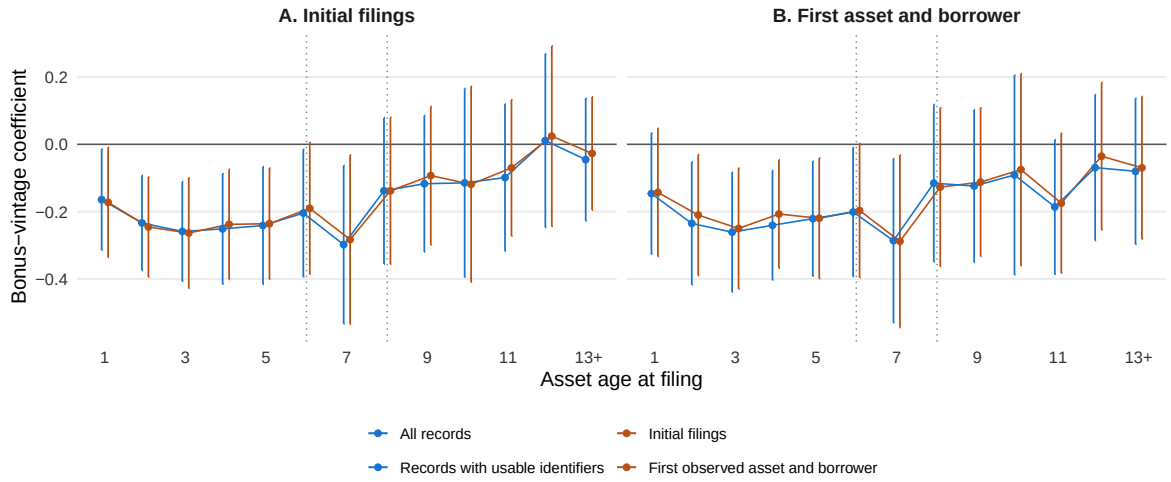
Notes: Each column reports a PPML regression for purchase vintages 2003–10 and asset ages 1–7. The coefficient compares the surrounding 2003–04 and 2008–10 bonus vintages with the 2005–07 lapse. Columns (2)–(4) require the equipment-group purchase vintage to appear at age zero; columns (3)–(4) control for its log age-zero count. Column (4) completes supported cells with zero transactions. Standard errors in parentheses cluster by purchase vintage. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ using t_{G-1}.

Table D.5: The Bonus-Vintage Transaction Gap by Asset Age

	Estimate	Standard error	p-value
Age 1	-0.164^{**}	0.076	0.034
Age 2	-0.233^{***}	0.071	0.002
Age 3	-0.259^{***}	0.074	0.001
Age 4	-0.251^{***}	0.083	0.004
Age 5	-0.241^{***}	0.088	0.008
Age 6	-0.204^{**}	0.095	0.036
Age 7	-0.298^{**}	0.118	0.014
Age 8	−0.138	0.109	0.211
Age 9	−0.117	0.102	0.256
Age 10	−0.115	0.141	0.420
Age 11	−0.099	0.110	0.373
Age 12	0.011	0.130	0.932
Age 13+	−0.045	0.092	0.621
Ages 2–7 jointly equal zero	2.463	—	0.033
Ages 8+ jointly equal zero	1.258	—	0.289
Age coefficients 2–7 equal one another	1.245	—	0.299
Equal-weight mean, ages 2–7	-0.248	0.079	0.003
Equal-weight mean, ages 8+	-0.084	0.101	0.408
Equal-weight mean ages 2–7 minus ages 8+	-0.164	0.084	0.055

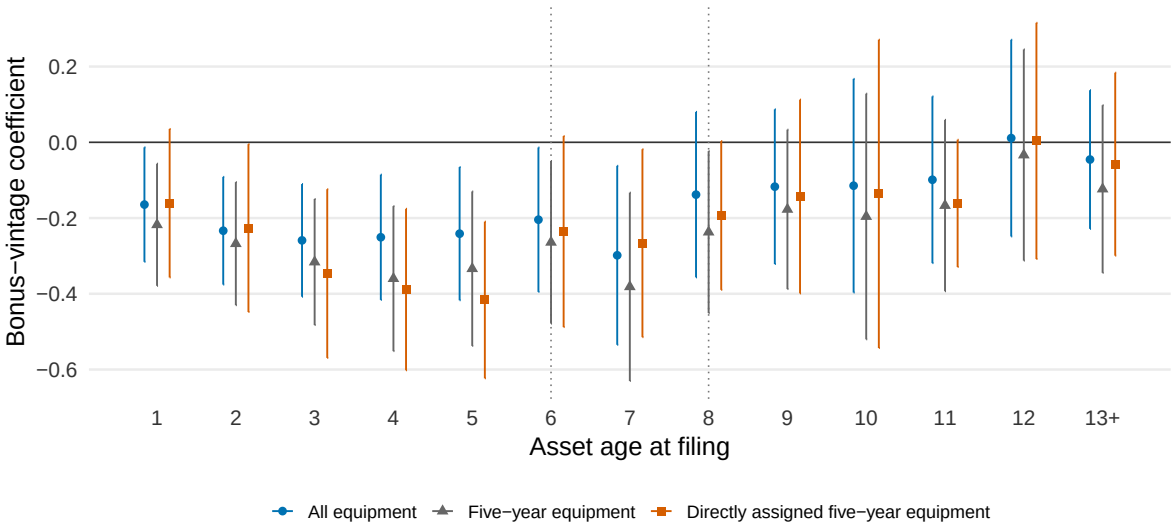
Notes: Coefficients correspond to the main age-profile figure. PPML includes equipment group × raw-age and filing-year fixed effects; age 13+ pools policy interactions only. The three joint tests report cluster-robust Wald F statistics in the estimate column. Equal-weight contrasts use six coefficients at ages 2–7 and six at ages 8–12 plus 13+. Standard errors cluster by purchase vintage; confidence intervals and contrast tests use t_{G-1} inference.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ using t_{G-1}.

Figure D.1: Bonus Age Profiles Under Alternative Transaction Counts



Notes: PPML estimates use filings from 2001–2017, with equipment-group-by-exact-age and filing-year fixed effects. Both specifications use identical cells within each panel. First asset–borrower appearances are defined using the available filing history. Ages 13 and older share one bonus coefficient. Bars report 95% confidence intervals using purchase-vintage clustering and t_{G-1} inference.

Figure D.2: Bonus Age Profiles and Recovery-Class Assignments



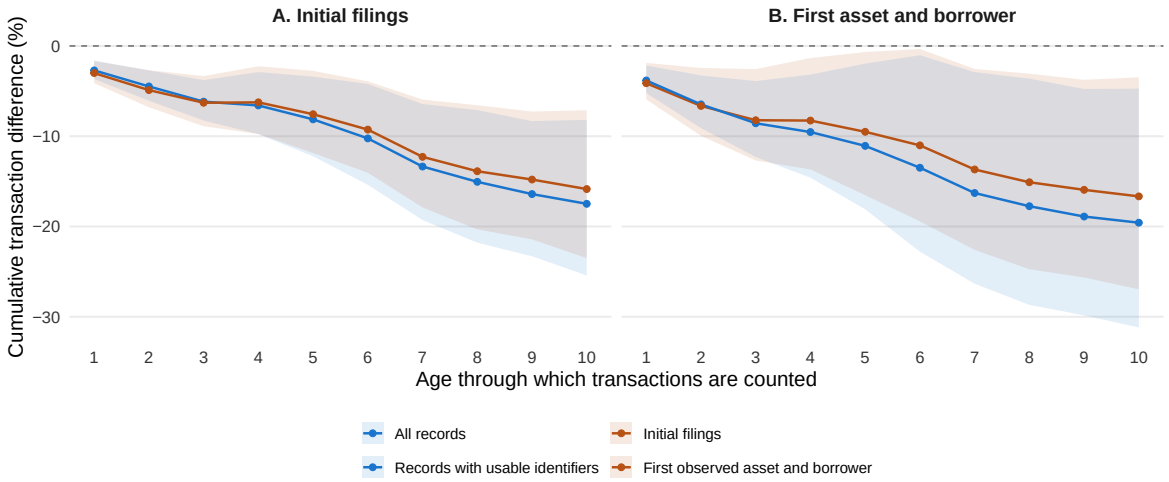
Notes: Direct five-year assignments use asset-specific rules for trailers, trucks and vans, and computer and office equipment. PPML estimates use filings from 2001–2017, with equipment-group-by-exact-age and filing-year fixed effects. Ages 13 and older share one bonus coefficient. Dotted lines mark ages six and eight. Bars report 95% confidence intervals using purchase-vintage clustering and t_{G-1} inference.

Table D.6: Bonus-Vintage Age Profiles and Recovery-Class Assignment

	All equipment	Five-year equipment	Directly assigned five-year equipment
Mean, ages 2–7	-0.248	-0.321	-0.313
	(0.079)	(0.085)	(0.085)
p-value	0.003	< 0.001	< 0.001
Mean, ages 8+	-0.084	-0.156	-0.115
	(0.101)	(0.105)	(0.110)
p-value	0.408	0.145	0.303
Ages 2–7 minus ages 8+	-0.164	-0.165	-0.199
	(0.084)	(0.097)	(0.102)
p-value	0.055	0.095	0.055
Equipment items	185,770	113,659	64,121
Equipment groups	3,088	2,242	1,488
Count cells	62,679	43,641	24,237
Purchase-vintage clusters	66	64	62

Notes: Columns use all equipment, five-year equipment, and equipment assigned five-year status by asset-specific rules: trailers, trucks and vans, and computer and office equipment. PPML regressions use filing years 2001–2017 and equipment-group × exact-age and filing-year fixed effects. Means weight age coefficients equally; ages 13 and older share one coefficient. Standard errors cluster by original purchase vintage; p-values use t_{G-1}.

Figure D.3: Cumulative Bonus Transaction Differences Under Alternative Counts



Notes: PPML follows purchase vintages 2001–2007 through ages 1–10, using identical cohorts and cells within each panel, including supported zeros. Regressions include equipment-group-by-age and filing-year fixed effects, without initial cohort-size controls. Within panels, curves share weights and an age-ten no-bonus denominator from the model with that control. Bands report conditional 95% intervals using seven purchase-vintage clusters and t_6 inference, holding weights and fitted fixed effects fixed.

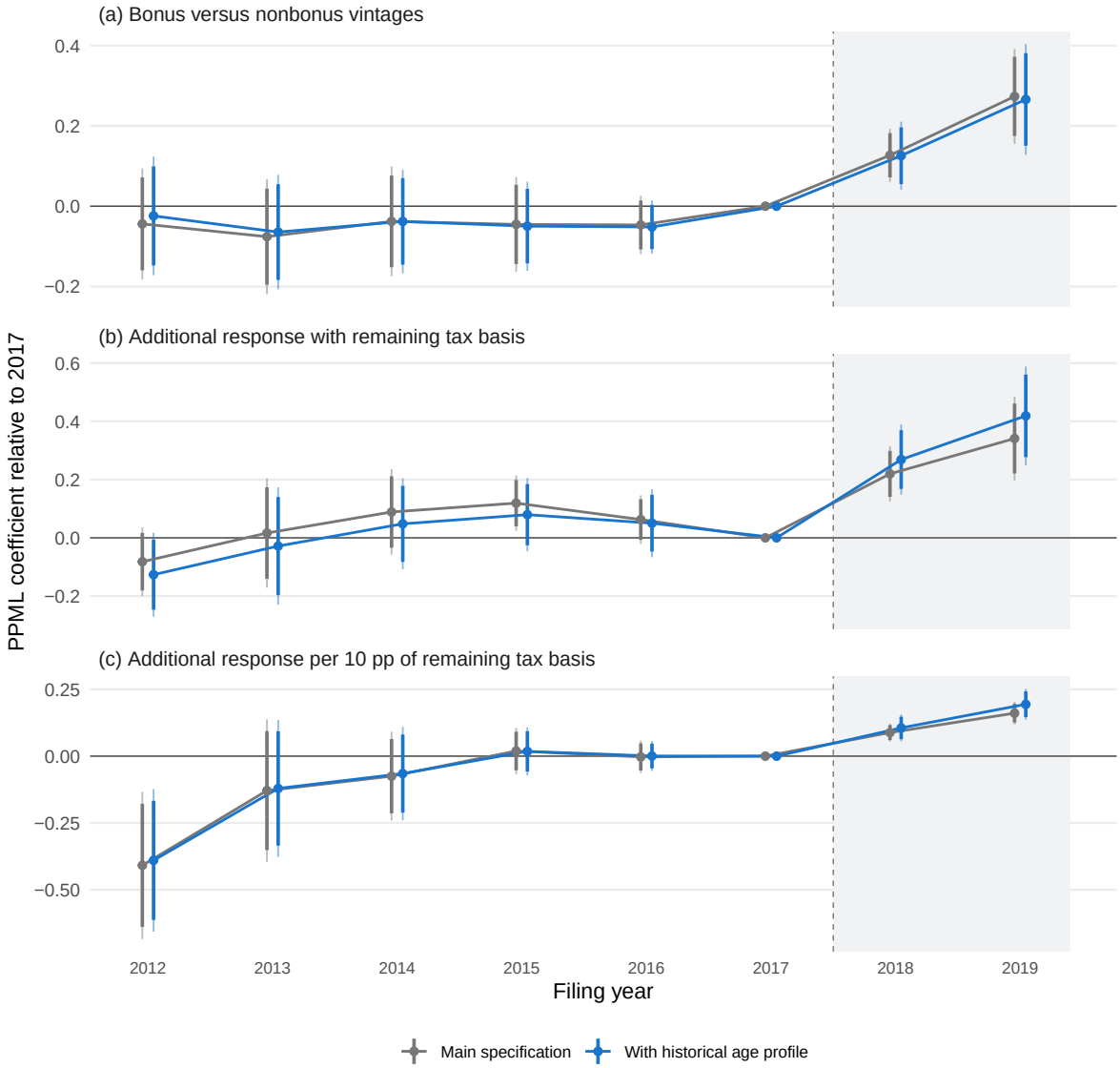
D.2 Evidence from the Tax Cuts and Jobs Act

Table D.7: TCJA Estimates with Bonus-Specific Aging Estimated Before the Reform

	Pooled		Positive gap		Continuous gap	
	Main	Age-adjusted	Main	Age-adjusted	Main	Age-adjusted
Bonus vintage × Post-TCJA	0.170***	0.143**	0.091	−0.008	0.084*	0.020
	(0.043)	(0.063)	(0.056)	(0.097)	(0.042)	(0.075)
Post × Positive basis gap			0.242***	0.349***		
			(0.071)	(0.099)		
Post × Basis gap (10 pp)					0.141***	0.176***
					(0.022)	(0.032)
Total post response: positive-gap vintages			0.333***	0.341***		
			(0.049)	(0.052)		
Equipment-group fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Equipment type × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Exact-age fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Fixed pre-reform bonus-age profile	No	Yes	No	Yes	No	Yes
Equipment items	286,478	286,478	286,478	286,478	286,478	286,478
Estimation cells	24,354	24,354	24,354	24,354	24,354	24,354
Purchase-vintage clusters	66	66	66	66	66	66

Notes: PPML estimates of transaction counts, 2001–2019; Post denotes 2018–2019. The second column in each pair adds a bonus-specific age profile estimated through 2016 and held fixed. Standard errors in parentheses cluster by purchase vintage; adjusted standard errors use 399 purchase-vintage multiplier draws reestimating both stages. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure D.4: TCJA Estimates Accounting for Historical Aging of Bonus Vintages



Notes: Annual PPML estimates corresponding to columns (1)–(3) of Table 2, with and without a bonus-specific age profile estimated through 2016 and held fixed. Estimates are relative to 2017. Ages 13 and above share the estimate for ages 13–14. Thick and thin bars show 90% and 95% confidence intervals. Original standard errors cluster by purchase vintage; adjusted errors use 399 purchase-vintage multiplier draws reestimating both stages. Intervals use t_{65} critical values.

Table D.8: TCJA: Alternative Transaction Counts

	All records	Initial filings	Serial number or VIN	First asset–borrower appearance
A. Pooled comparison				
Bonus vintage × Post-TCJA	0.170***	0.148***	0.012	0.028
	(0.043)	(0.041)	(0.053)	(0.052)
B. Positive basis gap				
Bonus vintage × Post-TCJA	0.091	0.069	0.074	0.088
	(0.056)	(0.054)	(0.078)	(0.079)
Post × Positive basis gap	0.242***	0.254***	0.016	0.013
	(0.071)	(0.064)	(0.074)	(0.076)
Total post response: positive-gap vintages	0.333***	0.323***	0.090	0.101*
	(0.049)	(0.044)	(0.057)	(0.056)
C. Continuous basis gap				
Bonus vintage × Post-TCJA	0.084*	0.069	0.076	0.087
	(0.042)	(0.042)	(0.062)	(0.062)
Post × Basis gap (10 pp)	0.141***	0.137***	0.013	0.011
	(0.022)	(0.022)	(0.027)	(0.027)
Equipment-group fixed effects	Yes	Yes	Yes	Yes
Equipment type × year fixed effects	Yes	Yes	Yes	Yes
Asset-age fixed effects	Yes	Yes	Yes	Yes
Initial cohort-size control	No	No	No	No
Equipment items	286,478	255,747	152,292	141,916
Estimation cells	24,354	23,470	17,493	17,209
Purchase-vintage clusters	66	66	65	65

Notes: PPML, 2001–2019; post: 2018–2019. Model-year vintages end in 2016. Each panel includes the pre-reform bonus indicator and, where applicable, the basis-gap variable. The basis gap is measured after the 2017 deduction; positive gap exceeds 1% of cost. Serial-number/VIN records require consistent machine and borrower links. First appearances use that same subset and its available history, including age-zero records. Standard errors clustered by purchase vintage are in parentheses; tests use t_{G-1}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table D.9: TCJA: Excluding Age-One Equipment

	Pooled		Binary gap		Continuous gap	
	All ages	Ages 2+	All ages	Ages 2+	All ages	Ages 2+
Bonus vintage (pre-TCJA)	-0.317^{***}	-0.326^{***}	-0.302^{***}	-0.310^{***}	-0.305^{***}	-0.311^{***}
	(0.083)	(0.089)	(0.080)	(0.085)	(0.081)	(0.087)
Bonus vintage × Post-TCJA	0.170^{***}	0.176^{***}	0.091	0.096^{*}	0.084^{*}	0.088^{**}
	(0.043)	(0.044)	(0.056)	(0.055)	(0.042)	(0.042)
Post × Positive basis gap			0.242^{***}	0.268^{***}		
			(0.071)	(0.072)		
Post × Basis gap (10 pp)					0.141^{***}	0.173^{***}
					(0.022)	(0.038)
Pre-reform basis-gap control	No	No	Yes	Yes	Yes	Yes
Equipment-group fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Equipment type × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Asset-age fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Initial cohort-size control	No	No	No	No	No	No
Equipment items	286,478	260,301	286,478	260,301	286,478	260,301
Estimation cells	24,354	23,340	24,354	23,340	24,354	23,340
Purchase-vintage clusters	66	66	66	66	66	66

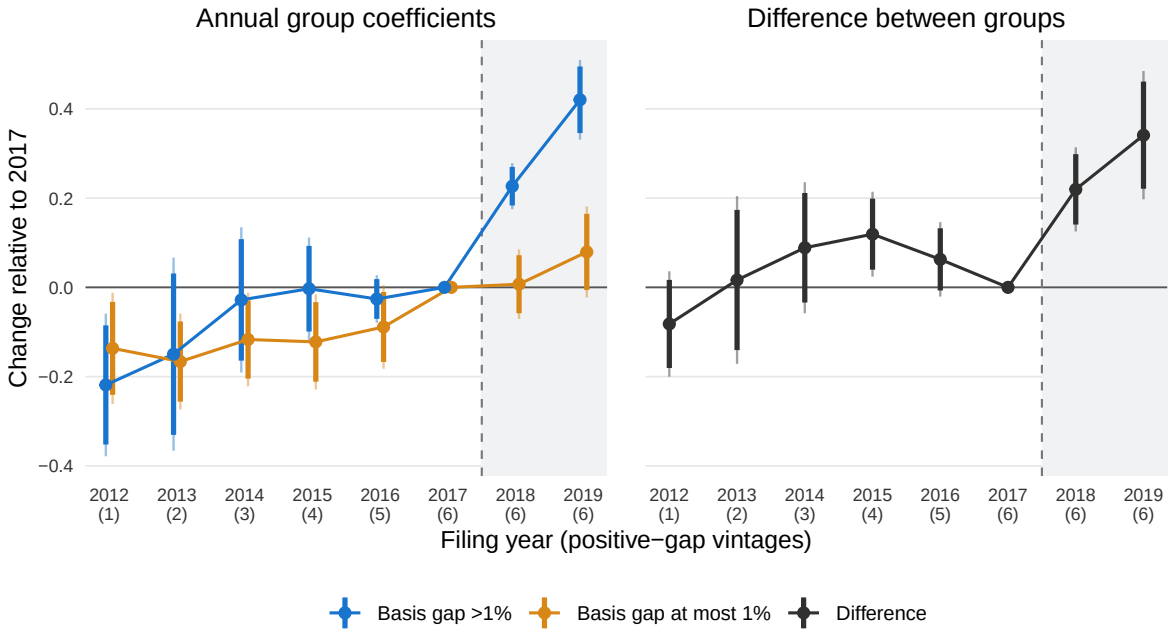
Notes: PPML, 2001–2019; post: 2018–2019. Each specification compares ages 1–50 with ages 2–50, using model-year vintages through 2016. The basis gap is the bonus-induced reduction in original basis after the 2017 deduction; positive gap exceeds 1% of cost. Standard errors clustered by purchase vintage are in parentheses; tests use t_{65}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table D.10: TCJA Responses Under Alternative Recovery Schedules

	Pooled	Positive gap	Gap (10 pp)
	(1)	(2)	(3)
Maintained assignments	0.170***	0.242***	0.141***
	(0.043)	(0.071)	(0.022)
Five-year nonfarm schedule	0.170***	0.235***	0.154***
	(0.043)	(0.061)	(0.021)
Seven-year nonfarm schedule	0.170***	0.312***	0.155***
	(0.043)	(0.075)	(0.028)
Farm schedule	0.170***	0.312***	0.132***
	(0.043)	(0.075)	(0.024)
Equipment items	286,478	286,478	286,478
Estimation cells	24,354	24,354	24,354
Vintage clusters	66	66	66
Equipment group FE	X	X	X
Equipment type × year FE	X	X	X
Asset-age FE	X	X	X

Notes: PPML, 2001–2019; post: 2018–2019. Observations and fixed effects are unchanged. Alternative exposure assignments apply to use-dependent construction, logging, and multiple-use equipment. Nonfarm schedules use 200-percent declining balance; the farm schedule uses seven-year 150-percent declining balance, except five-year in 2009. Positive gap exceeds 1% of original cost at end-2017. Columns report post interactions for bonus status, positive gap, and gap per 10 percentage points. Standard errors cluster by purchase vintage; stars use t_{65}: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure D.5: TCJA: Annual Transactions by Original Basis Gap

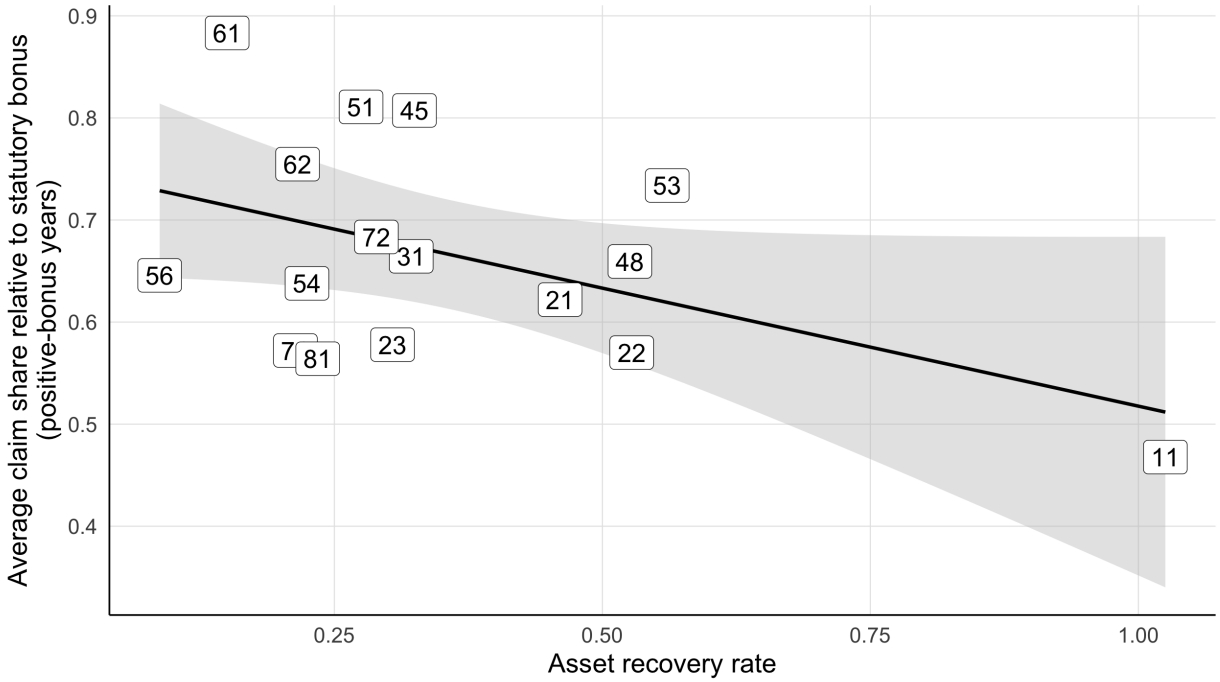


Notes: Annual PPML estimates use filings from 2001–2019 and model-year vintages through 2016, with equipment-group, equipment-type-by-year, and exact-age fixed effects. The basis gap is measured after the 2017 deduction; positive gap exceeds 1% of original cost. Estimates are relative to 2017. Parentheses show contributing positive-gap vintages. Thick and thin bars report 90% and 95% confidence intervals, respectively, using purchase-vintage clustering and t_{65} inference.

E Statistics of Income Evidence on Bonus Take-Up

Take-up of bonus depreciation is not automatic. Kitchen and Knittel (2011) report that roughly half of all bonus is claimed. In principle, if the mechanism in the paper holds, then industries with more liquid secondary markets for equipment should see lower take-up. That's because the return to reselling is higher when secondary markets are deeper. I test that by merging industry-level reports of bonus depreciation take-up from the Statistics of Income using Form 4562 with industry-level measures of investment irreversibility from Kermani and Ma (2022). Figure E.1 reports the correlation between the two variables. As expected, the relationship is negative; industries with deeper secondary markets for used capital and hence higher reversibility exhibit lower take-up of bonus depreciation.

Figure E.1: Bonus Depreciation Claims and Asset Tradeability



Notes: Each point is a two-digit NAICS industry. The vertical axis is the industry's average special-allowance claim share relative to the statutory bonus rate during years with positive bonus depreciation. The horizontal axis is the PPE recovery rate from Kermani and Ma (2022). The fitted line is unweighted.

Table E.1 reports panel regressions investigating the interaction between exposure to bonus and the ease with which industries can resell their capital. Column (1) implies that increasing bonus from zero to 50 percent is associated with a 1.35-percentage-point smaller increase in take-up in an industry with a ten-percentage-point higher asset recovery rate. The estimate is similar after adding controls and remains negative when the sample is restricted to positive-bonus years. Excluding agriculture also yields a similar estimate, although it is less precise. These results are consistent with bonus being less valuable when firms can recover their remaining deductions through an early sale.

Table E.1: Bonus Depreciation Claims and Asset Tradeability

	Full Sample		Bonus Years	Excl. Agriculture
	(1)	(2)	(3)	(4)
$\theta_t \times \ell_i$	-0.269*	-0.280*	-0.329**	-0.285
	(0.074)	(0.082)	(0.172)	(0.209)
Exact wild-cluster p-value	0.073	0.067	0.044	0.216
SOI claiming-capacity controls		X		
Industry linear trends	X	X	X	X
Year FE	X	X	X	X
Industry FE	X	X	X	X
Observations	316	298	236	295
Industry clusters	16	16	16	15

Notes: The dependent variable is bonus depreciation take-up. The reported coefficient is on the statutory bonus rate, θ_t, interacted with the industry asset recovery rate, ℓ_i. All specifications include industry and year fixed effects and industry-specific linear trends. Column (2) adds controls for losses relative to sales, the share of profitable tax returns, sales relative to capital, and sales growth. Column (3) includes only years with positive bonus; column (4) excludes agriculture. The sample covers 2000–2021, excluding 2008. Standard errors clustered by industry are in parentheses. Stars follow the wild-cluster bootstrap p-values reported in the table.

F Evidence from Compustat

I use firm-level data from Compustat to measure capital expenditures and cash proceeds from sales of property, plant, and equipment. In terms of the model, capital expenditures (item `capx`) are the closest empirical counterpart to new investment I, while the proceeds from used capital sales (item `sppe`) correspond to qS, the used price multiplied by the quantity of capital sold. Compustat therefore allows me to examine the joint behavior of capital formation and the value of capital sales. These data have been used extensively to study reallocation (Eisfeldt and Rampini 2006; Eisfeldt and Shi 2018), but they have a major shortcoming because I cannot separately recover used capital prices and quantities.

I construct two separate samples to study the responses of investment and used capital sales to bonus, as well as to the Tax Cuts and Jobs Act. In both samples, I set missing or negative capital sale proceeds to zero.

First, I study bonus depreciation using U.S.-domiciled nonfinancial firms with positive sales, assets, and net PPE and nonnegative capital expenditures. I match each firm to its four-digit industry exposure to bonus depreciation as constructed by Zwick and Mahon (2017). The sample spans 1995–2016 and contains 102,466 firm-years from 11,674 firms. I winsorize capital sale proceeds at the 3rd and 97th percentiles but do not winsorize capital expenditures in the baseline.

Second, I study the Tax Cuts and Jobs Act using a comparison between American and foreign

firms. I retain the same basic firm restrictions and additionally require at least four usable firm-years and nonzero pre-reform variation in sppe. I also require each four-digit NAICS industry to contain at least two foreign firm-years. Within the U.S. and foreign groups, I winsorize capital sales and investment at the 97th percentile, preserving zeros, and sales, assets, and net PPE at the 3rd and 97th percentiles. The sample spans 2011–2019 and contains 17,480 firm-years from 1,620 U.S. and 660 foreign firms.

F.1 Bonus Depreciation

This section studies the behavior of firm-level investment and capital sales across bonus depreciation episodes.

I begin by studying bonus depreciation for new capital, which was repeatedly varied by policymakers from 2001–2016. While the model shows that bonus can raise investment, the response of capital sales is ambiguous. To identify the effect of bonus depreciation on each outcome, I follow the same strategy as Zwick and Mahon (2017). Firms vary in their production technologies; some firms rely on long-lived equipment, while others rely to a greater extent on shorter-lived equipment. That creates predetermined variation in exposure to bonus depreciation because the net present value of a dollar of ordinary depreciation deductions is lower for firms with longer-lived equipment, making bonus more valuable to those firms (House and Shapiro 2008). Zwick and Mahon (2017) provide a measure of this exposure at the four-digit NAICS industry level using pre-bonus investment expenditures to form a composite net present value of depreciation deductions z_N. Applying bonus rate θ_t, that becomes

$$\widetilde{z}_{N,t} = \theta_t + (1 - \theta_t) z_N.$$

Consequently, I identify the effect of bonus by comparing changes within firms from more- and less-exposed industries while absorbing permanent firm differences and aggregate year effects. The identifying assumption is that, absent changes in bonus depreciation, capital expenditures and capital sales would not have changed differentially across industries according to this predetermined exposure.

For the nonnegative outcomes, I estimate

$$E\left[Y_{i,t} \mid \widetilde{z}_{N(i),t}, \alpha_i, \gamma_t\right] = \exp\left(\beta \widetilde{z}_{N(i),t} + \alpha_i + \gamma_t\right), \tag{F.1}$$

where α_i and γ_t denote firm and year fixed effects. I estimate (F.1) using PPML and estimate the corresponding extensive-margin specification using a linear probability model. Standard errors are clustered by four-digit NAICS industry, the level at which exposure varies.

Table F.1 reports the results for four different outcomes. Columns 1, 3, and 4 decompose capital sale proceeds into the unconditional amount, an indicator for reporting positive proceeds, and proceeds conditional on positive sales. The fifth column estimates the corresponding investment response, which Zwick and Mahon (2017) already show is significantly positive. The results indicate that capital sales and capital investment move in opposite directions. Whereas a one-percentage-point increase in exposure to bonus $\widetilde{z}$ raises investment by approximately 3.5 percent, it reduces unconditional capital sale proceeds by approximately 11 percent. Moreover, this difference seems driven by the extensive margin because the probability that a firm reports

positive proceeds falls by 0.64 percentage points, whereas it is not possible to distinguish the intensive-margin used capital sale response from zero, although the sign is consistent.[24]

Table F.1: Bonus Depreciation, Capital Sale Proceeds, and Investment

	Capital Sale Proceeds (sppe)				Investment (capx)	
	Baseline	Quadratic Trends	Any Proceeds	Positive Proceeds	Baseline	Quadratic Trends
	(1)	(2)	(3)	(4)	(5)	(6)
$\tilde{z}$	-11.858***	-4.809**	-0.641***	-2.470	3.402**	1.082**
	(2.286)	(2.214)	(0.242)	(1.580)	(1.684)	(0.477)
Estimator	PPML	PPML	LPM	PPML	PPML	PPML
Sample	All	All	All	Sellers	All	All
Firm FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Industry trends		X				X
Observations	69,445	69,445	102,466	30,917	102,210	102,210

Notes: Capital sale proceeds are cash proceeds from sales of property, plant, and equipment (sppe); investment is capital expenditures (capx). $\tilde{z}$ is the present value of depreciation deductions from Zwick and Mahon (2017). Columns (2) and (6) include two-digit NAICS-specific quadratic time trends. The sample consists of U.S. nonfinancial Compustat firms from 1995–2016. Parenthesized values are standard errors clustered by four-digit NAICS industry. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Toward supporting a parallel trends interpretation of the results, I also include specifications for both capital sales (column 2) and investment (column 6) that include quadratic trends for two-digit NAICS industries. This guards against the possibility that industries with different exposure to bonus were concentrated in broad sectors experiencing different underlying trends. The estimates attenuate but retain their signs and statistical significance. I find that firms more exposed to bonus reduced their capital sales by 4.8 percent more and increased investment by approximately 1.1 percent.

The negative response of capital sale proceeds remains when I exclude observations with missing or negative reported proceeds and when I restrict the sample to firms with proceeds observed in every sample year, although the extensive-margin estimates become less precise under these restrictions.

24. I do not report an extensive-margin investment result because practically all Compustat firms report positive investment.

Table F.2: Bonus Depreciation: Timing of Investment and Capital Sales

	Capital Sale Proceeds (sppe)			Investment (capx)
	Unconditional	Any Proceeds	Positive Proceeds	Unconditional
	(1)	(2)	(3)	(4)
Panel A: Inherited-cohort timing				
$\tilde{z}_{t-1}$	-10.134***	-0.577**	-1.466*	
	(1.628)	(0.269)	(0.862)	
$\tilde{z}_t$				3.242**
				(1.632)
Panel B: Joint current and lagged exposure				
$\tilde{z}_t$	-8.600***	-0.371**	-2.483	2.779**
	(2.277)	(0.148)	(1.708)	(1.186)
$\tilde{z}_{t-1}$	-4.344***	-0.343*	-0.048	0.743
	(0.732)	(0.192)	(0.770)	(0.761)
Sum	-12.944***	-0.714**	-2.531*	3.522*
	(2.262)	(0.319)	(1.450)	(1.904)
Estimator	PPML	LPM	PPML	PPML
Firm FE	X	X	X	X
Year FE	X	X	X	X
Panel A observations	64,497	96,323	28,926	96,070
Panel B observations	64,497	96,323	28,926	96,070

Notes: Panel A isolates the model's inherited-cohort timing: capital sale outcomes use the previous year's bonus generosity, $\tilde{z}_{t-1}$, while investment uses current generosity, $\tilde{z}_t$, over the same lag-available years, 1996–2016. Panel B includes current and lagged generosity jointly, allowing for the contemporaneous equilibrium response; Sum reports their linear combination. All specifications include firm and year fixed effects. Parenthesized values are standard errors clustered by four-digit NAICS industry. Columns (1), (3), and (4) are estimated by Poisson pseudo-maximum likelihood, and column (2) is a linear probability model. Column (3) conditions on positive proceeds. Capital sale proceeds are winsorized at the 3rd and 97th percentiles. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

F.2 Tax Cuts and Jobs Act

In the main text, I argue that the Tax Cuts and Jobs Act released capital through extension of full bonus to used capital. This subsection examines the response of capital sale proceeds to the combined reform using firm-level data from Compustat.

Following Crawford and Markarian (2024) and Chodorow-Reich et al. (2025), I construct a variety of tests comparing US-domiciled firms, which are exposed to the reform, to a control

group of foreign firms. For the first test, I estimate

$$\mathbb{E}\left[Y_{it} \mid \mathrm{US}_i, \mathrm{Post}_t, \alpha_i, \gamma_t\right] = \exp\left[\alpha_i + \gamma_t + \beta\left(\mathrm{US}_i \times \mathrm{Post}_t\right)\right], \tag{F.2}$$

where Post_t indicates fiscal years 2018 and 2019, α_i are firm fixed effects, and γ_t are year fixed effects. I estimate (F.2) by PPML for capital sale proceeds and investment and use a linear probability model for whether the firm reports positive proceeds. Standard errors are clustered by firm. The identifying assumption is that, absent the TCJA, outcomes for U.S. and foreign firms represented in the same industries would have changed similarly. Because some foreign firms have U.S. operations, any exposure of the control group to the reform would tend to attenuate the estimated difference. The results in Table F.3 show that investment and capital sale proceeds both increased following TCJA, drawing a sharp contrast to the effects of bonus. The investment results are similar to those obtained by a variety of studies also showing higher investment following TCJA (Chodorow-Reich et al. 2025; Kennedy et al. 2023; Hartley, Hassett, and Rauh 2025; Crawford and Markarian 2024).

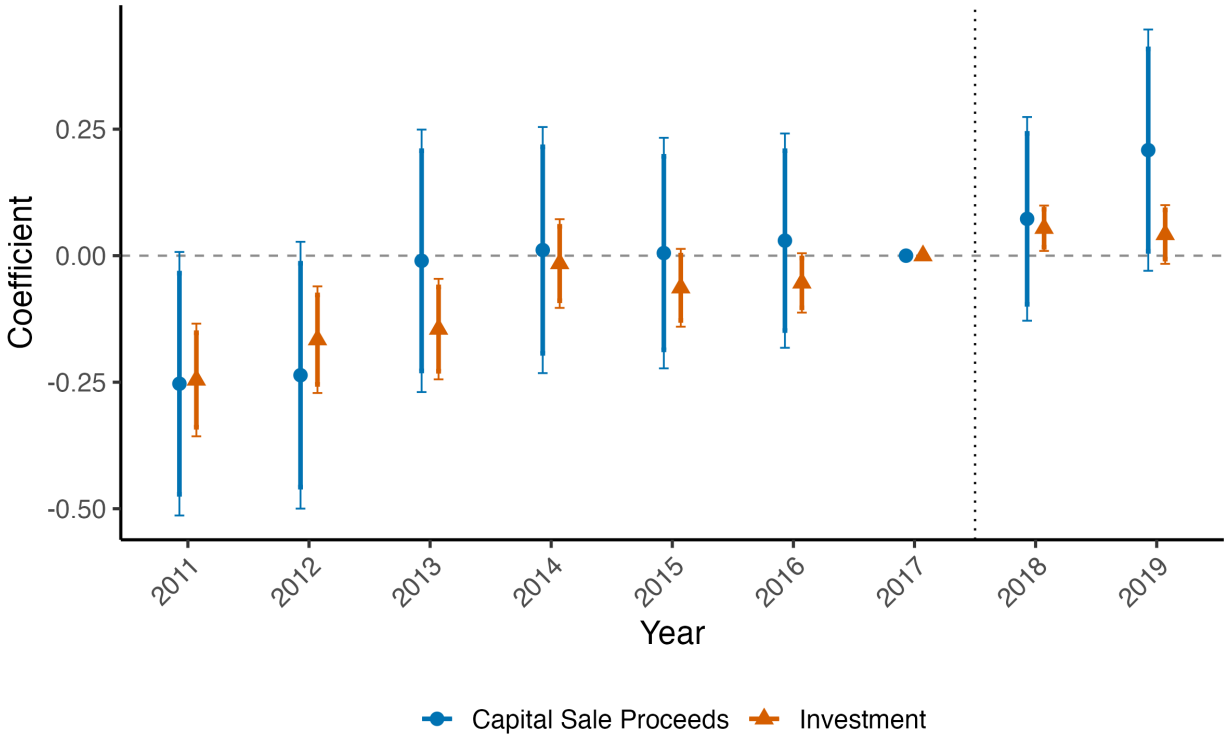
Table F.3: TCJA, Capital Sale Proceeds, and Investment

	Capital Sale Proceeds (sppe)			Investment (capx)
	Unconditional	Any Proceeds	Positive Proceeds	Unconditional
	(1)	(2)	(3)	(4)
Post-TCJA × US	0.196**	-0.015	0.123*	0.144***
	(0.084)	(0.019)	(0.069)	(0.033)
Estimator	PPML	LPM	PPML	PPML
Firm FE	X	X	X	X
Year FE	X	X	X	X
Observations	17,480	17,480	10,310	17,471

Notes: Capital sale proceeds are cash proceeds from sales of property, plant, and equipment (sppe); investment is capital expenditures (capx). Post-TCJA indicates fiscal years 2018 and later, and US identifies U.S.-domiciled firms; foreign-domiciled Compustat firms form the control group. The sample spans 2011–2019. Parenthesized values are standard errors clustered by firm. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The difference-in-differences analysis in Table F.3 relies on parallel trends for the comparison to be valid. Appendix Figure F.1 gives an event study version of the table. The capital sale proceeds coefficients are close to zero from 2013 through 2016 and are jointly insignificant, consistent with parallel trends over that interval. The 2011 and 2012 coefficients are more negative, however, making the evidence less supportive over the full sample period. The investment coefficients instead rise toward zero before enactment and are jointly different from zero, so the investment event study does not support the same interpretation. Both outcomes turn positive after enactment, although the year-specific sales estimates are imprecise.

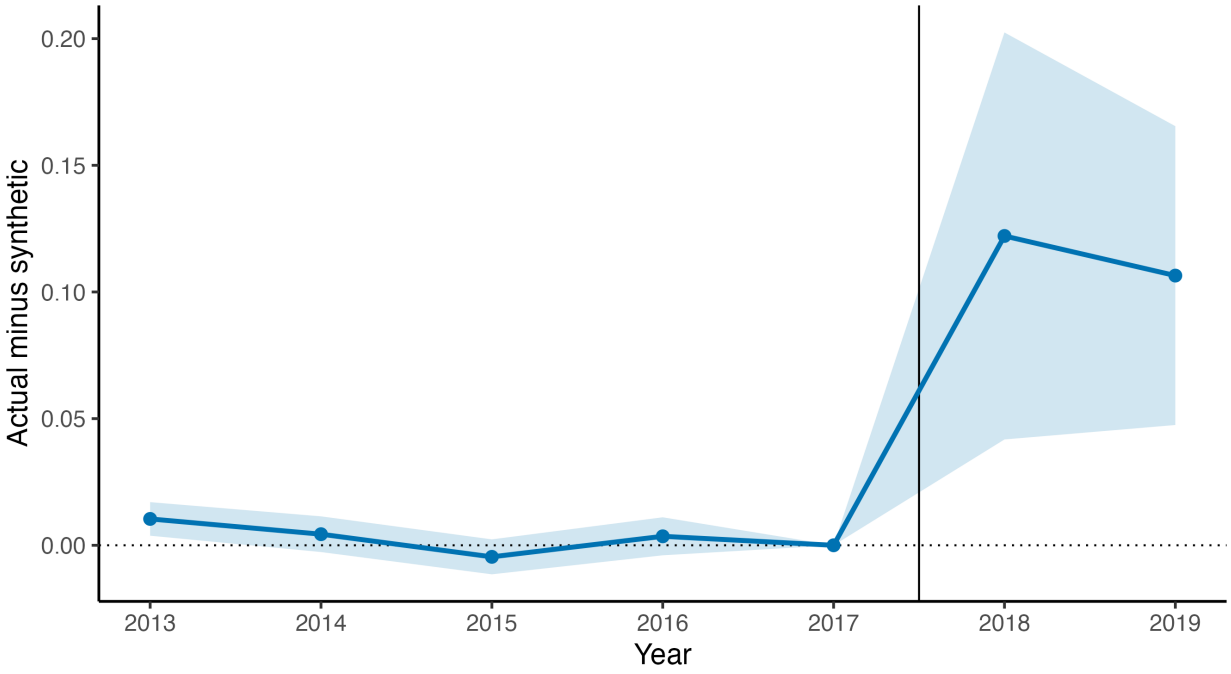
Figure F.1: Event Study of Capital Sales and Investment Around TCJA



Notes: This figure plots event-study coefficients for U.S. firms relative to non-U.S. firms around the TCJA. Outcomes are capital sales (sppe) and capital expenditures (capx), estimated by Poisson pseudo-maximum likelihood with firm and year fixed effects. The reference year is 2017. Thick bars report 90% confidence intervals and thin bars report 95% confidence intervals from standard errors clustered by firm. The vertical dotted line marks enactment of the TCJA.

In another test, I construct a firm-specific synthetic control using foreign firms from the same three-digit industry. I construct a synthetic U.S. firm by matching foreign firms on pre-TCJA capital sales, investment, revenue, the stock of property, plant, and equipment, and total assets. Figure F.2 shows that among the firms with close matches, the actual-minus-synthetic gap is approximately zero throughout 2013–2017 and increases considerably in both post-reform years. Compared to their synthetic counterparts, U.S. firms increase capital sales by roughly 0.1 inverse-hyperbolic sine units.

Figure F.2: U.S. Capital Sales Relative to Synthetic Control around the 2017 Tax Cuts and Jobs Act



Note: This figure plots coefficients from a regression of the firm-level gap between actual and synthetic capital sales, asinh(SPPE) − asinh($SPPE^{synth}$), on year indicators with firm fixed effects, with 2017 as the reference year. Each U.S. firm's synthetic control is built from foreign firms listed on U.S. exchanges in the same 3-digit NAICS industry, and the sample keeps firms whose mean absolute pre-period gap is below 0.10. The shaded region is the 95% confidence interval from standard errors clustered by firm. The vertical line marks the enactment of the TCJA.

Supplemental Appendix

This supplement develops a variant of the model with entry and reports supporting evidence on firm entry.

S.1 Reallocation and Entry in General Equilibrium

S.1.1 Environment

There are two types of firms: incumbents which own new and used capital, and entrants which can only purchase used capital.

Incumbents

I study a unit mass of identical incumbents in discrete time $t = 0, 1, \ldots$, which discount future cash flows using discount factor $\beta \in (0, 1)$. The output good is the numeraire. A representative firm produces output by combining "new" and "old" machines. At the beginning of period t, it purchases

$$I_t = K_t^n$$

units of new capital, which are immediately available for production and are elastically supplied at a price of one. In period $t + 1$, this capital becomes old capital:

$$K_{t+1}^o = K_t^n = I_t.$$

Each physical unit of old capital provides $\phi \in (0, 1)$ efficiency units of capital. After producing as old capital in period $t + 1$, the cohort is retired in period $t + 2$ and has no productive or resale value. Thus, the lifecycle of capital purchased in period t is

$$\underbrace{I_t = K_t^n}_{\substack{\text{new capital} \\ \text{produces at } t}} \quad \Longrightarrow \quad \underbrace{K_{t+1}^o = K_t^n}_{\substack{\text{old capital} \\ \phi K_{t+1}^o \text{ efficiency units at } t+1}} \quad \Longrightarrow \quad \underbrace{0}_{\text{retired at } t+2}.$$

At the beginning of period t, the firm may sell some of its inherited old capital to another firm at a price q_t per physical machine, which the firm takes as given. Because $K_t^o = K_{t-1}^n$, if the firm sells S_t units, it retains

$$\widetilde{K}_t^o = K_t^o - S_t = K_{t-1}^n - S_t \geq 0. \tag{S.1}$$

In words, the total stock of old capital K_t^o, the amount of that sold S_t, and the remainder $\widetilde{K}_t^o$ are measured in physical units, while $\phi \widetilde{K}_t^o$ measures the efficiency units supplied by retained old capital. Putting that together, output is

$$Y_t = F\left(K_t^n, \phi \widetilde{K}_t^o\right) = F\left(K_t^n, \phi\left(K_{t-1}^n - S_t\right)\right).$$

I impose standard regularity conditions on the production technology.

Assumption 1 (Technology)**.** *The production function $F : \mathbb{R}^2_+ \to \mathbb{R}_+$ is twice continuously differentiable on $\mathbb{R}^2_{++}$, strictly increasing, and concave. Additionally, the Hessian of F is negative definite, and F satisfies the Inada conditions in each input.*

I impose no sign restriction on F_{12}. New capital and the efficiency services of retained old capital may therefore be complements, $F_{12} > 0$, or substitutes, $F_{12} < 0$. These curvature conditions make the local comparative statics well-defined at an interior allocation.

The firm's tax treatment has three relevant components. First, operating income is taxed at rate $\tau \in [0, 1)$.

Second, the firm faces a tax depreciation schedule. Its ordinary schedule, which is the stylized model counterpart to MACRS, is

$$\bar{d}_0, \quad \bar{d}_1, \quad \bar{d}_2 \geq 0, \qquad \bar{d}_0 + \bar{d}_1 + \bar{d}_2 = 1.$$

If the firm purchases a unit of capital in period t, a fraction $\bar{d}_0$ would be deducted immediately, $\bar{d}_1$ in the following period, and the remainder is deducted upon retirement. I am interested in how bonus depreciation, which brings those deductions forward in time, affects the firm's decisions about investment and sales of capital. I model bonus depreciation as

$$d_0(\theta) = \theta + (1 - \theta)\bar{d}_0, \qquad d_j(\theta) = (1 - \theta)\bar{d}_j, \quad j = 1, 2. \tag{S.2}$$

When $\theta = 0$, the firm faces the ordinary depreciation schedule and when $\theta = 1$, all new investment is immediately tax-deductible. Intermediate values pull a proportional amount of depreciation forward to the present.

Third, I assume that the firm can use all deductions at rate τ and that gains and losses from asset sales are taxed symmetrically. Suppose a machine purchased in the previous period is sold in period t. Then its remaining basis immediately before sale is $d_1(\theta) + d_2(\theta)$. Mirroring the actual treatment of recapture in the tax code, the seller's after-tax proceeds from selling the capital are

$$(1 - \tau)q_t + \tau \left[d_1(\theta) + d_2(\theta) \right].$$

Conversely, if the old capital is retained rather than sold, the firm deducts $d_1(\theta)$ in period t, leaving $d_2(\theta)$ to be deducted when the firm retires the capital in period $t + 1$.

Optimality. Suppose $q_t = q$ is fixed over time. The firm's Bellman becomes

$$\begin{aligned} V(K^n_{t-1}) = \max_{\substack{K^n_t \geq 0 \\ 0 \leq S_t \leq K^n_{t-1}}} \Big\{ & (1 - \tau)F(K^n_t, \phi\widetilde{K}^o_t) - (1 - \tau d_0(\theta))K^n_t \\ & + \big[(1 - \tau)q + \tau(d_1(\theta) + d_2(\theta))\big]S_t + \tau d_1(\theta)\widetilde{K}^o_t \\ & + \beta\big[\tau d_2(\theta)\widetilde{K}^o_t + V(K^n_t)\big]\Big\}, \end{aligned} \tag{S.3}$$

subject to (S.1). At an interior stationary allocation, write $K^o = \widetilde{K}^o_t$ for the quantity of old capital retained by the firm. The optimality conditions are

$$F_1(K^n, \phi K^o) = 1 + \frac{\tau(1-\beta)}{1-\tau}\big(d_1(\theta) + d_2(\theta)\big) - \beta q, \tag{S.4}$$

$$\phi F_2(K^n, \phi K^o) = q + \frac{\tau(1-\beta)}{1-\tau} d_2(\theta). \tag{S.5}$$

Equation (S.4) governs investment in new capital, while (S.5) governs the decision to sell or retain old capital. The optimality condition for new investment states that its marginal product is equal to its marginal cost. While the marginal cost involves two standard terms—the unit price of new investment and the cost of delayed tax deductions—it also involves a term βq. This captures the option value of being able to sell capital in the following period, which encourages additional investment today. The second optimality condition relates the marginal product of retaining a unit of old capital in the firm to its marginal cost, which is the foregone proceeds from selling the capital. That includes both the sale price q as well as the tax benefit of recognizing d_2 one period earlier by selling.[25] Studying the optimality conditions individually reveals a tension. When bonus goes up, $d_1(\theta)$ and $d_2(\theta)$ decline. Holding old capital fixed, the first change reduces the marginal cost of new investment, which stimulates more capital creation. But holding new capital fixed, the reduction in $d_2(\theta)$ reduces the benefit of selling before retirement by effectively increasing depreciation recapture on sale. Because capital sales satisfy

$$S = K^n - K^o,$$

the effect on sales depends on which response is larger. To characterize these responses, I first assume a regularity condition which guarantees that bonus increases investment in new capital holding q fixed, provided $\tau > 0$, $\theta < 1$, and $d_1(\theta) > 0$. This assumption follows a large literature which illustrates this empirically (House and Shapiro 2008; Kitchen and Knittel 2011; Zwick and Mahon 2017). The proof below shows that, in order for investment to rise with bonus, the following condition is sufficient.

Assumption 2 (Investment response)**.** *At the stationary allocations considered below,*

$$F_{12} \geq \phi F_{22}.$$

Intuitively, bonus lowers the marginal cost of new investment and retained old capital. When the two inputs are complements, the resulting increase in retention reinforces the investment response; when they are substitutes, the retention response reduces the marginal product of new capital and partially offsets the direct investment incentive. Assumption 2 rules out substitution strong enough to reverse that response.

S.1.2 Entrants

To close the used capital market, I assume that each period a unit mass of potential entrants decides whether to operate. An active entrant purchases U physical units of used capital from

25. The deduction d_1 is received in the current period whether the machine is sold or retained and therefore cancels.

incumbents and produces for one period according to $h(\phi U)$, after which the machines retire. Thus, each physical unit supplies the same ϕ efficiency units whether it is retained by the incumbent or sold to an entrant. This reflects evidence from Ma, Murfin, and Pratt (2022) that the buyers of used capital are young, financially constrained firms who struggle to acquire new capital, though of course it is a simplification. The assumption of homogeneous technology is likewise a simplification, though it may be reasonable in the relevant settings. In the UCC data I use, the most active industries are in agriculture, trucking, and related fields, while Kermani and Ma (2022) show that industries like agriculture have the most liquid secondary markets for used capital. Those industries have relatively homogeneous entrants.

Entrants do not accumulate productive capital or become incumbents. Although their decision is repeated each period, the entrant problem is static, so I suppress time subscripts throughout this section. Each potential entrant draws a deductible fixed operating cost c from a continuous distribution G on $\mathbb{R}_+$; that cost governs entry. To ensure that entrants are well-behaved, I make assumptions about both h and G:

Assumption 3 (Entrant Regularity)**.** *h and G are governed by the following regularity conditions:*

(i) The function $h : \mathbb{R}_+ \to \mathbb{R}_+$ is continuous on $\mathbb{R}_+$, twice continuously differentiable and strictly increasing on $\mathbb{R}_{++}$, and satisfies $h''(x) < 0$ for every $x > 0$, $h(0) = 0$, and

$$\lim_{x \downarrow 0} h'(x) = \infty, \qquad \lim_{x \to \infty} h'(x) = 0.$$

(ii) G is continuously differentiable with positive density over its support, with $G(0) = 0$ and $0 < G(c) < 1$ for every finite $c > 0$.

Entrants are taxed at the same corporate tax rate as incumbents. When an entrant purchases an asset, it receives a new tax basis equal to its purchase price q. The entrant may deduct a share $\theta^U \in [0, 1]$ of q in the current period and the remainder when the capital is retired in the following period. I hold θ^U fixed when studying bonus depreciation for new capital and consider comparative statics for it below. Conditional on operating, the entrant solves

$$\pi(c) = \max_{U \geq 0} \left\{ (1 - \tau) \left[h(\phi U) - c \right] - q(1 - \tau\theta^U)U + \beta\tau(1 - \theta^U)qU \right\}. \tag{S.6}$$

An active entrant's interior choice satisfies

$$\phi h'(\phi U) = q \left[1 + \frac{\tau(1 - \beta)}{1 - \tau} (1 - \theta^U) \right]. \tag{S.7}$$

The right-hand side is the tax-adjusted cost of used capital.

The purchase price is q, while the second term reflects the cost of postponing the deduction for the share $1 - \theta^U$. Holding τ and θ^U fixed leaves the tax adjustment unchanged. Thus, bonus depreciation for new capital does not shift entrant demand directly; it affects entrants through the equilibrium used price q.

Because c is a fixed cost, it does not affect the optimal choice of U, and all active entrants purchase the same quantity. Using (S.7), the marginal entrant's zero-profit condition is

$$\bar{c} = h(\phi U) - \phi h'(\phi U) U. \tag{S.8}$$

An entrant operates if and only if $c \leq \bar{c}$. Entrants below the cutoff earn positive profits, while only the marginal entrant earns zero. The active mass, and hence the entry rate, is therefore $\lambda = G(\bar{c})$, and aggregate demand for used capital is λU.

S.1.3 Stationary Competitive Equilibrium

To obtain transparent analytical comparative statics, I consider constant policies and evaluate the incumbent's dynamic problem along a stationary path. Prices and allocations are then constant, while the incumbent continues to account for the future resale value of new investment.

Definition 2 (Stationary competitive equilibrium). Given constant policy parameters (τ, θ, θ^U), a stationary competitive equilibrium consists of a set of allocations $\{K^n, K^o, S, U, \bar{c}, \lambda\}$ and a price q such that, taking q as given, the constant choices $K_t^n = K^n$ and $S_t = S$ solve the incumbent's problem from the stationary inherited stock $K_t^o = K^n$, with retained old capital $K^o = K^n - S$.

Conditional on operating, each entrant optimally chooses U, the entry cutoff is given by (S.8), and $\lambda = G(\bar{c})$. Finally, the used capital market clears:

$$S = \lambda U. \tag{S.9}$$

Given policies, individual optimization maps each used price q into incumbent sales $S(q)$ and aggregate entrant demand

$$D(q) \equiv \lambda(q)U(q).$$

The equilibrium price equates these two quantities. To ensure that the resulting supply and demand schedules have a unique interior crossing, I impose the following condition.

Assumption 4 (Stationary used capital market)**.** *The set of used prices at which the stationary incumbent and entrant problems admit finite interior solutions is an interval. At every incumbent allocation induced by a price in this interval,*

$$F_{11} + \phi(1+\beta)F_{12} + \beta\phi^2 F_{22} < 0, \tag{S.10}$$

where the derivatives of F are evaluated at $(K^n, \phi K^o)$. Moreover, aggregate used capital demand exceeds incumbent sales at some price in this interval, while incumbent sales exceed aggregate demand at a higher price.

Together with the entrant regularity conditions, the assumption guarantees that the supply of used assets slopes upward and the demand for them slopes downward. Along a stationary path, q is both the current sale price of old capital and the anticipated resale price of new investment. Since q enters the incumbent's two optimality conditions, investment and retention therefore adjust together whenever the price changes. Condition (S.10) governs their net effect on incumbent sales by ensuring that incumbent supply slopes upward, which holds automatically when new and retained old capital are substitutes and limits the strength of their complementarity otherwise. By contrast, the regularity conditions for entrants ensure that a higher q raises the tax-adjusted cost of used capital, reducing purchases per entrant, the entry cutoff, and participation, so that aggregate demand slopes downward. The final part of the assumption ensures that the two schedules cross in the interior, which yields the following result.

Lemma 3 (Stationary equilibrium)**.** *Under Assumptions 1, 3, and 4, there exists a unique interior stationary competitive equilibrium. In particular,*

$$K^n > 0, \qquad K^o > 0, \qquad 0 < S < K^n, \qquad U > 0, \qquad 0 < \lambda < 1.$$

Proof. Fix the stationary policy parameters and a used price q in the interval specified by Assumption 4. Under that assumption, incumbent and entrant choices are finite and interior. Strict concavity of F implies that the two incumbent optimality conditions admit at most one interior stationary allocation at each price.

Differentiating (S.4) and (S.5) with respect to q gives

$$\begin{pmatrix} F_{11} & \phi F_{12} \\ \phi F_{12} & \phi^2 F_{22} \end{pmatrix} \begin{pmatrix} K^n_q \\ K^o_q \end{pmatrix} = \begin{pmatrix} -\beta \\ 1 \end{pmatrix}.$$

Assumption 1 implies

$$F_{11} < 0, \qquad F_{22} < 0, \qquad F_{11}F_{22} - F_{12}^2 > 0.$$

Inverting the system gives

$$K^n_q = -\frac{F_{12} + \beta\phi F_{22}}{\phi\left(F_{11}F_{22} - F_{12}^2\right)}, \tag{S.11}$$

$$K^o_q = \frac{F_{11} + \beta\phi F_{12}}{\phi^2\left(F_{11}F_{22} - F_{12}^2\right)}, \tag{S.12}$$

$$S_q = -\frac{F_{11} + \phi(1+\beta)F_{12} + \beta\phi^2 F_{22}}{\phi^2\left(F_{11}F_{22} - F_{12}^2\right)} > 0, \tag{S.13}$$

where the final inequality follows from Assumption 4. The nonsingular Hessian also makes stationary incumbent supply continuously differentiable in q.

For entrants, strict concavity and the Inada conditions imply that (S.7) determines a unique conditional choice $U(q) > 0$. Differentiation gives

$$U_q = \frac{1 + \dfrac{\tau(1-\beta)}{1-\tau}(1-\theta^U)}{\phi^2 h''(\phi U)} < 0. \tag{S.14}$$

The marginal entrant's cutoff is

$$\bar{c} = h(\phi U) - \phi h'(\phi U)U,$$

so

$$\frac{d\bar{c}}{dU} = -\phi^2 U h''(\phi U) > 0. \tag{S.15}$$

Strict concavity and $h(0) = 0$ also imply

$$h(\phi U) - \phi U h'(\phi U) > 0$$

for every $U > 0$. Hence $\bar{c} > 0$, and Assumption 3 gives

$$0 < \lambda = G(\bar{c}) < 1.$$

Aggregate entrant demand is

$$D(q) = G(\bar{c}(U(q)))U(q).$$

Differentiating yields

$$D_q = \left[G(\bar{c}) + UG'(\bar{c}) \frac{d\bar{c}}{dU} \right] U_q < 0. \tag{S.16}$$

Excess demand $D(q) - S(q)$ is continuous and strictly decreasing on the admissible price interval. The crossing condition in Assumption 4 gives opposite signs at two prices in this interval. The intermediate value theorem gives an interior clearing price, and strict monotonicity makes it unique. The unique conditional choices then give a unique stationary allocation.

At the clearing price, $U > 0$, $\bar{c} > 0$, and $0 < \lambda < 1$. Market clearing gives $S = \lambda U > 0$. Interiority of the incumbent choice gives $K^n > 0$ and $K^o > 0$. Since $S = K^n - K^o$, it follows that $0 < S < K^n$.

Finally, the incumbent Hessian is nonsingular and $h''(\phi U) < 0$, so individual choices are continuously differentiable in prices and policies. Moreover,

$$D_q - S_q < 0.$$

The implicit function theorem applied to market clearing gives a continuously differentiable equilibrium price and allocation under sufficiently small admissible policy changes. Interiority is preserved locally. □

The equilibrium remains interior and varies smoothly under sufficiently small admissible policy changes. This allows the stationary equilibrium conditions to be differentiated in the comparative statics below.

S.1.4 Reallocation and Entry in General Equilibrium

I start by showing that a perturbation to the economy moves reallocation and entry in the same direction.

Proposition 4 (Reallocation and entry)**.** *Under a local change in the model's policy parameters around an interior stationary equilibrium, S, U, $\bar{c}$, and λ move in the same direction.*

Proof. Intuition for the result comes from recognizing that the entrant side of the market can be summarized by purchases per entrant U. Since entrants are homogeneous except over fixed costs, all active participants choose the same quantity of used capital U. The cutoff condition (S.8) then maps U into the fixed cost of the marginal entrant, which earns no profits, and concavity implies

$$\frac{d\bar{c}}{dU} = -\phi^2 U h''(\phi U) > 0.$$

A higher U raises the cutoff $\bar{c}$ and therefore the mass of active entrants $\lambda = G(\bar{c})$, so aggregate entrant demand λU is monotonically increasing in U. By market clearing, capital sales, purchases

per entrant, the entry cutoff, and the entry rate must all move in the same direction. □

Consequently, once the general equilibrium response of reallocation is known, the response of entry follows from Proposition 4.

Given the linkage between reallocation and entry, I now show how the partial equilibrium response of reallocation to bonus depreciation translates into general equilibrium. In general equilibrium, this response survives, but it is attenuated. To see that, let

$$S_\theta^{PE} \equiv \left.\frac{\partial S(q,\theta)}{\partial \theta}\right|_{q \text{ fixed}}$$

denote the partial equilibrium response of reallocation. Because market clearing requires $S(q,\theta) = D(q)$, differentiating gives

$$\frac{dq}{d\theta} = \frac{S_\theta^{PE}}{D_q - S_q}, \qquad \frac{dS}{d\theta} = \frac{D_q}{D_q - S_q} S_\theta^{PE}, \tag{S.17}$$

where $0 < D_q/(D_q - S_q) < 1$ since $S_q > 0$ and $D_q < 0$. Thus, the equilibrium price response attenuates but does not reverse the partial equilibrium response of capital sales.

Proposition 5 (Bonus depreciation and entry)**.** *Suppose $\tau > 0$, $\theta < 1$, and $d_1(\theta) > 0$. Around an interior stationary equilibrium, under Assumptions 1, 2, 3, and 4, a marginal increase in bonus depreciation, holding τ and θ^U fixed, raises investment in new capital. When the lock-in channel dominates, bonus reduces capital sales, purchases per entrant, the entry cutoff, and the entry rate, while raising the used price. This occurs if and only if*

$$d_2(\theta)F_{11} - \phi d_1(\theta)F_{12} - \phi^2\big[d_1(\theta) + d_2(\theta)\big]F_{22} < 0, \tag{S.18}$$

where the derivatives of F are evaluated at $(K^n, \phi K^o)$. When the capital stock expansion channel dominates, the inequality is reversed: capital sales, purchases per entrant, the entry cutoff, and the entry rate rise, while the used price falls.

Proof. The bonus schedule implies

$$d_j'(\theta) = -\frac{d_j(\theta)}{1-\theta}, \qquad j = 1, 2.$$

Differentiating (S.4) and (S.5) with respect to θ, holding q fixed, gives

$$\begin{pmatrix} F_{11} & \phi F_{12} \\ \phi F_{12} & \phi^2 F_{22} \end{pmatrix} \begin{pmatrix} \left.\frac{\partial K^n}{\partial \theta}\right|_q \\ \left.\frac{\partial K^o}{\partial \theta}\right|_q \end{pmatrix} = -\frac{\tau(1-\beta)}{(1-\tau)(1-\theta)} \begin{pmatrix} d_1(\theta) + d_2(\theta) \\ d_2(\theta) \end{pmatrix}. \tag{S.19}$$

Assumption 1 implies

$$F_{11} < 0, \qquad F_{22} < 0, \qquad F_{11}F_{22} - F_{12}^2 > 0.$$

Inverting the system gives

$$\left.\frac{\partial K^n}{\partial \theta}\right|_q = \frac{\tau(1-\beta)}{(1-\tau)(1-\theta)} \frac{d_2(\theta)F_{12} - \phi\left[d_1(\theta) + d_2(\theta)\right]F_{22}}{\phi\left(F_{11}F_{22} - F_{12}^2\right)}, \tag{S.20}$$

and, using $S = K^n - K^o$,

$$S_\theta^{PE} = \frac{\tau(1-\beta)}{(1-\tau)(1-\theta)} \frac{d_2(\theta)F_{11} - \phi d_1(\theta)F_{12} - \phi^2\left[d_1(\theta) + d_2(\theta)\right]F_{22}}{\phi^2\left(F_{11}F_{22} - F_{12}^2\right)}. \tag{S.21}$$

The factor multiplying the numerator in (S.21) is positive. Equation (S.17) preserves its sign and reverses it for the price response. Proposition 4 then gives the corresponding responses of purchases per entrant, the entry cutoff, and entry. At equality in (S.18), these variables and the used price have no first-order response.

It remains to establish the equilibrium investment response. Assumption 2 implies

$$\begin{aligned} &d_2(\theta)F_{12} - \phi\left[d_1(\theta) + d_2(\theta)\right]F_{22} \\ &\quad = d_2(\theta)\left(F_{12} - \phi F_{22}\right) - \phi d_1(\theta)F_{22} > 0. \end{aligned}$$

Hence (S.20) gives a positive investment response holding q fixed.

Differentiating the incumbent conditions with respect to q also gives

$$\frac{\partial K^n}{\partial q} = -\frac{F_{12} + \beta\phi F_{22}}{\phi\left(F_{11}F_{22} - F_{12}^2\right)},$$

and

$$S_q = -\frac{F_{11} + \phi(1+\beta)F_{12} + \beta\phi^2 F_{22}}{\phi^2\left(F_{11}F_{22} - F_{12}^2\right)}.$$

Substituting these derivatives and the fixed-price bonus responses yields

$$S_q \left.\frac{\partial K^n}{\partial \theta}\right|_q - \frac{\partial K^n}{\partial q} S_\theta^{PE} = \frac{\tau(1-\beta)\left[d_1(\theta) + (1+\beta)d_2(\theta)\right]}{(1-\tau)(1-\theta)\phi^2\left(F_{11}F_{22} - F_{12}^2\right)} > 0.$$

The total investment response can be written as

$$\begin{aligned} \frac{dK^n}{d\theta} &= \left.\frac{\partial K^n}{\partial \theta}\right|_q + \frac{\partial K^n}{\partial q}\frac{dq}{d\theta} \\ &= \frac{S_q \left.\dfrac{\partial K^n}{\partial \theta}\right|_q - \dfrac{\partial K^n}{\partial q} S_\theta^{PE}}{S_q - D_q} - \frac{D_q}{S_q - D_q} \left.\frac{\partial K^n}{\partial \theta}\right|_q. \end{aligned}$$

Both terms in the final line are positive because $S_q - D_q > 0$, $D_q < 0$, and the fixed-price investment response is positive. Investment therefore rises in both sales-response cases. □

If lock-in dominates, then bonus contracts incumbent supply. The resulting excess demand

from entrants raises q, which moderates the initial contraction but is not enough to overcome it. Consequently, if lock-in dominates, then entry declines as well. The cumulative UCC evidence in the main text is consistent with a decline in total reallocation, while Compustat provides supporting evidence of lower sale proceeds.

Proposition 6 (Accelerated depreciation for used capital)**.** *Suppose $\tau > 0$, $\beta < 1$, and $\theta^U < 1$. Around an interior stationary equilibrium satisfying Assumptions 1, 3, and 4, a marginal increase in θ^U, holding τ and θ fixed, raises the used price, capital sales, purchases per entrant, the entry cutoff, and the entry rate. It raises investment in new capital if and only if*

$$F_{12} + \beta\phi F_{22} < 0.$$

If this inequality is reversed, new investment falls; at equality it has no first-order response.

Proof. Holding q fixed and differentiating (S.7) with respect to θ^U gives

$$\left.\frac{\partial U}{\partial \theta^U}\right|_q = -\frac{\tau(1-\beta)}{1-\tau}\frac{q}{\phi^2 h''(\phi U)} > 0.$$

Since aggregate demand is strictly increasing in U,

$$\left.\frac{\partial D}{\partial \theta^U}\right|_q = \left[G(\bar{c}) + UG'(\bar{c})\frac{d\bar{c}}{dU}\right]\left.\frac{\partial U}{\partial \theta^U}\right|_q > 0.$$

The reform does not enter incumbent supply directly. Differentiating market clearing therefore gives

$$\frac{dq}{d\theta^U} = \frac{\left.\frac{\partial D}{\partial \theta^U}\right|_q}{S_q - D_q} > 0.$$

It follows that

$$\frac{dS}{d\theta^U} = S_q\frac{dq}{d\theta^U} > 0.$$

Proposition 4 then gives the positive responses of U, $\bar{c}$, and λ.

Finally, used-purchase expensing affects incumbent investment only through the equilibrium price:

$$\frac{dK^n}{d\theta^U} = -\frac{F_{12} + \beta\phi F_{22}}{\phi\left(F_{11}F_{22} - F_{12}^2\right)}\frac{dq}{d\theta^U}.$$

Since $dq/d\theta^U > 0$, the stated condition is necessary and sufficient for investment to rise. □

S.2 Consequences for Firm Entry

The evidence in the main text shows that (i) bonus episodes induce greater investment and reduce capital sales and (ii) bonus-vintage capital has fewer cumulative transactions than nonbonus capital. By Proposition 4, that evidence suggests that entry should decline because reallocation and

entry are linked. Theory also suggests that extending bonus to used purchases should increase entry. This section explores that with data from the Business Dynamics Statistics.

S.2.1 Data

I use the Business Dynamics Statistics (BDS), published annually by the Census Bureau from the Longitudinal Business Database, to study entry. The BDS provides industry-level counts of firms, establishments, entry, and exit for the near-universe of U.S. employer businesses. I use it to test the effects of bonus depreciation on entry, focusing on the 4-digit NAICS level. Within the BDS there are several options for defining entry. For example, one could look at the startup rate for firms or establishments, or one could analyze a stock-flow measure. Guided by the model, in which entry is not necessarily from newly created legal entities, I focus on a stock-flow measure defined by the flow of firms into an industry's active population rather than only the number of firms legally born that year. Letting $F_{N,t}$ denote the number of firms operating in industry N and $D_{N,t}$ the number of firm deaths, the entry rate is

$$\frac{F_{N,t} - F_{N,t-1} + D_{N,t}}{F_{N,t-1}}.$$

At the four-digit industry level, this includes both de novo startups and the net entry of continuing firms into an industry, which is consistent with the broader participation margin in the model. By contrast, the BDS count of age-zero firms captures only de novo startups and therefore imposes a narrower definition of entry than the model requires. To study the effects of bonus on entry, I merge the BDS panel with industry-level bonus depreciation exposure from Zwick and Mahon (2017). This yields 4,984 industry-years across 227 industries from 1996–2017. I also link the data to a time-invariant measure of secondary asset market liquidity from Kermani and Ma (2022).

S.2.2 Bonus Depreciation

The empirical effect of bonus depreciation on entry nevertheless need not be negative. Entrants purchase only used equipment in the model, whereas in practice some potential entrants can purchase new equipment and benefit directly from bonus depreciation. The estimates therefore capture the net effect of these competing forces. The used capital mechanism also motivates a more specific test: whether any decline in entry is larger in industries where capital is more readily traded on secondary markets.

I test these predictions using panel local projections of the four-digit NAICS industry entry rate on the present value of depreciation deductions $\tilde{z}_{N,t}$ from Zwick and Mahon (2017). I estimate

$$y_{N,t+h} - y_{N,t-1} = \alpha_N + \gamma_t + \beta_h \tilde{z}_{N,t} + \text{Controls} + \varepsilon_{N,t}, \tag{S.1}$$

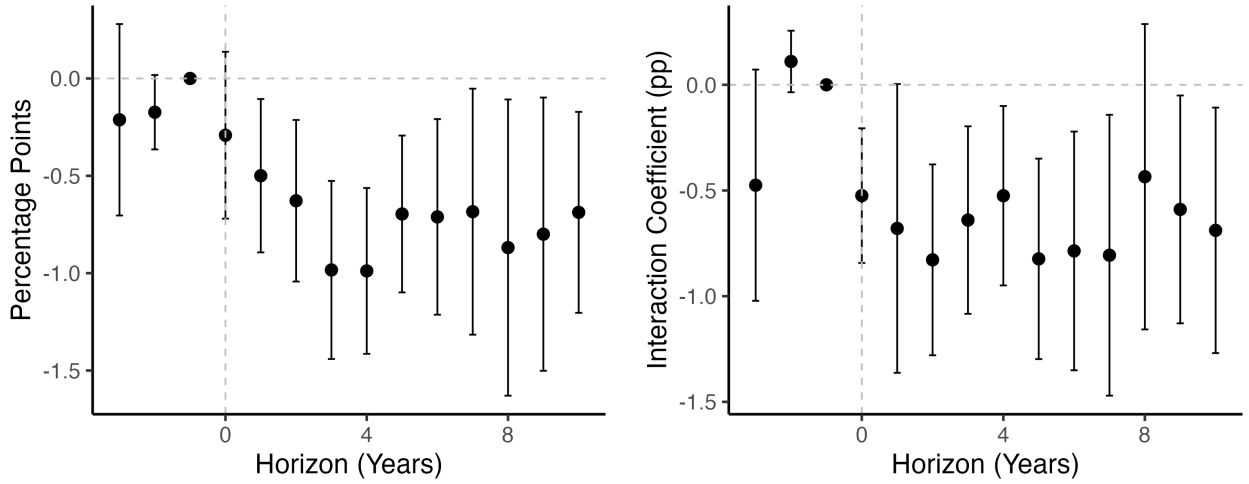
where $y_{N,t}$ is the stock-flow entry rate defined above, α_N and γ_t are industry and year fixed effects, and the controls include two-digit NAICS quadratic time trends. The identifying variation comes from national changes in bonus depreciation interacted with predetermined differences in the assets used by each industry. The identifying assumption is therefore that industries with different exposure would otherwise have experienced similar changes in entry after accounting for the fixed effects and industry trends. Standard errors are clustered by four-digit NAICS industry.

Because bonus policies are persistent, the coefficients trace the dynamic reduced-form response to greater depreciation generosity rather than the effect of an isolated one-year policy shock.

Figure S1 reports the results. Panel (a) shows that neither of the two pre-period coefficients is individually significant at the five-percent level. The entry response is negative on impact but initially imprecise. After one year, it becomes significantly negative, declines to -1 at horizons three and four, and remains significantly negative thereafter. Since identification comes from cross-sectional variation in exposure across industries, a coefficient of -1 means that an industry experiencing a one-percentage-point larger increase in $\widetilde{z}$ experiences a one-percentage-point larger decline in its entry rate, relative to a less-exposed industry. The delayed response is also consistent with the timing of the secondary market mechanism.

While the estimate is economically large, it is consistent with other evidence on the effects of tax policy on entry. In the context of our sample, a 50-percent bonus episode would produce an interquartile difference of approximately 0.73 percentage points in the increase in $\widetilde{z}$. At the three-year horizon, a coefficient of roughly -0.98 would imply a 0.72-percentage-point lower entry rate in a more-exposed industry, relative to a mean entry rate of 7.9 percent. This corresponds to a relative decline of approximately 9 percent. By comparison, Sapollnik and Swonder (2025) find that state corporate tax cuts raise entry by approximately 10 percent after five years, while Fairlie et al. (2026) find that large corporate tax cuts raise employer startups by approximately 5 percent. Thus, my estimate is similar in absolute magnitude, although it has the opposite sign. Nevertheless, that may result from differences in both the policy experiments and level of variation.

Figure S1: Bonus Depreciation and Firm Entry



(a) Entry response

(b) Asset tradeability interaction

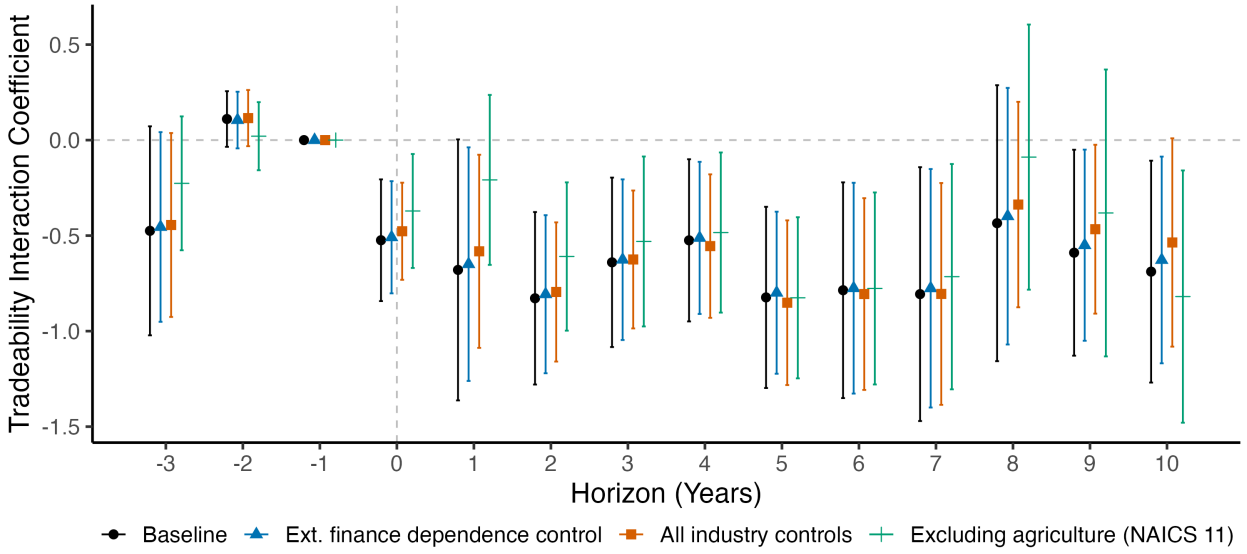
Notes: Panel (a) plots coefficients from panel local projections of the four-digit NAICS entry rate on the present value of depreciation deductions $\tilde{z}_{N,t}$. Panel (b) plots coefficients on the interaction of $\tilde{z}_{N,t}$ with the demeaned PPE recovery rate from Kermani and Ma (2022), expressed as a fraction. Panel (a) includes 227 industries, while panel (b) includes the 186 industries with recovery-rate data. All specifications include industry and year fixed effects and two-digit NAICS quadratic time trends. Bars report 95% confidence intervals using standard errors clustered by four-digit NAICS industry.

The baseline response is consistent with the model, but it does not by itself establish that used

capital is the operative channel. Panel (b) therefore asks whether the decline is larger in industries where secondary markets are more important. I measure asset tradeability using the PPE recovery rate from Kermani and Ma (2022), which reports the fraction of asset value recoverable upon liquidation. Industries with high recovery rates use standardized capital that retains substantial value in resale, while industries with low recovery rates use more specialized capital with limited secondary market value. The interaction is negative across the post-reform horizons and is statistically significant over much of the medium run. The two pre-period interaction coefficients have opposite signs and neither is statistically distinguishable from zero at the five-percent level. Thus, the decline in entry is larger in industries where capital is more readily traded, consistent with the secondary market mechanism.

Recovery rates may also be related to other features of an industry's capital and firms. I therefore estimate specifications that add interactions with average firm size, capital duration, entry cyclicality, and external finance dependence, both individually and jointly. Figure S2 reports representative specifications, and the medium-run tradeability interaction remains negative.

Figure S2: The Entry–Tradeability Interaction: Robustness



Note: This figure reports the coefficient on $\tilde{z}$ interacted with the PPE recovery rate from the panel local projection in Figure S1b. The specifications move from the baseline to a control for external finance dependence, controls for all four competing characteristics, and a sample excluding agriculture (NAICS 11). The competing characteristics are external finance dependence, average firm size, capital duration, and entry cyclicality; external finance dependence is the Rajan–Zingales measure constructed from U.S. Compustat firms over 1980–2001. All specifications include $\tilde{z}$, industry and year fixed effects, and two-digit NAICS quadratic trends. Bars report 95% confidence intervals from standard errors clustered by four-digit industry; horizon −1 is the omitted period.

Altogether, I find that industries more exposed to bonus see larger declines in entry, and those declines are larger in industries with more tradeable capital, although these patterns are more suggestive than conclusive. This pattern is consistent with the model's prediction that reduced circulation makes used capital more costly and discourages entry by marginal firms.

S.2.3 TCJA and Firm Entry

This section examines the effects of the Tax Cuts and Jobs Act on firm entry, which provides a second, weaker test of the entry mechanism. In addition to cutting the corporate tax rate and increasing bonus depreciation for new capital, the Act extended full expensing to purchases of used equipment. Used capital expensing shifts demand outward in the model and therefore increases reallocation and entry, although the other provisions of the Act can generate offsetting effects. Figure S3a shows that entry rises in 2018 and 2019 in industries where the reform most reduced the preexisting tax disadvantage of used equipment. With only two post-reform years before the pandemic, the estimates are imprecise, so I treat this evidence as directional rather than conclusive. Moreover, quantitative evidence from Sedlacek and Sterk (2019) and Zeida (2022) points toward higher entry through other channels like higher returns to entrepreneurship post-TCJA, which I cannot distinguish from here.

By extending full expensing to used purchases, the TCJA narrowed the tax disadvantage of buying used rather than new capital. I measure this change using the tax-adjusted acquisition cost of each type of capital. Let z_N^0 denote the present value of ordinary MACRS deductions for industry N, and let

$$z_N(\theta) = \theta + (1-\theta)z_N^0, \qquad p_N(\theta,\tau) = 1 - \tau z_N(\theta).$$

The pre-TCJA wedge is

$$g_N^{\text{pre}} = 1 - \frac{p_N(0.5, 0.35)}{p_N(0, 0.35)}.$$

Full expensing of both new and used purchases makes the corresponding post-TCJA wedge zero. I therefore define $\Delta g_N = g_N^{\text{pre}} - g_N^{\text{post}} = g_N^{\text{pre}}$, which is larger in industries where the reform did more to lower the relative cost of used capital. I construct z_N^0 following the approach of Zwick and Mahon (2017) by weighting each BEA industry's fixed investment from 2014 through 2016, and mapping the resulting exposure to four-digit NAICS industries.

Let $\widehat{\Delta g}_N$ denote this exposure after demeaning and dividing by its cross-industry standard deviation. I estimate

$$y_{N,t} = \alpha_N + \gamma_t + \sum_{s \neq 2017} \beta_s \widehat{\Delta g}_N \mathbf{1}\{t = s\} + \eta_{n(N)} t + \varepsilon_{N,t}, \tag{S.2}$$

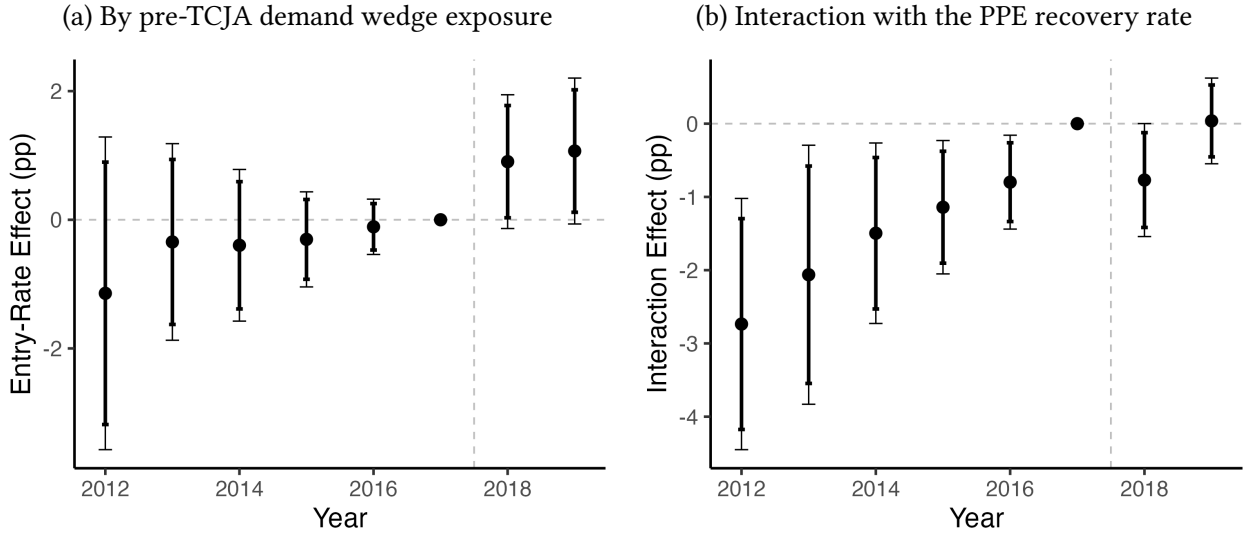
where $\eta_{n(N)} t$ denotes two-digit NAICS linear trends. Standard errors are clustered by four-digit NAICS industry. The sample contains 201 industries from 2012 through 2019.

Results. Panel (a) of Figure S3 plots β_s. The coefficients from 2013 through 2016 are close to zero; the 2012 coefficient is more negative but imprecise. They rise to 0.9 percentage points in 2018 and 1.1 percentage points in 2019 for a one-standard-deviation change in the wedge. Both post-TCJA estimates are statistically significant at the 10% level but not the 5% level. With only two post-reform years before COVID, I interpret this evidence as suggestive.

Panel (b) asks whether the response is larger in industries with more tradeable capital. I add year interactions with the demeaned recovery rate and with its interaction with $\widehat{\Delta g}_N$. I scale the interaction to a one-standard-deviation change in the wedge and a ten-percentage-point difference in the recovery rate. The pretrends are negative and converge toward 2017 as TCJA

approaches. Because this differential pretrend undermines a parallel-trends interpretation, the post-TCJA interaction does not provide a credible separate test of the tradeability mechanism.

Figure S3: Firm Entry Around the TCJA



Notes: Panel (a) reports β_s from (S.2). Panel (b) reports year-specific interactions between $\widehat{\Delta g}_N$ and the demeaned PPE recovery rate from Kermani and Ma (2022); the regression also includes the corresponding year interactions with each variable separately. Both specifications include four-digit NAICS and year fixed effects and two-digit NAICS linear trends. Panel (a) includes 201 industries, while panel (b) includes the 168 industries with recovery-rate data. The reference year is 2017. Thick bars report 90% confidence intervals and thin bars report 95% confidence intervals from standard errors clustered by four-digit NAICS industry.